

Part 3 The Global Trade and Investment Environment

6

INTERNATIONAL TRADE THEORY

Learning Objectives

After reading this chapter, you will be able to:

- LO6-1** Understand why nations trade with each other.
- LO6-2** Summarize the different theories explaining trade flows between nations.
- LO6-3** Recognize why many economists believe that unrestricted free trade between nations will raise the economic welfare of countries that participate in a free trade system.
- LO6-4** Explain the arguments of those who maintain that government can play a proactive role in promoting national competitive advantage in certain industries.
- LO6-5** Understand the important implications that international trade theory holds for management practice.



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opening case

THE RISE OF TAIWAN

Taiwan has benefited more than most countries from globalization. This island nation of some 24 million 100 miles off the coast of mainland China is home to some of the world's largest and most profitable electronics enterprises, including TSMC, the leading global manufacturer of advanced semiconductors, and Foxconn, Apple's primary subcontractor for the iPhone. Living standards are high, with a GDP PPP per capita of around \$65,000, putting Taiwan on the same level as the Netherlands and Sweden.

It wasn't always this way. Back in the 1950s, Taiwan was one of the world's poorer nations. It had an agrarian economy. The island had suffered hardships under Japanese occupation from the late 1800s until 1945. At the end of the second world war, the island was liberated and returned to China. When the Chinese Communist Party gained control of mainland China in 1949 after years of civil war, the defeated government of the Republic of China fled to the island, where it imposed a military dictatorship.

During the 1950s, the Taiwanese government placed tariffs on merchandised imports of around 30 to 40 percent, high by today's standards. The goals were to raise revenue (import tariffs are a tax on importers) and to protect domestic producers from foreign competitors. A further impediment to imports was a licensing system that limited the amount of foreign goods Taiwanese firms could import. Exports were low, at about 5 percent of GDP, and consisted primarily agricultural products such as rice and sugar. The value of the local currency, the Taiwanese dollar, was pegged to the U.S. dollar at a fixed exchange rate. The combination of high inflation in Taiwan along with a strong U.S. dollar (which pulled up the Taiwanese dollar with it) meant that Taiwanese exports were priced out of international markets. For Taiwan, the consequences of these policies included very low economic growth and large fiscal and trade deficits.

Confronted by ongoing economic malaise, in the late 1950s the government decided to change the country's economic policy. At the urging of a foreign-trained Taiwanese economist, the government moved progressively toward an economy that was open to international trade. The value of the Taiwanese dollar against the U.S. dollar was sharply devalued, making Taiwanese exports cheaper. The government established export processing zones. Companies that set up factories in those zones were allowed to import intermediate goods, such as component parts, at much lower tariff rates. This made them more efficient.

The government expected these reforms to result in higher agricultural exports from existing producers. While this did occur, to the government's surprise, there was also a significant response from new exporters—new producers who entered the market and started to sell things overseas that they had never sold before. To begin with, these were basic processed products, such as plywood or cotton cloth. Over time, however, existing and new producers moved up market into finished goods, adding machinery, automobile parts, audio video communication equipment, automobiles, and eventually, high-end electronics. What also helped Taiwan climb the ladder of economic progress was inward investment by foreign multinationals, who started to see the island as a base for producing and exporting goods to the surrounding region. The result was a surge in both imports and exports, and much higher economic growth.

As the economy grew, and the benefits of trade became evident, the government lowered import tariffs across the board and removed many of the import licensing requirements. In 1979, Taiwan ended the peg against the U.S. dollar and allowed its currency to float freely, with the value being determined by market forces. Today, the average tariff on imports of industrial goods into Taiwan is 4.13 percent (although it remains at 15 percent for agricultural products, where local farmers have successfully lobbied for some protection). Trade as a percentage of GDP has increased from 28.6 percent in 1961 to 134 percent in 2022, making Taiwan one of the most trade-dependent nations on earth. The gains from expanded international trade have propelled the Taiwanese economy forward, enabling it to join the ranks of the world's most wealthy nations measured by GDP PPP per capita.

Sources: Taiwan Trade Data from ceicdata.com/en/; Chad Brown, "Taiwan's Risky Trade Opening and How it Paid Off," *Trade Talks*, October 25, 2022; Douglass Irwin, "How Economic Ideas Led to Taiwan's Shift to Export Promotion in 1958," Working Paper, Peterson Institute for International Economics, July 2021.

Introduction

Over the past 70 years, the amount of international trade flowing across national borders has grown dramatically. While historically much of the growth in trade was due to rising cross-border trade for physical goods—including commodities, agricultural products, and manufactured goods such as semiconductor chips—more recently, the value of services that are traded across borders has accelerated. Globally traded services include financial services, entertainment, software offerings, telecommunications services, computing services, and education. Over the last decade, cross-border trade in services has been growing at three times the rate of cross-border trade in manufactured goods.

The motivation for trading both physical goods and services is to realize what economists refer to as the **gains from trade**. When economists talk about the *gains from trade*, they mean that trade allows countries to specialize in the production (and export) of goods and services that they can produce most efficiently, while importing from other nations those goods and services that they cannot produce so efficiently. By increasing the efficiency of resource utilization in the global economy, economists theorize that international trade results in greater economic growth and provides economic benefits to *all* countries that participate in a global trading system. Greater trade in goods and services is a rising tide that lifts all boats. Based on this theory, and the considerable amount of empirical evidence that supports it (see the opening case for an example), economists have long pushed for lower barriers to cross-border trade, initially in physical goods and more recently in services (where substantial barriers often remain).

The opening case discusses how one country, Taiwan, benefited from a switch in policies, away from those that discouraged international trade and toward those that embraced international trade and opened up the economy to greater imports and exports. Over five decades, Taiwan went from being a poor nation with an agrarian economy to a developed nation with a high-technology economy, in part because it realized the gains from trade.

For most of the last 70 years, the arguments in favor of promoting trade through lower barriers to the flow of goods and services across borders were broadly accepted by policymakers in the most economically advanced nations of the world. Under the leadership of countries such as the United States, United Kingdom, Germany, and Japan, countries around the world progressively lowered barriers to cross-border trade in physical goods and, more recently, to trade in services. Policymakers advocated free trade, and to a large extent, they were successful in pushing the world toward a multilateral free trade regime policed by the World Trade Organization. Since 2017, however, this trend has partially reversed. Initially this was because the United States, under the leadership of President Donald Trump, pulled back from its historic commitment to building a multilateral free trade regime and, instead, adopted a posture that might more accurately be described as *managed trade*, where what can be traded, and how much, is determined not just by market forces, but also by government mandates and negotiated agreements. For example, in early 2020 the Trump administration signed a trade deal with China under which the Chinese promised to buy \$200 billion of specific U.S. goods and services, including \$40 billion of agricultural products, in return for the Trump administration agreeing not to impose additional import tariffs on Chinese goods. While some hoped that a new administration in Washington might shift back toward policies promoting free trade, those hopes have been dashed. Far from revoking Trump's trade policies, the Biden administration has continued to pursue them and, if anything, has escalated the trade conflict with China and potentially created conflict with the European Union as well (we will discuss what has happened here in more detail in Chapter 7).

The shift in policy away from free trade toward managed trade and greater protectionism has pushed trade policy back onto center stage in the political arena. The tariffs and other restraints on trade imposed by the United States, and retaliatory tariffs imposed by other nations—including, most notably, China—have restricted the growth of international trade; disrupted or threatened to disrupt global supply chains; raised costs for many companies that source inputs from other nations; restricted export markets; and, economists would argue, resulted in lower economic growth as the gains from trade are reduced. For their part, those in the Trump and Biden administrations who promote these policies argue that, despite any short-term pain from this policy shift, in the long run it will yield significant economic gains and enhance the national security of the United States. Politicians in China and members of the European Union are now making similar arguments with respect to their own nations, suggesting that the world has indeed pivoted away from embracing free trade as a guiding principle for economic policy.

To fully understand the arguments here, and to appreciate the impact that international trade policy can have on businesses both large and small, we need first to develop a thorough understanding of the different approaches toward trade in goods and services. That is the primary function of the current chapter. Herein we focus on the theoretical foundations of trade policy. We will also look at what the economic evidence tells us about the relationship between trade policies and economic growth. In Chapter 7, we chart the development of the world trading system, discuss different aspects of trade policy, and look at how trade policy is managed by national and global institutions. In Chapter 8, we discuss the reasons for foreign direct investment and the

gains from trade

The economic benefits realized when countries specialize in the production (and export) of goods and services that they can produce most efficiently, while importing goods and services that they cannot produce so efficiently from other nations.

government policies adopted to manage foreign investment. We look at foreign direct investment because it is often either an alternative to trade or undertaken to support greater cross-border trade. In Chapter 9, we look at the reasons for creating trading blocs such as the European Union and NAFTA (and its successor, the USMCA), and we discuss how these transnational agreements have worked out in practice. By the time you have finished these four chapters, you should have a very solid understanding of the international trade and investment environment, and you will understand the extremely important impact that trade and investment policies have upon the practice of international business.

An Overview of Trade Theory

We open this chapter with a discussion of mercantilism. Propagated in the sixteenth and seventeenth centuries, the theory of mercantilism advocated that countries should simultaneously encourage exports and discourage imports. Although mercantilism is an old and largely discredited doctrine, its echoes remain in modern political debate and in the trade policies of many countries. Next, we look at Adam Smith's theory of absolute advantage. Proposed in 1776, Smith's theory was the first to explain why unrestricted free trade is beneficial to a country. **Free trade** refers to a situation in which a government does not attempt to influence through quotas or tariffs what its citizens can buy from another country or what they can produce and sell to another country. Smith argued that the invisible hand of the market mechanism, rather than government policy, should determine what a country imports and what it exports. His arguments imply that such a laissez-faire stance toward trade was in the best interests of a country. Building on Smith's work are two additional theories that we review. One is the theory of comparative advantage, advanced by the nineteenth-century English economist David Ricardo. This theory is the intellectual basis of the modern argument for unrestricted free trade. In the twentieth century, Ricardo's work was refined by two Swedish economists, Eli Heckscher and Bertil Ohlin, whose theory is known as the Heckscher-Ohlin theory.

free trade

The absence of barriers to the free flow of goods and services between countries.

THE GAINS FROM TRADE



Understand why nations trade with each other.

The great strength of the theories of Smith, Ricardo, and Heckscher-Ohlin is that they identify with precision the specific benefits of international trade—the gains from trade. Common sense suggests that some international trade is beneficial. For example, nobody would suggest that Iceland should grow its own oranges. Iceland can benefit from trade by exchanging some of the products that it can produce at a low cost (fish) for some products that it cannot produce at all (oranges). Thus, by engaging in international trade, Icelanders are able to add oranges to their diet of fish.

The theories of Smith, Ricardo, and Heckscher-Ohlin go beyond this commonsense notion, however, to show why a country benefits from engaging in international trade *even for products it is able to produce for itself*. This is a difficult concept for people to grasp. For example, many people in the United States believe that American consumers should buy products made in the United States by American companies whenever possible to help save American jobs from foreign competition. The same kind of nationalistic sentiments can be observed in many other countries.

However, the theories of Smith, Ricardo, and Heckscher-Ohlin tell us that a country's economy may gain if its citizens buy certain products from other nations that could be produced at home. The gains arise because international trade allows a country to *specialize* in the

manufacture and export of products that can be produced most efficiently in that country, while importing products that can be produced more efficiently in other countries.



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Trade Tutorials

In this chapter, we discuss benefits and costs associated with free trade, discuss the benefits of international trade, and explain the pattern of international trade in today's world economy. The general idea is that international trade theories explain why it can be beneficial for a country to engage in trade across country borders, even though countries are at different stages of development, have different product needs, and produce different types of products. International trade theory assumes that countries—through their governments, laws, and regulations—encourage or discourage trade across borders. As related to this chapter, check out the globalEDGE™ “trade

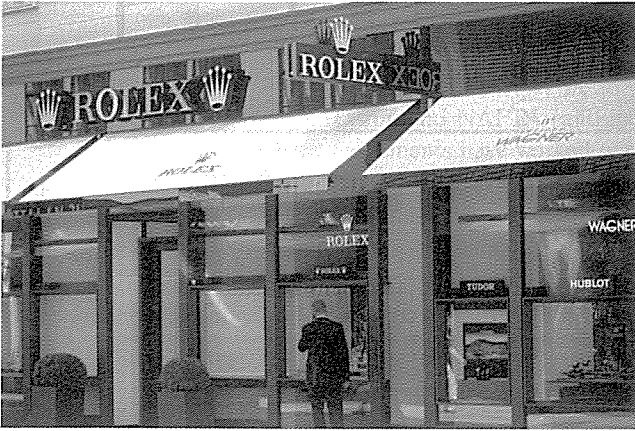
tutorials” section, where lots of information, data, and tools are compiled related to trading internationally (globaleledge.msu.edu/global-resources/trade-tutorials). The potpourri of trade resources includes export tutorials, online course modules, a glossary, a free trade agreement tariff tool, and much more. The glossary includes lots of terms related to trade. For example, “trade surplus” is defined as a situation in which a country's exports exceeds its imports (i.e., it represents a net inflow of domestic currency from foreign markets). The opposite is called trade deficit and is considered a net outflow, but how is it really defined? The globalEDGE™ glossary can help.

Thus, U.S. specialization in the production and export of commercial jet aircraft may make sense because the efficient production of commercial jet aircraft requires resources that are abundant in the United States, such as a highly skilled labor force and cutting-edge technological know-how. On the other hand, U.S. importation of textiles from Bangladesh may make sense because the efficient production of textiles requires a relatively cheap labor force—and cheap labor is not abundant in the United States.

Of course, this economic argument is often difficult for segments of a country's population to accept. With their future threatened by imports, U.S. textile companies and their employees have tried hard to persuade the government to limit the importation of textiles by demanding quotas and tariffs. Although such import controls may benefit particular groups, such as textile businesses and their employees, the theories of Smith, Ricardo, and Heckscher-Ohlin suggest that the economy and populace as a whole is hurt by such action. One of the key insights of international trade theory is that limits on imports are often in the interests of domestic producers but not domestic consumers.

THE PATTERN OF INTERNATIONAL TRADE

The theories of Smith, Ricardo, and Heckscher-Ohlin help explain the pattern (or geography) of international trade that we observe in the world economy. Some aspects of the pattern are easy to understand. Climate and natural resource endowments explain why Ghana exports cocoa, Brazil exports coffee, Saudi Arabia exports oil, and China exports crawfish. However, much of the observed pattern of international trade is more difficult to explain. For example, why does Japan export automobiles, consumer electronics, and machine tools? Why does Switzerland



A Rolex Group logo sits on display above a luxury wristwatch store in Vienna, Austria.

Bloomberg/Getty Images

new trade theory

The observed pattern of trade in the world economy may be due in part to the ability of firms in a given market to capture first-mover advantages.

factor endowments

A country's endowment with resources such as land, labor, and capital.

export chemicals, pharmaceuticals, watches, and jewelry? Why does Bangladesh export garments? Why does the United States export large commercial jet aircraft? David Ricardo's theory of comparative advantage offers an explanation in terms of international differences in productivity. The more sophisticated Heckscher-Ohlin theory emphasizes the interplay between the proportions in which the factors of production (such as land, labor, and capital) are available in different countries and the proportions in which they are needed for producing particular goods. This explanation rests on the assumption that countries have varying endowments of the various factors of production. Tests of this theory, however, suggest that it is a less powerful explanation of real-world trade patterns than once thought.

One early response to the failure of the Heckscher-Ohlin theory to explain the observed pattern of international trade was the product life-cycle theory. Proposed by Raymond Vernon, this theory suggests that early in their life cycle, most new products are produced in and exported from the country in which

they were developed. As a new product becomes widely accepted internationally, however, production starts in other countries. As a result, the theory suggests, the product may ultimately be exported back to the country of its original innovation.

In a similar vein, during the 1980s, economists such as Paul Krugman developed what has come to be known as the new trade theory. **New trade theory** (for which Krugman won the Nobel Prize in economics in 2008) stresses that in some cases, countries specialize in the production and export of particular products not because of underlying differences in factor endowments but because in certain industries the world market can support only a limited number of firms. (This is argued to be the case for the commercial aircraft industry, and in the most advanced segment of the global semiconductor industry.) In such industries, firms that enter the market first are able to build a competitive advantage that is subsequently difficult to challenge. Thus, the observed pattern of trade between nations may be due in part to the ability of firms within a given nation to capture first-mover advantages. The United States is a major exporter of commercial jet aircraft because American firms such as Boeing were first movers in the world market. Boeing built a competitive advantage that has subsequently been difficult for firms from countries with equally favorable **factor endowments** to challenge (although Europe's Airbus has succeeded in doing that). Similarly, the United States still has a major position in the development of semiconductor technology and exporting advanced semiconductors, primarily because American firms based in Silicon Valley, such as Intel, captured first-mover advantages. In a work related to the new trade theory, Michael Porter developed a theory referred to as the theory of national competitive advantage. This attempts to explain why particular nations achieve international success in particular industries. In addition to factor endowments, Porter points out the importance of country factors such as domestic demand, domestic rivalry, and a network of local suppliers in explaining a nation's dominance in the production and export of particular products.

TRADE THEORY AND GOVERNMENT POLICY

Although all these theories agree that international trade is beneficial to a country, they lack agreement in their recommendations for government policy, as we shall see in the following sections. Mercantilism makes a case for government involvement in promoting exports and limiting imports. The theories of Smith, Ricardo, and Heckscher-Ohlin form part of the case for unrestricted free trade. The argument for unrestricted free trade is that both import controls and export incentives (such as subsidies) are self-defeating and result in wasted resources. Both the new trade theory and Porter's theory of national competitive advantage can be interpreted as justifying some limited government intervention to support the development of certain export-oriented industries. We discuss the pros and cons of this argument, known as strategic trade policy, as well as the pros and cons of the argument for unrestricted free trade, in Chapter 7.



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Mercantilism



Summarize the different theories explaining trade flows between nations.

The first theory of international trade, mercantilism, emerged in England in the mid-sixteenth century. The principle assertion of mercantilism was that gold and silver were the mainstays of national wealth and essential to vigorous commerce. At that time, gold and silver were the currency of trade between countries; a country could earn gold and silver by exporting goods. Conversely, importing goods from other countries would result in an outflow of gold and silver to those countries. The main tenet of **mercantilism** was that it was in a country's best interests to maintain a trade surplus, to export more than it imported. By doing so, a country would accumulate gold and silver and, consequently, increase its national wealth, prestige, and power. As the English mercantilist writer Thomas Mun put it in 1630:

The ordinary means therefore to increase our wealth and treasure is by foreign trade, wherein we must ever observe this rule: to sell more to strangers yearly than we consume of theirs in value.¹

Consistent with this belief, the mercantilist doctrine advocated government intervention to achieve a surplus in the balance of trade. The mercantilists saw no virtue in a large volume of trade. Rather, they recommended policies to maximize exports and minimize imports. To achieve this, imports were limited by tariffs and quotas, while exports were subsidized.

The classical economist David Hume pointed out an inherent inconsistency in the mercantilist doctrine in 1752. According to Hume, if England had a balance-of-trade surplus with France (it exported more than it imported), the resulting inflow of gold and silver would swell the domestic money supply and generate inflation in England. In France, however, the outflow of gold and silver would have the opposite effect. France's money supply would contract, and its prices would fall. This change in relative prices between France and England would encourage the French to buy fewer English goods (because they were becoming more expensive) and the English to buy more French goods (because they were becoming cheaper). The result would be a deterioration in the English balance of trade and an improvement in France's trade balance, until the English surplus was eliminated. Hence, according to Hume, in the long run, no country could sustain a surplus on the balance of trade and so accumulate gold and silver as the mercantilists had envisaged.

The flaw with mercantilism was that it viewed trade as a zero-sum game in which the size of the economic pie was fixed. (A **zero-sum game** is one in which a gain by one country results in a loss by another.) Economists Adam Smith and David Ricardo showed the limitations of this approach and demonstrated that trade is a positive-sum game, or a situation in which all countries can benefit. Despite this, the mercantilist doctrine is by no means dead. For example, former U.S. President Donald Trump appeared to advocate neo-mercantilist policies.² Neo-mercantilists equate political power with economic power and economic power with a balance-of-trade surplus. Critics argue that several nations have

mercantilism

An economic philosophy advocating that countries should simultaneously encourage exports and discourage imports.

zero-sum game

A situation in which an economic gain by one country results in an economic loss by another.

adopted a neo-mercantilist strategy that is designed to simultaneously boost exports and limit imports.³ For example, they charge that China long pursued a neo-mercantilist policy, deliberately keeping its currency value low against the U.S. dollar in order to sell more goods to the United States and other developed nations, and thus amass a trade surplus and foreign exchange reserves (see the accompanying Country Focus).

COUNTRY FOCUS

Is China Manipulating Its Currency in Pursuit of a Neo-Mercantilist Policy?

China's rapid rise in economic power has been built on export-led growth. For decades, the country's exports have been growing faster than its imports. This has led some critics to claim that China is pursuing a neo-mercantilist policy, trying to amass record trade surpluses and foreign currency that will give it economic power over developed nations. By the end of 2014, its foreign exchange reserves exceeded \$3.8 trillion, some 60 percent of which were held in U.S.-denominated assets such as U.S. Treasury bills. Observers worried that if China ever decided to sell its holdings of U.S. currency, that action would depress the value of the dollar against other currencies and increase the price of imports into the United States.

The U.S. trade deficit with China has been a particular cause for concern. In 2017, this reached a record \$375 billion. At the same time, China has long resisted attempts to let its currency float freely against the U.S. dollar. Many have claimed that China's currency has been too cheap and that this keeps the prices of China's goods artificially low, which fuels the country's exports. China, the critics charge, is guilty of currency manipulation.

So is China manipulating the value of its currency to keep exports artificially cheap? The facts of the matter are less clear than the rhetoric. China actually started to allow the value of the yuan (China's currency) to appreciate against the dollar in July 2005, albeit at a slow pace. In July 2005, one U.S. dollar purchased 8.11 yuan. By January 2014 one U.S. dollar purchased 6.05 yuan, which implied a 25 percent increase in the price of Chinese

exports, not what one would expect from a country that was trying to keep the price of its exports low through currency manipulation.

Moreover, in 2015 and 2016, the rate of growth in China started to slow significantly. China's stock market fell sharply, and capital started to leave the country, with investors selling yuan and buying U.S. dollars. To stop the yuan from *declining* in value against the U.S. dollar, China began to spend about \$100 billion of its foreign exchange reserves *every month* to buy yuan on the open market. Far from allowing its currency to decline against the U.S. dollar, thereby giving a boost to its exports, China was trying to prop up its value, running down its foreign exchange reserves by \$2 trillion in the process. This action seems inconsistent with the charges that the country is pursuing a neo-mercantilist policy by artificially depressing the value of its currency. In recognition of these developments, in late 2017 the U.S. Treasury Department declined to name China a currency manipulator and moderated its criticism of the country's foreign exchange policies. On the other hand, the Treasury said that it remained concerned by the lack of progress in reducing China's bilateral trade surplus with the United States.

Sources: S. H. Hanke, "Stop the Mercantilists," *Forbes*, June 20, 2005, p. 164; G. Dyer and A. Balls, "Dollar Threat as China Signals Shift," *Financial Times*, January 6, 2006; R. Silk, "China's Foreign Exchange Reserves Jump Again," *The Wall Street Journal*, October 15, 2013; T. Jeffrey, "U.S. Merchandise Trade Deficit with China Hit Record in 2015," *cnsnews.com*, February 9, 2016; "Trump's Chinese Currency Manipulation," *The Wall Street Journal*, December 7, 2016; E. Holodny, "The Treasury Department Backs Down on Some of Its Criticisms of China's Currency Policies," *Business Insider*, October 18, 2017; A. Swanson, "U.S.-China Trade Deficit Hits Record, Fueling Trade Fight," *The New York Times*, February 6, 2018.



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Absolute Advantage



LO6-2

Summarize the different theories explaining trade flows between nations.

In his 1776 landmark book *The Wealth of Nations*, Adam Smith attacked the mercantilist assumption that trade is a zero-sum game. Smith argued that countries differ in their ability to produce goods efficiently. In his time, the English, by virtue of their superior manufacturing processes, were the world's most efficient textile manufacturers. Due to the combination of favorable climate, good soils, and accumulated expertise, the French had the world's most efficient wine industry. The English had an *absolute advantage* in the production of textiles, while the French had an *absolute advantage* in the production of wine. Thus, a country has an **absolute advantage** in the production of a product when it is more efficient than any other country at producing it.

According to Smith, countries should specialize in the production of goods for which they have an absolute advantage and then trade these goods for those produced by other countries. In Smith's time, this suggested the English should specialize in the production of textiles, while the French should specialize in the production of wine. England could get all the wine it needed by selling its textiles to France and buying wine in exchange. Similarly, France could get all the textiles it needed by selling wine to England and buying textiles in exchange. Smith's basic argument, therefore, is that a country should never produce goods at home that it can buy at a lower cost from other countries. Smith demonstrates that by specializing in the production of goods in which each has an absolute advantage, both countries benefit by engaging in trade.

Consider the effects of trade between two countries, Ghana and South Korea. The production of any good (output) requires resources (inputs) such as land, labor, and capital. Assume that Ghana and South Korea both have the same amount of resources and that these resources can be used to produce either rice or cocoa. Assume further that 200 units of resources are available in each country. Imagine that in Ghana it takes 10 resources to produce 1 ton of cocoa and 20 resources to produce 1 ton of rice. Thus, Ghana could produce 20 tons of cocoa and no rice, 10 tons of rice and no cocoa, or some combination of rice and cocoa between these two extremes. The different combinations that Ghana could produce are represented by the line GG' in Figure 6.1. This is referred to as Ghana's *production possibility frontier (PPF)*. Similarly, imagine that in South Korea it takes 40 resources to produce 1 ton of cocoa and 10 resources to produce 1 ton of rice. Thus, South Korea could produce 5 tons of cocoa and no rice, 20 tons of rice and no cocoa, or some combination between these two extremes. The different combinations available to South Korea are represented by the line KK' in Figure 6.1, which is South Korea's PPF. Clearly, Ghana has an absolute advantage in the production of cocoa. (More resources are needed to produce a ton of cocoa in South Korea than in Ghana.) By the same token, South Korea has an absolute advantage in the production of rice.

Now consider a situation in which neither country trades with any other. Each country devotes half its resources to the production of rice and half to the production of cocoa. Each country must also consume what it produces. Ghana would be able to produce 10 tons of cocoa and 5 tons of rice (point A in Figure 6.1), while South Korea would be able to produce 10 tons of rice and 2.5 tons of cocoa (point B in Figure 6.1). Without trade, the combined production of both countries would be 12.5 tons of cocoa (10 tons in Ghana plus 2.5 tons in South Korea) and 15 tons of rice (5 tons in Ghana and 10 tons in South Korea). If each country were to specialize in producing the good for which it had an absolute advantage and then trade with the other for the good it lacks, Ghana could produce 20 tons of cocoa, and South Korea could produce

absolute advantage

A country has an absolute advantage in the production of a product when it is more efficient than any other country at producing it.

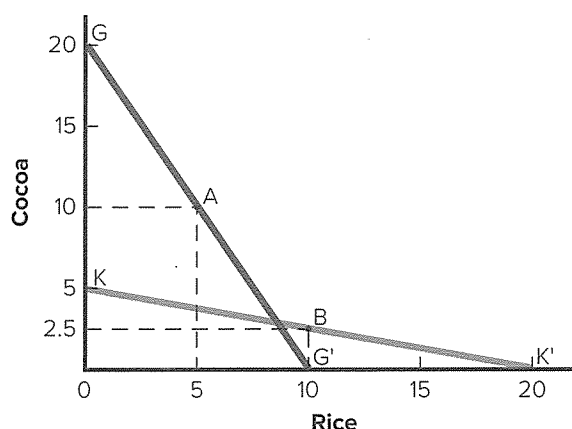


FIGURE 6.1 The theory of absolute advantage.

20 tons of rice. Thus, by specializing, the production of both goods could be increased. Production of cocoa would increase from 12.5 tons to 20 tons, while production of rice would increase from 15 tons to 20 tons. The increase in production that would result from specialization is therefore 7.5 tons of cocoa and 5 tons of rice. Table 6.1 summarizes these figures.

By engaging in trade and swapping 1 ton of cocoa for 1 ton of rice, producers in both countries could consume more of both cocoa and rice. Imagine that Ghana and South Korea swap cocoa and rice on a one-to-one basis; that is, the price of 1 ton of cocoa is equal to the price of 1 ton of rice. If Ghana decided to export 6 tons of cocoa to South Korea and import 6 tons of rice in return, its final consumption after trade would be 14 tons of cocoa and 6 tons of rice. This is 4 tons more cocoa than it could have consumed before specialization and trade and 1 ton more rice.

TABLE 6.1 Absolute Advantage and the Gains from Trade

	Resources Required to Produce 1 Ton of Cocoa and Rice	
	Cocoa	Rice
Ghana	10	20
South Korea	40	10
Production and Consumption without Trade		
Ghana	10.0	5.0
South Korea	2.5	10.0
Total production	12.5	15.0
Production with Specialization		
Ghana	20.0	0.0
South Korea	0.0	20.0
Total production	20.0	20.0
Consumption after Ghana Trades 6 Tons of Cocoa for 6 Tons of South Korean Rice		
Ghana	14.0	6.0
South Korea	6.0	14.0
Increase in Consumption as a Result of Specialization and Trade		
Ghana	4.0	1.0
South Korea	3.5	4.0

Similarly, South Korea's final consumption after trade would be 6 tons of cocoa and 14 tons of rice. This is 3.5 tons more cocoa than it could have consumed before specialization and trade and 4 tons more rice. Thus, as a result of specialization and trade, output of both cocoa and rice would be increased, and consumers in both nations would be able to consume more. Thus, we can see that trade is a positive-sum game; it produces net gains for all involved.

Comparative Advantage



Summarize the different theories explaining trade flows between nations.

David Ricardo took Adam Smith's theory one step further by exploring what might happen when one country has an absolute advantage in the production of all goods.⁴ Smith's theory of absolute advantage suggests that such a country might derive no benefits from international trade. In his 1817 book *Principles of Political Economy*, Ricardo showed that this was not the case. According to Ricardo's theory of comparative advantage, it makes sense for a country to specialize in the production of those goods that it produces most efficiently and to buy the goods that it produces less efficiently from other countries, even if this means buying goods from other countries that it could produce more efficiently itself.⁵ While this may seem counterintuitive, the logic can be explained with a simple example.

Assume that Ghana is more efficient in the production of both cocoa and rice; that is, Ghana has an absolute advantage in the production of both products. In Ghana it takes 10 resources to produce 1 ton of cocoa and 13.33 resources to produce 1 ton of rice. Thus, given its 200 units of resources, Ghana can produce 20 tons of cocoa and no rice, 15 tons of rice and no cocoa, or any combination in between on its PPF (the line GG' in Figure 6.2). In South Korea, it takes 40 resources to produce 1 ton of cocoa and 20 resources to produce 1 ton of rice. Thus, South Korea can produce 5 tons of cocoa and no rice, 10 tons of rice and no cocoa, or any combination on its PPF (the line KK' in Figure 6.2). Again assume that without trade, each country uses half its resources to produce rice and half to produce cocoa. Thus, without trade, Ghana will produce 10 tons of cocoa and 7.5 tons of rice (point A in Figure 6.2), while South Korea will produce 2.5 tons of cocoa and 5 tons of rice (point B in Figure 6.2).

In light of Ghana's absolute advantage in the production of both goods, why should it trade with South Korea? Although Ghana has an absolute advantage in the production of both cocoa and rice, it has a *comparative advantage* in the production of cocoa: Ghana can produce 4 times

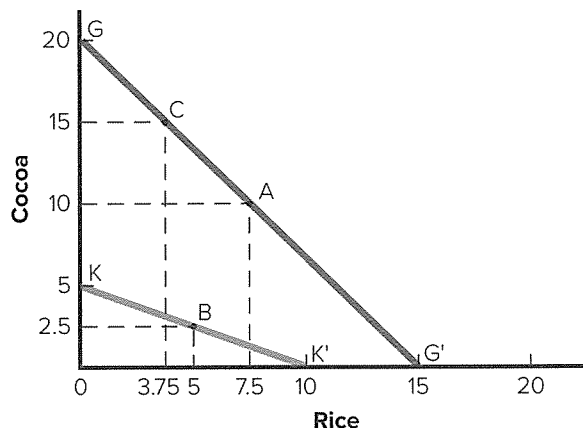


FIGURE 6.2 The theory of comparative advantage.

as much cocoa as South Korea, but only 1.5 times as much rice. Ghana is *comparatively* more efficient at producing cocoa than it is at producing rice.

Without trade, the combined production of cocoa will be 12.5 tons (10 tons in Ghana and 2.5 in South Korea), and the combined production of rice will also be 12.5 tons (7.5 tons in Ghana and 5 tons in South Korea). Without trade each country must consume what it produces. By engaging in trade, the two countries can increase their combined production of rice and cocoa, and consumers in both nations can consume more of both goods.

THE GAINS FROM TRADE

Imagine that Ghana exploits its comparative advantage in the production of cocoa to increase its output from 10 tons to 15 tons. This uses up 150 units of resources, leaving the remaining 50 units of resources to use in producing 3.75 tons of rice (point C in Figure 6.2). Meanwhile, South Korea specializes in the production of rice, producing 10 tons. The combined output of both cocoa and rice has now increased. Before specialization, the combined output was 12.5 tons of cocoa and 12.5 tons of rice. Now the output is 15 tons of cocoa and 13.75 tons of rice (3.75 tons in Ghana and 10 tons in South Korea). The source of the increase in production is summarized in Table 6.2.

Not only is output higher, but both countries also can now benefit from trade. If Ghana and South Korea swap cocoa and rice on a one-to-one basis, with both countries choosing to exchange 4 tons of their export for 4 tons of the import, both countries are able to consume more cocoa and rice than they could before specialization and trade (see Table 6.2). Thus, if Ghana exchanges 4 tons of cocoa with South Korea for 4 tons of rice, it is still left with 11 tons of cocoa, which is 1 ton more than it had before trade. The 4 tons of rice it gets from South Korea in exchange for its 4 tons of cocoa, when added to the 3.75 tons it now produces domestically, leave it with a total of 7.75 tons of rice, which is 0.25 ton more than it had before specialization.

TABLE 6.2 Comparative Advantage and the Gains from Trade

	Resources Required to Produce 1 Ton of Cocoa and Rice	
	Cocoa	Rice
Ghana	10	13.33
South Korea	40	20
Production and Consumption without Trade		
Ghana	10.0	7.5
South Korea	2.5	5.0
Total production	12.5	12.5
Production with Specialization		
Ghana	15.0	3.75
South Korea	0.0	10.0
Total production	15.0	13.75
Consumption after Ghana Trades 4 Tons of Cocoa for 4 Tons of South Korean Rice		
Ghana	11.0	7.75
South Korea	4.0	6.0
Increase in Consumption as a Result of Specialization and Trade		
Ghana	1.0	0.25
South Korea	1.5	1.0

Similarly, after swapping 4 tons of rice with Ghana, South Korea still ends up with 6 tons of rice, which is more than it had before specialization. In addition, the 4 tons of cocoa it receives in exchange is 1.5 tons more than it produced before trade. Thus, consumption of cocoa and rice can increase in both countries as a result of specialization and trade.

The basic message of the theory of comparative advantage is that *potential world production is greater with unrestricted free trade than it is with restricted trade*. Ricardo's theory suggests that consumers in all nations can consume more if no restrictions are placed on trade. This occurs even in countries that lack an absolute advantage in the production of any good. In other words, to an even greater degree than the theory of absolute advantage, *the theory of comparative advantage suggests that trade is a positive-sum game in which all countries that participate realize economic gains*. This theory provides a strong rationale for encouraging free trade. So powerful is Ricardo's theory that it remains a major intellectual weapon for those who argue for free trade.

QUALIFICATIONS AND ASSUMPTIONS



LOG-3

Recognize why many economists believe that unrestricted free trade between nations will raise the economic welfare of countries that participate in a free trade system.

The conclusion that free trade is universally beneficial is a rather bold one to draw from such a simple model. Our simple model includes many unrealistic assumptions:

1. We have assumed a simple world in which only two countries and two goods exist. The real world includes many countries and many goods.
2. We have assumed away transportation costs between countries.
3. We have assumed away differences in the prices of resources in different countries. We have said nothing about exchange rates, simply assuming that cocoa and rice could be swapped on a one-to-one basis.
4. We have assumed that resources can move freely from the production of one good to another within a country. In reality, this is not always the case.
5. We have assumed constant returns to scale; that is, that specialization by Ghana or South Korea has no effect on the amount of resources required to produce 1 ton of cocoa or rice. In reality, both diminishing and increasing returns to specialization exist. The amount of resources required to produce a good might decrease or increase as a nation specializes in production of that good.
6. We have assumed that each country has a fixed stock of resources and that free trade does not change the efficiency with which a country uses its resources. This static assumption makes no allowances for the dynamic changes in a country's stock of resources and in the efficiency with which the country uses its resources that might result from free trade.
7. We have assumed away the effects of trade on income distribution within a country.

Given these assumptions, can the conclusion that free trade is mutually beneficial be extended to the real world of many countries, many goods, positive transportation costs, volatile exchange rates, immobile domestic resources, nonconstant returns to specialization, and dynamic changes? Although a detailed extension of the theory of comparative advantage is beyond the scope of this book, economists have shown that the basic result derived from our simple model can be generalized to a world composed of many countries producing many different goods.⁶ Despite the shortcomings of the Ricardian model, research suggests that the basic proposition that countries will export the goods that they are most efficient at producing is borne out by the data.⁷

However, some economists associated with the “new trade theory” have argued that once all the assumptions are dropped, the case for unrestricted free trade, while still positive, loses some of its strength.⁸ We return to this issue later in this chapter and in the next when we discuss the new trade theory. In a recent and widely discussed analysis, the Nobel Prize-winning economist Paul Samuelson argued that, contrary to the standard interpretation, in certain circumstances the theory of comparative advantage predicts that a rich country might actually be *worse* off by switching to a free trade regime with a poor nation.⁹ We consider Samuelson’s critique in the next section.

EXTENSIONS OF THE RICARDIAN MODEL

Let us explore the effect of relaxing three of the assumptions identified earlier in the simple comparative advantage model. To do so, we will relax the assumptions that resources move freely from the production of one good to another within a country, that there are constant returns to scale, and that trade does not change a country’s stock of resources or the efficiency with which those resources are utilized.

Immobile Resources

In our simple comparative model of Ghana and South Korea, we assumed that producers (farmers) could easily convert land from the production of cocoa to rice, and vice versa. While this assumption may hold for some agricultural products, resources do not always shift quite so easily from producing one good to another. A certain amount of friction is involved. For example, embracing a free trade regime for an advanced economy such as the United States often implies that the country will produce less of some labor-intensive goods, such as textiles, and more of some knowledge-intensive goods, such as computer software or biotechnology products. Although the country as a whole will gain from such a shift, textile producers will lose. A textile worker in South Carolina is probably not qualified to write software for Microsoft. Thus, the shift to free trade may mean that the textile worker becomes unemployed or has to accept another less attractive job, such as working at a fast-food restaurant.

Resources do not always move easily from one economic activity to another. The process creates friction and human suffering, too. While the theory predicts that the benefits of free trade outweigh the costs by a significant margin, this prediction is of cold comfort to those who bear the costs. Accordingly, political opposition to the adoption of a free trade regime typically comes from those whose jobs are most at risk. In the United States, for example, textile workers and their unions have long opposed the move toward free trade precisely because this group has much to lose from free trade. Governments often ease the transition toward free trade by helping retrain those who lose their jobs as a result. The pain caused by the movement toward a free trade regime is a short-term phenomenon, while the gains from trade are both significant and enduring once the transition has been made.

Diminishing Returns

The simple comparative advantage model developed above assumes constant returns to specialization. By **constant returns to specialization**, we mean the units of resources required to produce a good (cocoa or rice) are assumed to remain constant no matter where one is on a country’s production possibility frontier (PPF). Thus, we assumed that it always took Ghana 10 units of resources to produce 1 ton of cocoa. However, assuming diminishing returns to specialization is more realistic. Diminishing returns to specialization occur when more units of resources are required to produce each additional unit. While 10 units of resources may be sufficient to increase Ghana’s output of cocoa from 12 tons to 13 tons, 11 units of resources may be needed to increase output from 13 to 14 tons, 12 units of resources to increase output from 14 tons to 15 tons, and so on. Diminishing returns imply a convex PPF for Ghana (see Figure 6.3), rather than the straight line depicted in Figure 6.2.

The assumption of diminishing returns is more realistic for two reasons. First, not all resources are of the same quality. As a country tries to increase its output of a certain good, it will most likely increasingly draw on more marginal resources whose productivity is not as great as those initially employed. The result is that it requires ever more resources to produce an equal increase in output.

constant returns to specialization

The units of resources required to produce a good are assumed to remain constant no matter where one is on a country’s production possibility frontier.

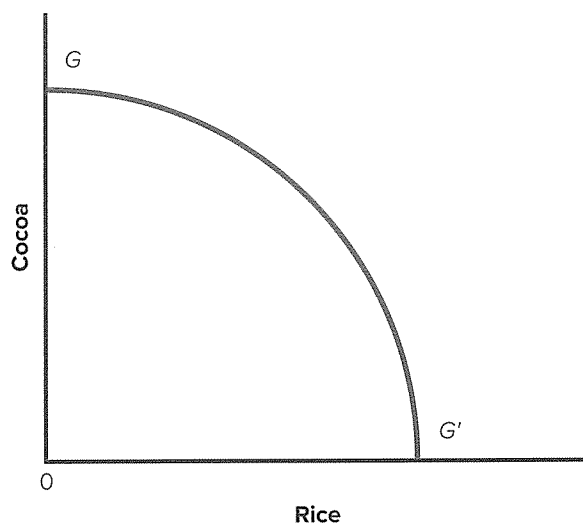


FIGURE 6.3 Ghana's PPF under diminishing returns.

For example, some land is more productive than other land. As Ghana tries to expand its output of cocoa, it might have to utilize increasingly marginal land that is less fertile than the land it originally used. As yields per acre decline, Ghana must use more land to produce 1 ton of cocoa.

A second reason for diminishing returns is that different goods use resources in different proportions. For example, imagine that growing cocoa uses more land and less labor than growing rice and that Ghana tries to transfer resources from rice production to cocoa production. The rice industry will release proportionately too much labor and too little land for efficient cocoa production. To absorb the additional resources of labor and land, the cocoa industry will have to shift toward more labor-intensive methods of production. The effect is that the efficiency with which the cocoa industry uses labor will decline, and returns will diminish.

Diminishing returns show that it is not feasible for a country to specialize to the degree suggested by the simple Ricardian model outlined earlier. Diminishing returns to specialization suggest that the gains from specialization are likely to be exhausted before specialization is complete. In reality, most countries do not specialize, but instead produce a range of goods. However, the theory predicts that specialization is worthwhile until that point where the resulting gains from trade are outweighed by diminishing returns. Thus, the basic conclusion that unrestricted free trade is beneficial still holds, although because of diminishing returns, the gains may not be as great as suggested in the constant returns case.

Dynamic Effects and Economic Growth



LO6-3

Recognize why many economists believe that unrestricted free trade between nations will raise the economic welfare of countries that participate in a free trade system.

The simple comparative advantage model assumed that trade does not change a country's stock of resources or the efficiency with which it utilizes those resources. This static assumption makes no allowances for the dynamic changes that might result from trade. If we relax this assumption, it becomes apparent that opening an economy to trade is likely to generate dynamic gains of two sorts.¹⁰ First, free trade might increase a country's stock of resources as increased supplies of labor and capital from abroad become available for use within the country. This has been occurring in eastern Europe, for example, since the early 1990s, with many Western businesses investing significant capital in the former communist countries.

Second, free trade might also increase the efficiency with which a country uses its resources. Gains in the efficiency of resource utilization could arise from a number of factors. For example, economies of large-scale production might become available as trade expands the size of the total market available to domestic firms. Trade might make better technology from abroad available to domestic firms; better technology can increase labor productivity or the productivity of land. (The so-called green revolution had this effect on agricultural outputs in developing countries.) Also, opening an economy to foreign competition might stimulate domestic producers to look for ways to increase their efficiency. Again, this phenomenon has arguably been occurring in the once-protected markets of eastern Europe, where many former state monopolies have had to increase the efficiency of their operations to survive in the competitive world market.

Dynamic gains in both the stock of a country's resources and the efficiency with which resources are utilized will cause a country's PPF to shift outward. This influence is illustrated in Figure 6.4, where the shift from PPF_1 to PPF_2 results from the dynamic gains that arise from free trade. As a consequence of this outward shift, the country in Figure 6.4 can produce more of both goods than it did before introduction of free trade. The theory suggests that opening an economy to free trade not only results in static gains of the type discussed earlier but also results in dynamic gains that stimulate economic growth. If this is so, then one might think that the case for free trade becomes stronger still, and in general it does. However, as noted, one of the leading economic theorists of the twentieth century, Paul Samuelson, argued that in some circumstances, dynamic gains can lead to an outcome that is not so beneficial.

Trade, Jobs, and Wages: The Samuelson Critique

Paul Samuelson's critique looks at what happens when a rich country—the United States—enters into a free trade agreement with a poorer country—China—that rapidly improves its productivity after the introduction of a free trade regime (i.e., there is a dynamic gain in the efficiency with which resources are used in the poor country). Samuelson's model suggests that in such cases, the lower prices that U.S. consumers pay for goods imported from China following the introduction of a free trade regime *may* not be enough to produce a net gain for the U.S. economy if the dynamic effect of free trade is to lower real wage rates in the United States. As he stated in a *New York Times* interview, "Being able to purchase groceries 20 percent cheaper at Wal-Mart (due to international trade) does not necessarily make up for the wage losses (in America)."¹¹

Samuelson was particularly concerned about the ability to offshore service jobs that traditionally were not internationally mobile, such as software debugging, call center jobs, accounting jobs, and even medical diagnosis of MRI scans. Advances in communications technology since the development of the World Wide Web in the early 1990s have made this offshoring possible, effectively expanding the labor market for these jobs to include educated people in places such as India, the Philippines, and China. When coupled with rapid advances in the productivity of foreign labor due to better education, the effect on middle-class wages in the United States, according

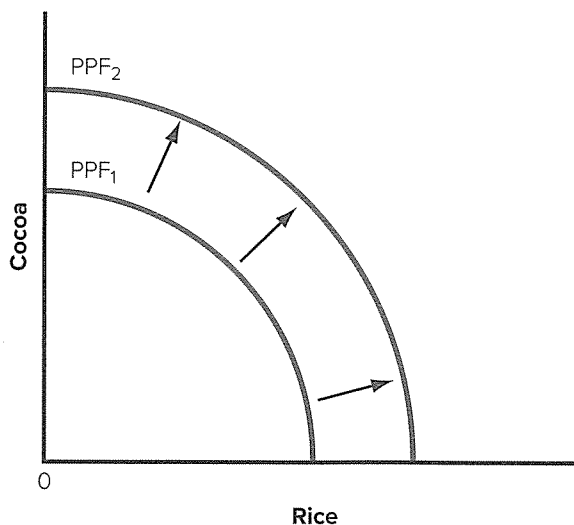


FIGURE 6.4 The influence of free trade on the PPF.

to Samuelson, may be similar to mass inward migration into the country: It will lower the market clearing wage rate, *perhaps* by enough to outweigh the positive benefits of international trade.

Having said this, we should note that Samuelson concedes that free trade has historically benefited rich countries (as data discussed later seem to confirm). Moreover, he notes that introducing protectionist measures (e.g., trade barriers) to guard against the theoretical possibility that free trade may harm the United States in the future may produce a situation that is worse than the disease the measures are trying to prevent. To quote Samuelson: “Free trade may turn out pragmatically to be still best for each region in comparison to lobbyist-induced tariffs and quotas which involve both a perversion of democracy and non-subtle deadweight distortion losses.”¹²

One notable recent study by MIT economist David Autor and his associates found evidence in support of Samuelson’s thesis. The study has been widely quoted in the media and cited by politicians. Autor and his associates looked at every county in the United States for its manufacturers’ exposure to competition from China.¹³ The researchers found that regions most exposed to China tended not only to lose more manufacturing jobs but also to see overall employment decline. Areas with higher exposure to China also had larger increases in workers receiving unemployment insurance, food stamps, and disability payments. The costs to the economy from the increased government payments amounted to two-thirds of the gains from trade with China. In other words, many of the ways trade with China has helped the United States—such as providing inexpensive goods to U.S. consumers—have been wiped out. Even so, like Samuelson the authors of this study argued that in the long run, free trade is a good thing. They note, however, that the rapid rise of China has resulted in some large adjustment costs that, in the short run, significantly reduce the gains from trade.

Other economists have dismissed Samuelson’s fears.¹⁴ While not questioning his analysis, they note that as a practical matter, developing nations are unlikely to be able to upgrade the skill level of their workforce rapidly enough to give rise to the situation in Samuelson’s model. In other words, they will quickly run into diminishing returns. However, such rebuttals are at odds with data suggesting that Asian countries are rapidly upgrading their educational systems. For example, about 56 percent of the world’s engineering degrees awarded in 2008 were in Asia, compared with 4 percent in the United States!¹⁵

Evidence for the Link Between Trade and Growth

Many economic studies have looked at the relationship between trade and economic growth.¹⁶ In general, these studies suggest that, as predicted by the standard theory of comparative advantage, countries that adopt a more open stance toward international trade enjoy higher growth rates than those that close their economies to trade. Jeffrey Sachs and Andrew Warner created a measure of how “open” to international trade an economy was and then looked at the relationship between “openness” and economic growth for a sample of more than 100 countries from 1970 to 1990.¹⁷ Among other findings, they reported

We find a strong association between openness and growth, both within the group of developing and the group of developed countries. Within the group of developing countries, the open economies grew at 4.49 percent per year, and the closed economies grew at 0.69 percent per year. Within the group of developed economies, the open economies grew at 2.29 percent per year, and the closed economies grew at 0.74 percent per year.¹⁸

A study by Wacziarg and Welch updated the Sachs and Warner data through the late 1990s. They found that over the period 1950–1998, countries that liberalized their trade regimes experienced, on average, increases in their annual growth rates of 1.5–2.0 percent compared to pre-liberalization times.¹⁹ An exhaustive survey of 61 studies published between 1967 and 2009 concluded: “The macroeconomic evidence provides dominant support for the positive and significant effects of trade on output and growth.”²⁰

More recently, economists have started to measure the impact of the tariff barriers erected by President Trump (see the Country Focus for details of Trump’s approach to trade). Among other barriers, Trump imposed tariff barriers on imports of aluminum and steel from other nations and also on imports of a wide range of goods from China. If the theory reviewed here is correct, we would expect these barriers to depress the economic growth rate of the United States. A study published in late 2019 by two economists who work for the U.S. Federal Reserve Board sheds some light on this.²¹ Although the authors caution that their results are preliminary,

they found that Trump's tariffs were associated with *reductions* in manufacturing employment and *increases* in producer prices in the United States. For manufacturing employment, they found that a small positive boost from protection from imports was more than offset by rising input costs due to the tariffs and from the imposition of retaliatory tariffs, both of which led to an overall decline in manufacturing employment. In other words, they found that the tariffs had the opposite of the intended effect on manufacturing employment. Moreover, they found that American importers passed on some of the increased costs of inputs to consumers in the form of higher prices, which left them with less money to spend on other goods and services and reduced consumer purchasing power. These provisional results are not good news for the Trump trade strategy—although they are consistent with what economists would expect—but clearly a fuller assessment will take more time and require more data.

The takeaway from the body of research data seems clear: Adopt an open economy and embrace free trade, and your nation will be rewarded with higher economic growth rates. Higher growth will raise income levels and living standards. This last point has been confirmed by a study that looked at the relationship between trade and growth in incomes. The study, undertaken by Jeffrey Frankel and David Romer, found that on average, a 1 percentage point increase in the ratio of a country's trade to its gross domestic product increases income per person by at least 0.5 percent.²² For every 10 percent increase in the importance of international trade in an economy, average income levels will rise by at least 5 percent. Despite the short-term adjustment costs associated with adopting a free trade regime, which can be significant, trade would seem to produce greater economic growth and higher living standards in the long run, just as the theory of Ricardo would lead us to expect.²³

COUNTRY FOCUS

“Trade Wars Are Good and Easy to Win”

At 3:50 a.m. on March 2, 2018, Donald Trump, the 45th president of the United States, tweeted “When a country [USA] is losing many billions of dollars on trade with virtually every country it does business with, trade wars are good and easy to win. When we are down \$100 billion with a certain country and they get cute, don't trade anymore—we win big. It's easy!”*

Trump's tweet was a response to backlash over his decision to impose a 25 percent tariff on imports of steel and a 10 percent tariff on imports of aluminum. The Trump administration claimed these tariffs were necessary to protect two industries that were important for national security. His critics had a different take. They argued that the tariffs would raise input costs for consumers of steel and aluminum, which included construction companies, manufacturers of construction equipment, appliance makers, auto manufacturers, makers of containers and packaging (e.g., beer cans), and aerospace companies. Among those hit by higher costs due to these tariffs would be two of the largest U.S. exporters: Boeing and Caterpillar Tractor. The critics also noted that only 140,000 people are employed in the steel and aluminum industries, whereas 6.5 million Americans are employed in industries that *use* steel and aluminum, where input prices had just gone up.

Trump's actions should not have been a surprise. Donald Trump long voiced strong opposition to trade deals



Ron Sachs - Pool/Getty Images News/Getty Images

designed to lower tariff barriers and foster the free flow of goods and services between the United States and its trading partners. During the presidential election campaign, he called the North American Free Trade Agreement (NAFTA) “the worst trade deal maybe ever signed anywhere.” Upon taking office, his administration launched a renegotiation of NAFTA, with the aim of making the treaty more favorable to the United States. As a candidate, he vowed to “kill” the Trans-Pacific Partnership (TPP), a free trade deal among 12 Pacific Rim countries, including the United States (but excluding China), negotiated by the Obama administration. In his first week in office, he signed an executive order

formally withdrawing the United States from the TPP. He even threatened to pull the United States out of the World Trade Organization (WTO) if the global trade body interfered with his plans to impose tariffs.

Trump's position seemed to be based on a belief that trade is a game that the United States needs to win. He appeared to equate winning with running a trade surplus and saw the persistent U.S. trade deficit as a sign of American weakness. In his words, "you only have to look at our trade deficit to see that we are being taken to the cleaners by our trading partners."^{**} He apparently believes that other countries have taken advantage of the United States in trade deals and that the result has been a sharp decline in manufacturing jobs in the United States. China and Mexico have been frequent targets of his criticisms, and he has argued that China's trade surplus with the United States is a result of that country's currency manipulation, which he believes has made Chinese exports artificially cheap. He seemed to think that the United States could win at the trade game by becoming a tougher negotiator and extracting favorable terms from foreign nations that want access to the U.S. market. He even characterized previous American trade negotiators as "stupid people," "political hacks and diplomats," and "saps," and suggested he should become "negotiator in chief."

In contrast to Donald Trump's espoused position, defenders of the established system argue that the pro-trade policies of the last 70 years were based upon a substantial body of economic theory and evidence that suggests free trade has a positive impact on the economic growth rate of *all* nations that participate in a free trade system. According to this world view, free trade doesn't destroy jobs; it creates jobs and raises national income. To be sure, some sectors will lose jobs when a nation moves to a free trade regime, but the argument is that jobs created elsewhere in the economy will more than compensate for such losses, and in aggregate, the nation will be better off.

The United States has long been the world's largest economy, largest foreign investor, and one of the three largest exporters (along with China and Germany). Due to exports and foreign direct investments in other countries, 43 percent of the sales of all American firms in the S&P 500 stock market index are made outside of U.S. borders. As a result of U.S. economic power, Americans' long adherence to free trade policies has helped to set the tone for the world trading system. In large part, the post–World War II international trading system, with its emphasis on lowering barriers to international trade and investment, was only possible because of vigorous U.S. leadership. With the ascendancy of Donald Trump to the presidency, that seemed to change. Pro-free traders argued that if Trump continued to push for more protectionist trade policies—and his rhetoric and cabinet picks suggested he would—the unintended consequences could include retaliation from the United States' trading partners, a trade war characterized by higher tariffs, a decline in the volume of world trade, job losses in the United States, and lower economic growth around the world. As evidence, they pointed to the last time such protectionist policies were implemented. That was in the early 1930s, when a trade war between nations deepened the Great Depression.

^{*}Donald John Trump, Twitter, March 2, 2018, <https://twitter.com/realdonaldtrump>.

^{**}Donald J. Trump and Dave Shiflett, *The America We Deserve: on Free Trade* (New York: Renaissance Books, 2000).

Sources: "Donald Trump on Free Trade," *On the Issues*, www.ontheissues.org/2016/Donald_Trump_Free_Trade.htm; K. Bradsher, "Trump's Pick on Trade Could Put China in a Difficult Spot," *The New York Times*, January 13, 2017; W. Mauldin, "Trump Threatens to Pull U.S. Out of World Trade Organization," *The Wall Street Journal*, July 24, 2016; "Trump's Antitrade Warriors," *The Wall Street Journal*, January 16, 2017; "Donald Trump's Trade Bluster," *The Economist*, December 10, 2016; C. Brown, "Trump's Steel and Aluminum Tariffs Are Counterproductive," Peterson Institute for International Economics, March 7, 2018.



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Heckscher–Ohlin Theory



LO6-2

Summarize the different theories explaining trade flows between nations.

Ricardo's theory stresses that comparative advantage arises from differences in productivity. Thus, whether Ghana is more efficient than South Korea in the production of cocoa depends on how productively it uses its resources. Ricardo stressed labor productivity and argued that differences in labor productivity between nations underlie the notion of comparative advantage. Swedish economists Eli Heckscher (in 1919) and Bertil Ohlin (in 1933) put forward a different explanation of comparative advantage. They argued that comparative advantage arises from differences in national factor endowments.²⁴ By factor endowments, they meant the extent to which a country is endowed with such resources as land, labor, and capital. Nations have varying factor endowments, and different factor endowments explain differences in factor costs; specifically, the more abundant a factor, the lower its cost. The Heckscher-Ohlin theory predicts that countries will export those goods that make intensive use of factors that are locally abundant, while importing goods that make intensive use of factors that are locally scarce. Thus, the Heckscher-Ohlin theory attempts to explain the pattern of international trade that we observe in the world economy. Like Ricardo's theory, the Heckscher-Ohlin theory argues that free trade is beneficial. Unlike Ricardo's theory, however, the Heckscher-Ohlin theory argues that the pattern of international trade is determined by differences in factor endowments, rather than differences in productivity.

The Heckscher-Ohlin theory has commonsense appeal. For example, the United States has long been a substantial exporter of agricultural goods, reflecting in part its unusual abundance of arable land. In contrast, China has excelled in the export of goods produced in labor-intensive manufacturing industries. This reflects China's relative abundance of low-cost labor. The United States, which lacks abundant low-cost labor, has been a primary importer of these goods. Note that it is relative, not absolute, endowments that are important; a country may have larger absolute amounts of land and labor than another country but be relatively abundant in one of them.

THE LEONTIEF PARADOX

The Heckscher-Ohlin theory has been one of the most influential theoretical ideas in international economics. Most economists prefer the Heckscher-Ohlin theory to Ricardo's theory because it makes fewer simplifying assumptions. Because of its influence, the theory has been subjected to many empirical tests. Beginning with a famous study published in 1953 by Wassily Leontief (winner of the Nobel Prize in economics in 1973), many of these tests have raised questions about the validity of the Heckscher-Ohlin theory.²⁵ Using the Heckscher-Ohlin theory, Leontief postulated that because the United States was relatively abundant in capital compared to other nations, the United States would be an exporter of capital-intensive goods and an importer of labor-intensive goods. To his surprise, however, he found that U.S. exports were less capital intensive than U.S. imports. Because this result was at variance with the predictions of the theory, it has become known as the *Leontief paradox*.

No one is quite sure why we observe the Leontief paradox. One possible explanation is that the United States has a special advantage in producing new products or goods made with innovative technologies. Such products may be less capital intensive than products whose technology has had time to mature and become suitable for mass production. Thus, the United States may be exporting goods that heavily use skilled labor and innovative entrepreneurship, such as computer software, while importing heavy manufacturing products that use large amounts of capital. Some empirical studies tend to confirm this.²⁶ Still, tests of the Heckscher-Ohlin theory using data for a large number of countries tend to confirm the existence of the Leontief paradox.²⁷

This leaves economists with a difficult dilemma. They prefer the Heckscher-Ohlin theory on theoretical grounds, but it is a relatively poor predictor of real-world international trade patterns. On the other hand, the theory they regard as being too limited, Ricardo's theory of comparative advantage, actually predicts trade patterns with greater accuracy. The best solution to this dilemma may be to return to the Ricardian idea that trade patterns are largely driven by international differences in productivity. Thus, one might argue that the United States exports commercial aircraft and imports textiles not because its factor endowments are especially suited to aircraft manufacture and not suited to textile manufacture, but because the United States is relatively more efficient at producing aircraft than textiles. A key assumption in the Heckscher-Ohlin theory is that technologies are the same across countries. This may not be the case. Differences in technology may lead to differences in productivity, which in turn, drives international trade patterns.²⁸ Thus, Japan's success in exporting automobiles from the 1970s onward has

been based not only on the relative abundance of capital but also on its development of innovative manufacturing technology that enabled it to achieve higher productivity levels in automobile production than other countries that also had abundant capital. Empirical work suggests that this theoretical explanation may be correct.²⁹ The new research shows that once differences in technology across countries are controlled for, countries do indeed export those goods that make intensive use of factors that are locally abundant, while importing goods that make intensive use of factors that are locally scarce. In other words, once the impact of differences of technology on productivity is controlled for, the Heckscher–Ohlin theory seems to gain predictive power.



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The Product Life-Cycle Theory



LO6-2

Summarize the different theories explaining trade flows between nations.

Raymond Vernon initially proposed the product life-cycle theory in the mid-1960s.³⁰ Vernon's theory was based on the observation that, for most of the twentieth century, a very large proportion of the world's new products had been developed by U.S. firms and sold first in the U.S. market (e.g., mass-produced automobiles, televisions, instant cameras, photocopiers, personal computers, and semiconductor chips). To explain this, Vernon argued that the wealth and size of the U.S. market gave U.S. firms a strong incentive to develop new consumer products. In addition, the high cost of U.S. labor gave U.S. firms an incentive to develop cost-saving process innovations.

Just because a new product is developed by a U.S. firm and first sold in the U.S. market does not necessarily mean that the product must be produced in the United States. It could be produced abroad at some low-cost location and then exported back into the United States. However, Vernon argued that most new products were initially produced in America. Apparently, the pioneering firms believed it was better to keep production facilities close to the market and to the firm's center of decision making, given the uncertainty and risks inherent in introducing new products. Also, the demand for most new products tends to be based on nonprice factors. Consequently, firms can charge relatively high prices for new products, which obviates the need to look for low-cost production sites in other countries.

Vernon went on to argue that early in the life cycle of a typical new product, while demand is starting to grow rapidly in the United States, demand in other advanced countries is limited to high-income groups. The limited initial demand in other advanced countries does not make it worthwhile for firms in those countries to start producing the new product, but it does necessitate some exports from the United States to those countries.

Over time, demand for the new product starts to grow in other advanced countries (e.g., Great Britain, France, Germany, and Japan). As it does, the possibility of foreign producers beginning to produce for their home markets becomes worthwhile. In addition, U.S. firms might set up production facilities in those advanced countries where demand is growing. Consequently, production within other advanced countries begins to limit the potential for exports from the United States.

As the market in the United States and other advanced nations matures, the product becomes more standardized, and price becomes the main competitive weapon. As this occurs, cost considerations start to play a greater role in the competitive process. Producers based in advanced countries where labor costs are lower than in the United States (e.g., Italy and Spain) might now be able to export to the United States. If cost pressures become intense, the process might not stop there. The cycle by which the United States lost its advantage to other advanced countries might be repeated once more, as developing countries (e.g., Thailand) begin to acquire a production advantage over advanced countries. Thus, the locus of global production initially switches from the United States to other advanced nations and then from those nations to developing countries.

The consequence of these trends for the pattern of world trade is that over time, the United States switches from being an exporter of the product to an importer of the product as production becomes concentrated in lower-cost foreign locations.

PRODUCT LIFE-CYCLE THEORY IN THE TWENTY-FIRST CENTURY

Historically, the product life-cycle theory seems to offer a reasonable explanation of international trade patterns. Consider photocopiers: The product was first developed in the early 1960s by Xerox in the United States and sold initially to U.S. users. Originally, Xerox exported photocopiers from the United States, primarily to Japan and the advanced countries of western Europe. As demand began to grow in those countries, Xerox entered into joint ventures to set up production in Japan (Fuji-Xerox) and Great Britain (Rank-Xerox). In addition, once Xerox's patents on the photocopier process expired, other foreign competitors began to enter the market (e.g., Canon in Japan and Olivetti in Italy). As a consequence, exports from the United States declined, and U.S. users began to buy some photocopiers from lower-cost foreign sources, particularly Japan. More recently, Japanese companies found that manufacturing costs are too high in their own country, so they have begun to switch production to developing countries such as Thailand. Thus, initially the United States and now other advanced countries (e.g., Japan and Great Britain) have switched from being exporters of photocopiers to importers. This evolution in the pattern of international trade in photocopiers is consistent with the predictions of the product life-cycle theory that mature industries tend to go out of the United States and into low-cost assembly locations.

However, the product life-cycle theory is not without weaknesses. Viewed from an Asian or European perspective, Vernon's argument that most new products are developed and introduced in the United States seems ethnocentric and dated. Although the theory that, during the U.S. dominance of the global economy (from 1945 to 1975), most new products were introduced in the United States may be true, important exceptions have always existed. These exceptions appear to have become more common in recent years. Many new products are now first introduced in Japan (e.g., video-game consoles) or South Korea (e.g., Samsung smartphones). Moreover, with the increased globalization and integration of the world economy discussed in Chapter 1, an increasing number of new products (e.g., tablet computers, smartphones, and digital cameras) are now introduced simultaneously in the United States and many European and Asian nations. This may be accompanied by globally dispersed production, with particular components of a new product being produced in those locations around the globe where the mix of factor costs and skills is most favorable (as predicted by the theory of comparative advantage). In sum, although Vernon's theory may be useful for explaining the pattern of international trade during the period of American global dominance, its relevance in the modern world seems more limited.



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New Trade Theory



LO6-2

Summarize the different theories explaining trade flows between nations.

The new trade theory began to emerge in the 1970s when a number of economists pointed out that the ability of firms to attain economies of scale might have important implications for international trade.³¹ **Economies of scale** are unit cost reductions associated with a large scale of output. Economies of scale have a number of sources, including the ability to spread fixed costs over a large volume and the ability of large-volume producers to utilize specialized employees and equipment that are more productive than less specialized employees and equipment. Economies of scale are a major source of cost reductions in many industries, from computer software to automobiles and from pharmaceuticals to aerospace. For example, Microsoft realizes economies of scale by spreading the fixed costs of developing new versions of its Windows operating system, which runs to about \$10 billion, over the 2 billion or so personal computers on which each new system is ultimately installed. Similarly, automobile companies realize economies of scale by producing a high volume of automobiles from an assembly line where each employee has a specialized task.

economies of scale

Cost advantages associated with large-scale production.

New trade theory makes two important points: First, through its impact on economies of scale, trade can increase the variety of goods available to consumers and decrease the average cost of those goods. Second, in those industries in which the output required to attain economies of scale represents a significant proportion of total world demand, the global market may be able to support only a small number of enterprises. Thus, world trade in certain products may be dominated by countries whose firms were first movers in their production.

INCREASING PRODUCT VARIETY AND REDUCING COSTS



LO6-3

Recognize why many economists believe that unrestricted free trade between nations will raise the economic welfare of countries that participate in a free trade system.

Imagine first a world without trade. In industries where economies of scale are important, both the variety of goods that a country can produce and the scale of production are limited by the size of the market. If a national market is small, not enough demand may exist to enable producers to realize economies of scale for certain products. Accordingly, those products may not be produced, thereby limiting the variety of products available to consumers. Alternatively, they may be produced but at such low volumes that unit costs and prices are considerably higher than they might be if economies of scale could be realized.

Now consider what happens when nations trade with each other. Individual national markets are combined into a larger world market. As the size of the market expands due to trade, individual firms may better be able to attain economies of scale. The implication, according to new trade theory, is that each nation may be able to specialize in producing a narrower range of products than it would in the absence of trade, yet by buying goods that it does not make from

other countries, each nation can simultaneously increase the *variety* of goods available to its consumers and *lower the costs* of those goods; thus, trade offers an opportunity for mutual gain even when countries do not differ in their resource endowments or technology.

Suppose there are two countries, each with an annual market for 1 million automobiles. By trading with each other, these countries can create a combined market for 2 million cars. In this combined market, due to the ability to better realize economies of scale, more varieties (models) of cars can be produced, and cars can be produced at a lower average cost, than in either market alone. For example, demand for a sports car may be limited to 55,000 units in each national market, while a total output of at least 100,000 per year may be required to realize significant scale economies. Similarly, demand for a minivan may be 80,000 units in each national market, and again a total output of at least 100,000 per year may be required to realize significant scale economies. Faced with limited domestic market demand, firms in each nation may decide not to produce a sports car, because the costs of doing so at such low volume are too great. Although they may produce minivans, the cost of doing so will be higher, as will prices, than if significant economies of scale had been attained. Once the two countries decide to trade, however, a firm in one nation may specialize in producing sports cars, while a firm in the other nation may produce minivans. The combined demand for 110,000 sports cars and 160,000 minivans allows each firm to realize scale economies. Consumers in this case benefit from having access to a product (sports cars) that was not available before international trade and from the lower price for a product (minivans) that could not be produced at the most efficient scale before international trade. Trade is thus mutually beneficial because it allows the specialization of production, the realization of scale economies, the production of a greater variety of products, and lower prices.

ECONOMIES OF SCALE, FIRST-MOVER ADVANTAGES, AND THE PATTERN OF TRADE

first-mover advantages

Advantages accruing to the first to enter a market.

A second theme in new trade theory is that the pattern of trade we observe in the world economy may be the result of economies of scale and first-mover advantages. **First-mover advantages** are the economic and strategic advantages that accrue to early entrants into an industry.³² The ability to capture scale economies ahead of later entrants, and thus benefit from a lower cost structure, is an important first-mover advantage. New trade theory argues that for those products where economies of scale are significant and represent a substantial proportion of world demand, the first movers in an industry can gain a scale-based cost advantage that later entrants find almost impossible to match. Thus, the pattern of trade that we observe for such products may reflect first-mover advantages. Countries may dominate in the export of certain goods because economies of scale are important in their production and because firms located in those countries were the first to capture scale economies, giving them a first-mover advantage.

For example, consider the commercial aerospace industry. In aerospace, substantial scale economies come from the ability to spread the fixed costs of developing a new jet aircraft over a large number of sales. Airbus spent some \$25 billion to develop its superjumbo jet, the 550-seat A380, which entered service in 2007. To recoup those costs and break even, Airbus will have to sell at least 250 A380 planes. By 2018, it had sold some 240 aircraft. If Airbus can sell more than 350 A380 planes, the venture will apparently be profitable. Total demand over the first 20 years of service for this class of aircraft was estimated to be between 400 and 600 units. Thus, at best, the global market could probably profitably support only one producer of jet aircraft in the superjumbo category. It follows that the European Union might come to dominate in the export of very large jet aircraft, primarily because a European-based firm, Airbus, was the first to produce a superjumbo jet aircraft and realize scale economies. Other potential producers, such as Boeing, might have been shut out of the market because they lack the scale economies that Airbus enjoys. Because it pioneered this market category, Airbus captured a *first-mover advantage* based on *scale economies* that was difficult for rivals to match, and that resulted in the European Union becoming the *leading exporter* of very large jet aircraft. (As it turns out, however, the super-jumbo market may not be big enough to support even one producer. In early 2019, Airbus announced that it will stop producing the A380 in 2021 due to weak demand.)

IMPLICATIONS OF NEW TRADE THEORY

New trade theory has important implications. The theory suggests that nations may benefit from trade even when they do not differ in resource endowments or technology. Trade allows a nation to specialize in the production of certain products, attaining scale economies and lowering the costs of producing those products, while buying products that it does not produce from other nations that specialize in the production of those other products. By this mechanism, the variety of products available to consumers in each nation is increased, while the average costs of those products should fall, as should their price, freeing resources to produce other goods and services.

The theory also suggests that a country may predominate in the export of a good simply because it was lucky enough to have one or more firms among the first to produce that good. Because they are able to gain economies of scale, the first movers in an industry may get a lock on the world market that discourages subsequent entry. First-movers' ability to benefit from increasing returns creates a barrier to entry. In the commercial aircraft industry, the fact that Boeing and Airbus are already in the industry and have the benefits of economies of scale discourages new entry and reinforces the dominance of America and Europe in the trade of midsize and large jet aircraft. This dominance is further reinforced because global demand may not be sufficient to profitably support another producer of midsize and large jet aircraft in the industry. So although Japanese firms might be able to compete in the market, they have decided not to enter the industry but to ally themselves as major subcontractors with primary producers (e.g., Mitsubishi Heavy Industries is a major subcontractor for Boeing on the 777 and 787 programs).

New trade theory is at variance with the Heckscher-Ohlin theory, which suggests a country will predominate in the export of a product when it is particularly well endowed with those factors used intensively in its manufacture. New trade theorists argue that the United States is a major exporter of commercial jet aircraft not because it is better endowed with the factors of production required to manufacture aircraft, but because one of the first movers in the industry, Boeing, was a U.S. firm. The new trade theory is not at variance with the theory of comparative advantage. Economies of scale increase productivity. Thus, the new trade theory identifies an important source of comparative advantage.

This theory is quite useful in explaining trade patterns. Empirical studies seem to support the predictions of the theory that trade increases the specialization of production within an industry, increases the variety of products available to consumers, and results in lower average prices.³³ With regard to first-mover advantages and international trade, a study by Harvard business historian Alfred Chandler suggests the existence of first-mover advantages is an important factor in explaining the dominance of firms from certain nations in specific industries.³⁴ The number of firms is very limited in many global industries, including the chemical industry, the heavy construction-equipment industry, the heavy truck industry, the tire industry, the consumer electronics industry, the jet engine industry, and the computer software industry.

Perhaps the most contentious implication of the new trade theory is the argument that it generates for government intervention and strategic trade policy.³⁵ New trade theorists stress the role of luck, entrepreneurship, and innovation in giving a firm first-mover advantages. According to this argument, the reason Boeing was the first mover in commercial jet aircraft manufacture—rather than firms such as Great Britain's De Havilland and Hawker Siddeley or Holland's Fokker, all of which could have been—was that Boeing was both lucky and innovative. One way Boeing was lucky is that De Havilland shot itself in the foot when its Comet jet airliner, introduced two years earlier than Boeing's first jet airliner, the 707, was found to be full of serious technological flaws. Had De Havilland not made some serious technological mistakes, Great Britain might have become the world's leading exporter of commercial jet aircraft. Boeing's innovativeness was demonstrated by its independent development of the technological know-how required to build a commercial jet airliner. Several new trade theorists have pointed out, however, that Boeing's research and development (R&D) was largely paid for by the U.S. government; the 707 was a spin-off from a government-funded military program (the entry of Airbus into the industry was also supported by significant government subsidies). Herein is a rationale for government intervention: By the sophisticated and judicious use of subsidies, could a government increase the chances of its domestic firms becoming first movers in newly emerging industries, as the U.S. government apparently did with Boeing (and the European Union did with Airbus)? If this is possible, and the new trade theory suggests it might be, we have an economic rationale for a

proactive trade policy that is at variance with the free trade prescriptions of the trade theories we have reviewed so far. We consider the policy implications of this issue in Chapter 7.



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National Competitive Advantage: Porter's Diamond



Summarize the different theories explaining trade flows between nations.

Michael Porter, the famous Harvard strategy professor who developed the five forces framework for analyzing an industry, has also written extensively on international trade.³⁶ Porter and his team looked at 100 industries in 10 nations. Like the work of the new trade theorists, Porter's work was driven by a belief that existing theories of international trade told only part of the story. For Porter, the essential task was to explain why a nation achieves international success in a particular industry. Why does Japan do so well in the automobile industry? Why does Switzerland excel in the production and export of precision instruments and pharmaceuticals? Why do Germany and the United States do so well in the chemical industry? These questions cannot be answered easily by the Heckscher-Ohlin theory, and the theory of comparative advantage offers only a partial explanation. The theory of comparative advantage would say that Switzerland excels in the production and export of precision instruments because it uses its resources very productively in these industries. Although this may be correct, this does not explain why Switzerland is more productive in this industry than Great Britain, Germany, or Spain. Porter tries to solve this puzzle.

Porter theorizes that four broad attributes of a nation shape the environment in which local firms compete, and these attributes promote or impede the creation of competitive advantage (see Figure 6.5). These attributes are:

- *Factor endowments*—a nation's position in factors of production, such as skilled labor or the infrastructure necessary to compete in a given industry.
- *Demand conditions*—the nature of home-country demand for the industry's product or service.
- *Related and supporting industries*—the presence or absence of supplier industries and related industries that are internationally competitive.
- *Firm strategy, structure, and rivalry*—the conditions governing how companies are created, organized, and managed and the nature of domestic rivalry.

Porter speaks of these four attributes as constituting the *diamond*. He argues that firms are most likely to succeed in industries or industry segments where the diamond is most favorable. He also argues that the diamond is a mutually reinforcing system. The effect of one attribute is contingent on the state of others. For example, Porter argues favorable demand conditions

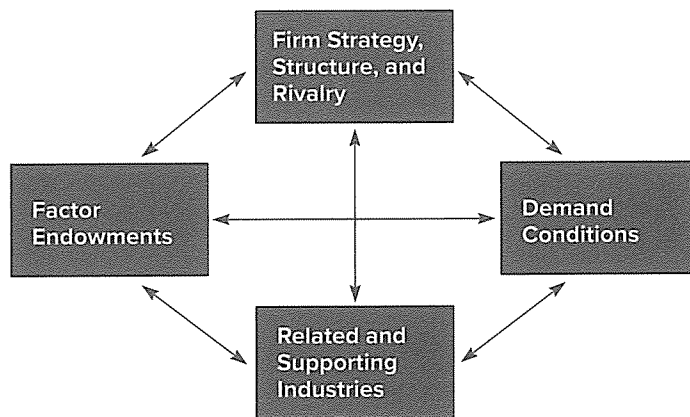


FIGURE 6.5 The determinants of national competitive advantage: Porter's diamond.

Source: Michael E. Porter, *The Competitive Advantage of Nations* (New York: Free Press, 1990; republished with a new introduction, 1998), p. 72.

will not result in competitive advantage unless the state of rivalry is sufficient to cause firms to respond to them.

Porter maintains that two additional variables can influence the national diamond in important ways: chance and government. Chance events, such as major innovations, can reshape industry structure and provide the opportunity for one nation's firms to supplant another's. Government, by its choice of policies, can detract from or improve national advantage. For example, regulation can alter home demand conditions, antitrust policies can influence the intensity of rivalry within an industry, and government investments in education can change factor endowments.

FACTOR ENDOWMENTS

Factor endowments lie at the center of the Heckscher–Ohlin theory. While Porter does not propose anything radically new, he does analyze the characteristics of factors of production. He recognizes hierarchies among factors, distinguishing between *basic factors* (e.g., natural resources, climate, location, and demographics) and *advanced factors* (e.g., communication infrastructure, sophisticated and skilled labor, research facilities, and technological know-how). He argues that advanced factors are the most significant for competitive advantage. Unlike the naturally endowed basic factors, advanced factors are a product of investment by individuals, companies, and governments. Thus, government investments in basic and higher education, by improving the general skill and knowledge level of the population and by stimulating advanced research at higher education institutions, can upgrade a nation's advanced factors.

The relationship between advanced and basic factors is complex. Basic factors can provide an initial advantage that is subsequently reinforced and extended by investment in advanced factors. Conversely, disadvantages in basic factors can create pressures to invest in advanced factors. An obvious example of this phenomenon is Japan, a country that lacks arable land and mineral deposits and yet through investment has built a substantial endowment of advanced factors. Porter notes that Japan's large pool of engineers (reflecting a much higher number of engineering graduates per capita than almost any other nation) has been vital to Japan's success in many manufacturing industries.

DEMAND CONDITIONS

Porter emphasizes the role home demand plays in upgrading competitive advantage. Firms are typically most sensitive to the needs of their closest customers. Thus, the characteristics of home demand are particularly important in shaping the attributes of domestically made products and in creating pressures for innovation and quality. Porter argues that a nation's firms gain competitive advantage if their domestic consumers are sophisticated and demanding. Such consumers

pressure local firms to meet high standards of product quality and to produce innovative products. For example, Porter notes that Japan's sophisticated and knowledgeable buyers of cameras helped stimulate the Japanese camera industry to improve product quality and to introduce innovative models.

RELATED AND SUPPORTING INDUSTRIES

The third broad attribute of national advantage in an industry is the presence of suppliers or related industries that are internationally competitive. The benefits of investments in advanced factors of production by related and supporting industries can spill over into an industry, thereby helping it achieve a strong competitive position internationally. Swedish strength in fabricated steel products (e.g., ball bearings and cutting tools) has drawn on strengths in Sweden's specialty steel industry. Technological leadership in the U.S. semiconductor industry provided the basis for U.S. success in personal computers and several other technically advanced electronic products. Similarly, Switzerland's success in pharmaceuticals is closely related to its previous international success in the technologically related dye industry.

One consequence of this process is that successful industries within a country tend to be grouped into clusters of related industries. This was one of the most pervasive findings of Porter's study. One such cluster Porter identified was in the German textile and apparel sector, which included high-quality cotton, wool, synthetic fibers, sewing machine needles, and a wide range of textile machinery. Such clusters are important because valuable knowledge can flow between the firms within a geographic cluster, benefiting all within that cluster. Knowledge flows occur when employees move between firms within a region and when national industry associations bring employees from different companies together for regular conferences or workshops.³⁷

FIRM STRATEGY, STRUCTURE, AND RIVALRY

The fourth broad attribute of national competitive advantage in Porter's model is the strategy, structure, and rivalry of firms within a nation. Porter makes two important points here. First, different nations are characterized by different management ideologies, which either help them or do not help them build national competitive advantage. For example, Porter noted the predominance of engineers in top management at German and Japanese firms. He attributed this to these firms' emphasis on improving manufacturing processes and product design. In contrast, Porter noted a predominance of people with finance backgrounds leading many U.S. firms. He linked this to U.S. firms' lack of attention to improving manufacturing processes and product design. He argued that the dominance of finance led to an overemphasis on maximizing short-term financial returns. According to Porter, one consequence of these different management ideologies was a relative loss of U.S. competitiveness in those engineering-based industries where manufacturing processes and product design issues are all-important (e.g., the automobile industry).

Porter's second point is that a strong association exists between vigorous domestic rivalry and the creation and persistence of competitive advantage in an industry. Vigorous domestic rivalry induces firms to look for ways to improve efficiency, which makes them better international competitors. Domestic rivalry creates pressures to innovate, to improve quality, to reduce costs, and to invest in upgrading advanced factors. All this helps create world-class competitors. Porter cites the case of Japan:

Nowhere is the role of domestic rivalry more evident than in Japan, where it is all-out warfare in which many companies fail to achieve profitability. With goals that stress market share, Japanese companies engage in a continuing struggle to outdo each other. Shares fluctuate markedly. The process is prominently covered in the business press. Elaborate rankings measure which companies are most popular with university graduates. The rate of new product and process development is breathtaking.³⁸

EVALUATING PORTER'S THEORY



LOG-4

Explain the arguments of those who maintain that government can play a proactive role in promoting national competitive advantage in certain industries.

Porter contends that the degree to which a nation is likely to achieve international success in a certain industry is a function of the combined impact of factor endowments, domestic demand conditions, related and supporting industries, and domestic rivalry. He argues that the presence of all four components is usually required for this diamond to boost competitive performance (although some exceptions exist). Porter also contends that government can influence each of the four components of the diamond—either positively or negatively. Factor endowments can be affected by subsidies, policies toward capital markets, policies toward education, and so on. Government can shape domestic demand through local product standards or with regulations that mandate or influence buyer needs. Government policy can influence supporting and related industries through regulation and influence firm rivalry through such devices as capital market regulation, tax policy, and antitrust laws.

If Porter is correct, we would expect his model to predict the pattern of international trade that we observe in the real world. Countries should be exporting products from those industries where all four components of the diamond are favorable, while importing in those areas where the components are not favorable. Is he correct? We simply do not know. Porter's theory has not been subjected to detailed empirical testing. Much about the theory rings true, but the same can be said for the new trade theory, the theory of comparative advantage, and the Heckscher-Ohlin theory. It may be that each of these theories, which complement each other, explains something about the pattern of international trade.



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360° View: Managerial Implications

LOCATION, FIRST-MOVER ADVANTAGES, GOVERNMENT POLICY, INVESTMENTS, AND STRATEGY



LOG-5

Understand the important implications that international trade theory holds for management practice.

Why does all this matter for business? The material discussed in this chapter has at least four main implications for international businesses: location implications, first-mover implications, government policy implications, and the implications for business investments and enterprise strategy that flow from changes in government policy.

Location

Underlying most of the theories we have discussed is the notion that different countries have particular advantages in different productive activities. Thus, from a profit perspective, it makes sense for a firm to disperse its productive activities to those countries where, according to the theory of international trade, they can be performed most efficiently. If design can be performed most efficiently in France, that is where design facilities should be located; if the manufacture of basic components can be performed most efficiently in Singapore, that is where they should be manufactured; and if final assembly can be performed most efficiently in China, that is where final assembly should be performed. The result is a global web of productive activities, with different activities being performed in different locations around the globe depending on considerations of comparative advantage, factor endowments, and the like. If the firm does not do this, it may find itself at a competitive disadvantage relative to firms that do.

First-Mover Advantages

According to the new trade theory, firms that establish a first-mover advantage with regard to the production of a particular new product may subsequently dominate global trade in that product. This advantage is particularly true in industries where the global market can profitably support only a limited number of firms, such as the aerospace market or the market for the most advanced microprocessors, but early commitments may also seem to be important in less concentrated industries. For the individual firm, the clear message is that investing substantial financial resources in trying to build a first-mover, or early mover, advantage pays off, even if that means several years of losses before a new venture becomes profitable. The idea is to preempt the available demand, gain cost advantages related to volume, build an enduring brand ahead of later competitors, and, consequently, establish a long-term sustainable competitive advantage. Although the details of how to achieve this are beyond the scope of this book, many publications offer strategies for exploiting first-mover advantages and for avoiding the traps associated with pioneering a market (first-mover disadvantages).³⁹

Government Policy

The theories of international trade also matter to international businesses because firms are major players on the international trade scene. Business firms produce exports, and business firms import the products of other countries. Because of their pivotal role in international trade, businesses can exert a strong influence on government trade policy, lobbying to promote either free trade or trade restrictions. The theories of international trade claim that promoting free trade is generally in the best interests of a country, although it may not always be in the best interest of an individual firm. Many firms recognize this and lobby for open markets.

For example, when the U.S. government announced its intention to place a tariff on Japanese imports of liquid crystal display (LCD) screens in the 1990s, IBM and Apple Computer protested strongly. Both IBM and Apple pointed out that (1) Japan was the lowest-cost source of LCD screens; (2) they used these screens in their own laptop computers; and (3) the proposed tariff, by increasing the cost of LCD screens, would increase the cost of laptop computers produced by IBM and Apple, thus making them less competitive in the world market. In other words, the tariff, designed to protect U.S. firms, would be self-defeating. In response to these pressures, the U.S. government reversed its posture.

Unlike IBM and Apple, however, businesses do not always lobby for free trade. In the United States, for example, restrictions on imports of steel have periodically been put into place in response to direct pressure by U.S. firms on the government (the latest example being in March 2018 when the Trump administration placed a 25 percent tariff on imports of foreign steel). In some cases, the government has responded to pressure from domestic companies seeking protection by getting foreign companies to agree to “voluntary” restrictions on their imports, using the implicit threat of more comprehensive formal trade barriers to get them to adhere to these agreements (historically, this has occurred in the automobile industry). In other cases, the

government used what are called “antidumping” actions to justify tariffs on imports from other nations (these mechanisms will be discussed in detail in Chapter 7).

As predicted by international trade theory, many of these agreements have been self-defeating, such as the voluntary restriction on machine tool imports agreed to in 1985. Shielded from international competition by import barriers, the U.S. machine tool industry had no incentive to increase its efficiency. Consequently, it lost many of its export markets to more efficient foreign competitors. Because of this misguided action, the U.S. machine tool industry shrank during the period when the agreement was in force. For anyone schooled in international trade theory, this outcome was not surprising.⁴⁰

Finally, Porter’s theory of national competitive advantage also contains policy implications. Porter’s theory suggests that it is in the best interest of business for a firm to invest in upgrading advanced factors of production (e.g., to invest in better training for its employees) and to increase its commitment to research and development. It is also in the best interests of business to lobby the government to adopt policies that have a favorable impact on each component of the national diamond. Thus, according to Porter, businesses should urge government to increase investment in education, infrastructure, and basic research (because all these enhance advanced factors) and to adopt policies that promote strong competition within domestic markets (because this makes firms stronger international competitors, according to Porter’s findings).

Changes in Government Policy, Investments, and Business Strategy

For most of the last three decades, government policy with regard to international trade has been stable and well understood. As we shall discuss in detail in the next chapter, an international order based on agreed rules that govern trade has been established under the auspices of the World Trade Organization. All major trading nations are members of this body. Together, the world’s nations have worked toward lowering barriers to cross-border trade in goods and services. These shared goals and acceptance of the prevailing rules-based order have created a predictable environment that has made cross-border business investments less risky. Businesses have known what the playing field looks like. In turn, as noted earlier, firms have configured their value chains to take advantage of this order. Productive activities have been dispersed to various locations around the world where they can be performed most efficiently. Thus, Apple might design its iPhone in California and write the software that runs the phone there as well, while outsourcing the production of hardware components to Germany, Malaysia, and Japan and assembling the phone in China, before shipping the finished product to major markets around the world.

The election of Donald Trump to the presidency of the United States, and the shift in trade policy that he initiated, represented a major challenge to the established order. The Biden administration has let many of Trump’s trade policies stay in place, such as the tariffs on Chinese imports. Due to these changes, the trade environment is less predictable than it was. Business investments made today may become unprofitable tomorrow if trade barriers are imposed. Long-established global supply chains, which for years made economic sense, may no longer be viable given changes in trade policy. Thus, for example, Apple’s decision to assemble iPhones in China, which seemed rational in 2007, may become uneconomic and a strategic liability if the United States continues to impose high tariffs on Chinese imports. Similarly, American firms such as Starbucks, which see China as a major growth opportunity, might find their ambitions thwarted if the Chinese react to American trade barriers by restricting inward investment into China.

The change in trade policy has created major uncertainty about the future, which raises the risks of cross-border investments to support international trade. The playing field is no longer well-known. The risks of stranded investment have increased. The value of an investment made a few years ago may now be significantly reduced. How should businesses respond? Although we will return to this issue in later chapters, we should note here that when investment risks increase, businesses typically do three things. First, they reduce their investments in risky projects, which in this context would probably mean lower investments in countries involved in a trade war. Second, they hedge their bets. Hedging bets might require a firm like Apple to source a component from several different locations and, perhaps, to assemble in multiple locations so that if a trade dispute and higher tariff barriers make one location less economical, the firm can shift some of its production to another location. Hedging bets might also require the firm to increase its domestic production, even if doing so raises costs. We should also note that

reducing investment risks, as well as hedging by sourcing from multiple locations, will reduce the efficiency of resource allocation in the global economy, depress and/or distort trade flows, and lower economic growth rates going forward. Third, to counteract all of this, business can and will lobby governments to encourage them to resolve disputes and create a more stable and predictable environment.

Key Terms

gains from trade, p. 167
 free trade, p. 168
 new trade theory, p. 170
 factor endowments, p. 170
 mercantilism, p. 171
 zero-sum game, p. 171
 absolute advantage, p. 173
 constant returns to specialization, p. 178

economies of scale, p. 187
 first-mover advantages, p. 188
 balance-of-payments accounts, p. 200
 current account, p. 201
 current account deficit, p. 201
 current account surplus, p. 201
 capital account, p. 201
 financial account, p. 201

Summary

This chapter reviewed a number of theories that explain why a country benefits from engaging in international trade and explained the pattern of international trade observed in the world economy. The theories of Smith, Ricardo, and Heckscher-Ohlin all make strong cases for unrestricted free trade. In contrast, the mercantilist doctrine and, to a lesser extent, the new trade theory can be interpreted to support government intervention to promote exports through subsidies and to limit imports through tariffs and quotas.

In explaining the pattern of international trade, this chapter shows that, with the exception of mercantilism, which is silent on this issue, the different theories offer largely complementary explanations. Although no one theory may explain the apparent pattern of international trade, taken together, the theory of comparative advantage, the Heckscher-Ohlin theory, the product life-cycle theory, the new trade theory, and Porter's theory of national competitive advantage do suggest which factors are important. Comparative advantage tells us that productivity differences are important; Heckscher-Ohlin tells us that factor endowments matter; the product life-cycle theory tells us that where a new product is introduced is important; the new trade theory tells us that increasing returns to specialization and first-mover advantages matter; Porter tells us that all these factors may be important insofar as they affect the four components of the national diamond. The chapter made the following points:

advantage and import goods in areas where other countries have absolute advantages.

3. The theory of comparative advantage suggests that it makes sense for a country to specialize in producing those goods that it can produce most efficiently, while buying goods that it can produce relatively less efficiently from other countries—even if that means buying from other countries goods that it could produce more efficiently itself.
4. The theory of comparative advantage suggests that unrestricted free trade brings about increased world production—that is, that trade is a positive-sum game.
5. The theory of comparative advantage also suggests that opening a country to free trade stimulates economic growth, which creates dynamic gains from trade. The empirical evidence seems to be consistent with this claim.
6. The Heckscher-Ohlin theory argues that the pattern of international trade is determined by differences in factor endowments. It predicts that countries will export those goods that make intensive use of locally abundant factors and will import goods that make intensive use of factors that are locally scarce.
7. The product life-cycle theory suggests that trade patterns are influenced by where a new product is introduced. In an increasingly integrated global economy, the product life-cycle theory seems to be less predictive than it once was.
8. New trade theory states that trade allows a nation to specialize in the production of certain goods, attaining scale economies and lowering the costs of producing those goods, while buying goods that it does not produce from other nations that are similarly specialized.

1. Mercantilists argued that it was in a country's best interests to run a balance-of-trade surplus. They viewed trade as a zero-sum game, in which one country's gains cause losses for other countries.
2. The theory of absolute advantage suggests that countries differ in their ability to produce goods efficiently. The theory suggests that a country should specialize in producing goods in areas where it has an absolute

By this mechanism, the variety of goods available to consumers in each nation is increased, while the average costs of those goods should fall.

9. New trade theory also states that in those industries where substantial economies of scale imply that the world market will profitably support only a few firms, countries may predominate in the export of certain products simply because they had a firm that was a first mover in that industry.
10. Some new trade theorists have promoted the idea of strategic trade policy. They argue that government, by the sophisticated and judicious use of subsidies, might be able to increase the chances of domestic firms becoming first movers in newly emerging industries.
11. Porter's theory of national competitive advantage suggests that the pattern of trade is influenced by four attributes of a nation: (a) factor endowments; (b) domestic demand conditions; (c) related and supporting industries; and (d) firm strategy, structure, and rivalry.
12. Theories of international trade are important to an individual business firm primarily because they can help the firm decide where to locate its various production activities.
13. Firms involved in international trade can and do exert a strong influence on government policy toward trade. By lobbying government, business firms can promote free trade or trade restrictions.

Critical Thinking and Discussion Questions

1. Mercantilism is a bankrupt theory that has no place in the modern world. Discuss.
2. Is free trade fair? Discuss.
3. Unions in developed nations often oppose imports from low-wage countries and advocate trade barriers to protect jobs from what they often characterize as "unfair" import competition. Is such competition "unfair"? Do you think that this argument is in the best interests of (a) the unions, (b) the people they represent, and/or (c) the country as a whole?
4. What are the potential costs of adopting a free trade regime? Do you think governments should do anything to reduce these costs? Why?
5. Reread the Country Focus, "Is China Manipulating Its Currency in Pursuit of a Neo-Mercantilist Policy?"
 - a. Do you think China is pursuing a currency policy that can be characterized as neo-mercantilist?
 - b. What should the United States and other countries do about this?
6. Is there a difference between the transference of high-paying white-collar jobs, such as computer programming and accounting, to developing nations, and low-paying blue-collar jobs? If so, what is the difference, and should government do anything to stop the flow of white-collar jobs out of the country to countries such as India?
7. Drawing upon the new trade theory and Porter's theory of national competitive advantage, outline the case for government policies that would build national competitive advantage in biotechnology. What kinds of policies would you recommend that the government adopt? Are these policies at variance with the basic free trade philosophy?
8. The world's poorest countries are at a competitive disadvantage in every sector of their economies. They have little to export; they have no capital; their land is of poor quality; they often have too many people given available work opportunities; and they are poorly educated. Free trade cannot possibly be in the interests of such nations. Discuss.



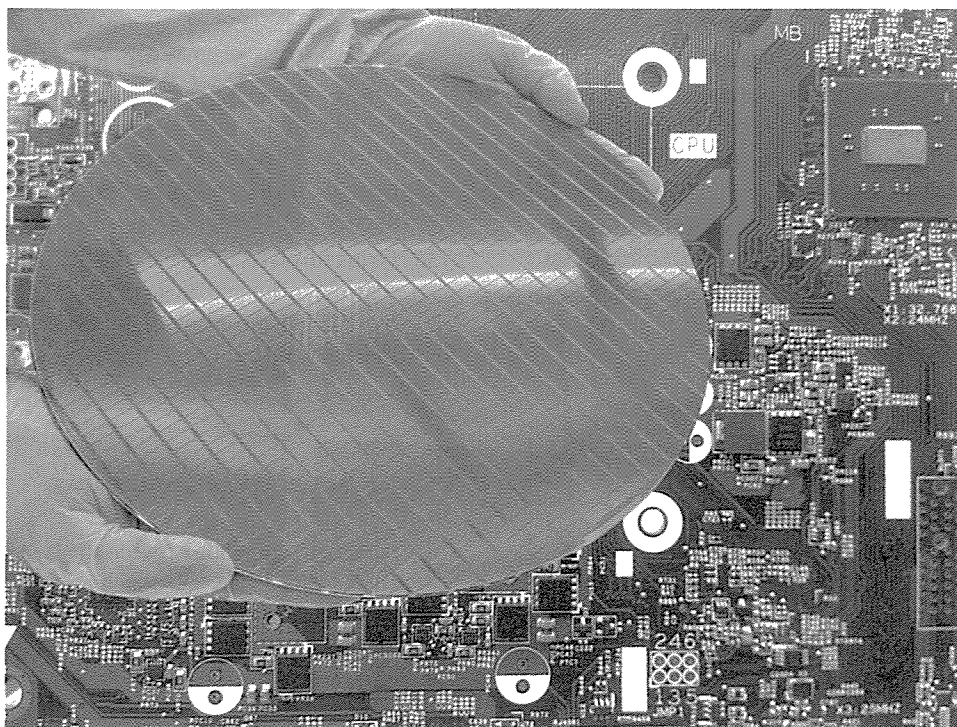
Research Task

globalEDGE.msu.edu

Use the globalEDGE™ website (globaledge.msu.edu) to complete the following exercises:

1. The *World Trade Organization International Trade Statistics* is an annual report that provides comprehensive, comparable, and updated statistics on trade in merchandise and commercial services. The report is an assessment of world trade flows by country, region, and main product or service categories. Using the most recent statistics available, identify the top 10 countries that lead in the export and import of merchandise trade, respectively. Which countries appear in the top 10 in both exports and imports? Can you explain why these countries appear at the top of both lists?
2. Food is an integral part of understanding different countries, cultures, and lifestyles. You run a chain of high-end premium restaurants in the United States, and you are looking for unique Australian wines you can import. However, you must first identify which *Australian suppliers* can provide you with premium wines. After searching through the Australian supplier directory, identify three to four companies that can be potential suppliers. Then, develop a list of criteria you would need to ask these companies about to select which one to work with.

CLOSING CASE



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Global Trade War in Semiconductors

The global semiconductor industry is a high-tech colossus. The microprocessor and memory chips that the industry produces are to be found everywhere: from computers to telecommunications equipment to mobile phones, automobiles, aircraft, and industrial machinery. The industry churns out about a trillion chips every year, or 128 for every person on the planet. Each iPhone contains over 25 chips, whereas an electric car can have 3,000 of them. In 2019, the industry generated around \$412 billion in revenues. When the COVID-19 pandemic swept the world in 2020, demand for semiconductors didn't fall with the resulting economic recession—demand jumped to \$435 billion as people worked from home and spent more time playing video games or streaming shows. Looking forward, the fusion of high-speed 5G communications networks, the Internet of Things (IoT), and cloud computing technology will continue to drive strong demand growth for semiconductor chips far into the future.

U.S. firms have long held a strong position in this business. The industry was born in Silicon Valley, California, during the 1960s, with Fairchild Semiconductor emerging as the first large enterprise. Today, some of the top firms in the world, including Intel, Nvidia, and Qualcomm, are based in California. The U.S. industry registered exports of \$75 billion in 2021 as it supplied chips to foreign device makers, particularly in Southeast Asia. This puts the industry on a par with automobiles as one of the most important U.S. export industries.

However, although exports are still robust, the geography of the industry has shifted considerably since the 1980s. U.S. firms still produce around 44 percent of all the semiconductor chips they design at home, but significant production has migrated to Japan, Taiwan, and South Korea. For high-end microprocessors in particular, only three firms in the world now produce efficiently at scale. One of these, Intel, has significant manufacturing activities in the United States. The other two are foreign entities: Taiwan Semiconductor Manufacturing Company (TSMC) and Samsung of South Korea. While Intel designs and makes its own chips, companies such as Nvidia and Qualcomm specialize in chip design, outsourcing production to contract manufacturing companies such as TSMC and Samsung. Thus, while the United States still produces 85 percent of all chip design software and around 50 percent of chip design intellectual property and manufacturing equipment, its share of global chip manufacturing has slipped from more than 50 percent in the 1990s to 12 percent today.

The shift of manufacturing to Southeast Asia has been driven by several factors. Governments outside the United States have often offered hefty financial incentives to chipmakers. This is particularly important given that the cost of a new fabrication facility can amount to \$5 billion or more. A leading-edge TSMC factory that opened in 2020 cost \$19.5 billion. Chip companies also

have been attracted by growing networks of suppliers outside the United States and an expanding workforce of skilled engineers capable of operating expensive manufacturing machinery.

Looking forward, semiconductor production is viewed across the globe as a national security priority because of the powerful role chips play not only in consumer technology, but also in the military and cyber-warfare. The sale of chips to China, in particular, has come under political scrutiny. Under the Trump administration, the United States placed new restrictions on China's industry, including banning Chinese telecom giant Huawei Technologies Co. and preventing some Chinese chipmakers from buying U.S. manufacturing equipment without a license. China is the world's largest chip importer, buying \$300 billion worth of foreign-made chips in 2019. As a producer, China is a bit player. Chinese firms supply only 5 percent of the world market, and Chinese chips are far less advanced, lagging behind their Taiwanese and U.S. counterparts by five years or more, experts say.

Moves by the Trump administration to block exports of semiconductor chips and technology to China galvanized the Chinese. In response, President Xi Jinping called for accelerating the development of critical industries, including semiconductors. In 2020, Chinese semiconductor companies raised nearly \$38 billion through public offerings, private placements, and asset sales. Six Chinese provinces and regions also pledged to invest about \$13 billion in semiconductors. An article in *The Economist* estimated that the Chinese government may be planning to pump \$100 billion in subsidies into Chinese chipmakers. Chinese universities are now prioritizing programs dedicated to training a new generation of semiconductor experts. In addition, China's cabinet raised the status of university degrees tied to semiconductors, promising more funding and prestige. Meanwhile, China's elite Peking, Tsinghua, and Fudan universities have started to channel additional resources into their semiconductor programs.

China seems determined to reduce its reliance on foreigners (and particularly the United States) for semiconductor technology and chips. As one Chinese chip expert noted, "It's about protecting the safety of your supply chain. You never know if you'll be next on the U.S. blacklist." If the Chinese succeed, the geography of international trade in semiconductors may shift once again, with Chinese producers increasing their global market share. If they are successful, some fear that the biggest loser would be the United States. Politicians in the United States have also been growing increasingly concerned that China might ultimately try to seize Taiwan, which it regards as a renegade province rather than an independent nation. If this occurs, TSMC, the world's largest semiconductor manufacturer, would fall under Chinese jurisdiction.

In response to these perceived threats, the Biden administration upped the ante in 2022 with the Chips and Science Act. Signed into law in August 2022 with significant bipartisan support, the Chips Act provides \$52 billion in subsidies to companies building computer chip factories and research facilities in the United States. Biden sold the bill to Congress as being essential for national security. Early signs suggested that the bill would spur construction of more than half a dozen major semiconductor manufacturing facilities in the United States, including new plants by TSMC and Intel, providing more local supply for this vital component of the information age. In October 2022, President Xi Jinping of China noted that the American policies would only speed up China's efforts to "resolutely win the battle in key core technologies." Not to be left out, in November 2022, EU countries agreed to a \$46.6 billion plan to subsidize the production of chip factories in Europe in order to reduce the region's reliance on both Asia and the United States. Ironically, in the midst of all this action by governments, demand for semiconductors slumped in late 2022 as the pandemic era boom in technology purchases gave way to a bust. This raised the question of how effective the subsidies would be in spurring new domestic production in a business that suddenly found itself with excess productive capacity?

Case Discussion Questions

1. Why do you think chip manufacturing has migrated to Asia in recent years? Has this migration been beneficial for American consumers? For American producers?
2. What trade theory best explains the geography of the global chip industry in the period prior to the 2020–2022 COVID pandemic?
3. How did the COVID pandemic change demand and supply conditions in the industry? What lessons did the pandemic teach the industry?
4. Why did the Trump administration place restrictions on the export of certain chips and technology to China? Do you think this was the correct thing to do? How would you expect China to respond?
5. Is the Biden Chips Act good for producers? Is it good for consumers? Is it necessary for the national security of the United States?
6. How might the growing conflict between China and the United States shape the global semiconductor industry going forward? What happens to the gains from trade in an industry where the geography of production is determined by subsidies and export restrictions?

Sources: *State of the U.S. Semiconductor Industry*, Semiconductor Industry Association, 2020; L. Lin, "Tech War with the U.S. Turbocharges China's Chip Development Resolve," *The Wall Street Journal*, November 16, 2020; A. Fitch and L. Santiago, "Why Fewer Chips Say 'Made in the USA,'" *The Wall Street Journal*, November 3, 2020; "Chipmaking Is Being Redesigned. Effects Will Be Far Reaching," *The Economist*, January 23, 2021; J. Whalen, "A New Era of Industrial Policy Kicks off with Signing of the Chips Act," *Washington Post*, August 9, 2022; "The American Chip Industry's \$1.5 Trillion Meltdown," *The Economist*, October 17, 2022.

Appendix

International Trade and the Balance of Payments

balance-of-payments accounts

National accounts that track both payments to and receipts from foreigners.

International trade involves the sale of goods and services to residents in other countries (exports) and the purchase of goods and services from residents in other countries (imports). A country's **balance-of-payments accounts** keep track of the payments to and receipts from other countries for a particular time period. These include payments to foreigners for imports of goods and services, and receipts from foreigners for goods and services exported to them. A summary copy of the U.S. balance-of-payments accounts for 2021 is given in Table A.1. In this appendix, we briefly describe the form of the balance-of-payments accounts, and we discuss whether a current account deficit, often a cause of much concern in the popular press, is something to worry about.

TABLE A.1 U.S. Balance-of-Payments Accounts, 2021

	Current Account	\$ Millions
Exports of goods, services, and income receipts (credits)		\$3,780,154
Goods		1,761,364
Services		795,273
Primary income receipts		1,052,080
Secondary income receipts		171,436
Imports of goods, services, and income (debits)		4,626,508
Goods		2,851,660
Services		550,025
Primary income payments		912,587
Secondary income payments		312,236
Capital Account		
Capital transfer receipts		3,864
Capital transfer debits		6,338
Financial Account		
Net U.S. acquisition of financial assets		1,278,599
Net U.S. incurrence of liabilities		1,977,294
Net financial derivatives		-41,902
Statistical discrepancy		108,231
Balances		
Balance on current account		-846,354
Balance on capital account		-2,474
Balance on primary income		139,492
Balance on secondary income		-140,800

Source: Bureau of Economic Analysis.

BALANCE-OF-PAYMENTS ACCOUNTS

Balance-of-payments accounts are divided into three main sections: the current account, the capital account, and the financial account (to confuse matters, what is now called the capital account until recently was part of the current account, and the financial account used to be called the capital account). The **current account** records transactions that pertain to four categories, all of which can be seen in Table A.1. The first category, goods, refers to the export or import of physical goods (e.g., agricultural foodstuffs, aircraft, autos, computers, chemicals). The second category is the export or import of services (e.g., intangible products such as banking and insurance services, royalty payments on intellectual property, sale of computer software services, and earnings from foreign tourists who visit the United States). The third category, primary income receipts or payments, refers to income from foreign investments or payments to foreign investors (e.g., interest and dividend receipts or payments). The third category also includes payments that foreigners have made to U.S. residents for work performed outside the United States and payments that U.S. entities make to foreign residents. The fourth category, secondary income receipts or payments, refers to the transfer of a good, service, or asset to the U.S. government or U.S. private entities, or the transfer to a foreign government or entity in the case of payments (this includes tax payments, foreign pension payments, cash transfers, etc.).

A **current account deficit** occurs when a country imports more goods, services, and income than it exports. A **current account surplus** occurs when a country exports more goods, services, and income than it imports. Table A.1 shows that in 2021 the United States ran a current account deficit of \$856.354 billion. This is often a headline-grabbing figure and is widely reported in the news media. The U.S. current account deficit reflects the fact that the United States imports far more physical goods than it exports. In 2021 the U.S. imported \$1.09 trillion more in physical goods than it exported. Offsetting this, the United States typically runs a surplus on trade in services and on income payments.

The 2006 current account deficit of \$803 billion was the largest on record until 2021, and was equivalent to about 6.5 percent of the country's \$13.8 trillion GDP. While the 2021 deficit was larger in absolute terms, the economy has grown substantially since 2006, so it came in at 3.72 percent of the country's GDP. Many people find the fact that the United States runs a persistent deficit on its current account to be disturbing, even though as a percentage of GDP the number is not large. The common assumption is that the high import of goods displaces domestic production, causes unemployment, and reduces the growth of the U.S. economy. However, the issue is more complex than this. Fully understanding the implications of a large and persistent deficit requires that we look at the rest of the balance-of-payments accounts.

The **capital account** records one-time changes in the stock of assets. As noted earlier, until recently, this item was included in the current account. The capital account includes capital transfers, such as debt forgiveness and migrants' transfers (the goods and financial assets that accompany migrants as they enter or leave the country). In the big scheme of things, this is a small figure, amounting to $-\$2.47$ billion in 2021.

The **financial account** (formerly the capital account) records transactions that involve the purchase or sale of assets. Thus, when a German firm purchases stock in a U.S. company or buys a U.S. bond, the transaction enters the U.S. balance of payments as a credit on the financial account. This is because capital is flowing into the country. When capital flows out of the United States, it enters the financial account as a debit.

The financial account is composed of a number of elements. The net U.S. acquisition of financial assets includes the change in foreign assets owned by the U.S. government (e.g., U.S. official reserve assets) and the change in foreign assets owned by private individuals and corporations (including changes in assets owned through foreign direct investment). As can be seen from Table A.1, in 2021 U.S. ownership of foreign assets increased by \$1.279 trillion, which tells us that the U.S. government and U.S. private entities were purchasing more foreign assets than they were selling. The net U.S. incurrence of liabilities refers to the change in U.S. assets owned by foreigners. In 2021, foreigners increased their holdings of U.S. assets by \$1.977 trillion, signifying that foreigners were net acquirers of U.S. stocks, bonds (including Treasury bills), and physical assets such as real estate.

A basic principle of balance-of-payments accounting is double-entry bookkeeping. Every international transaction automatically enters the balance of payments twice—once as a credit

current account

In the balance of payments, records transactions involving the export or import of goods and services.

current account deficit

The current account of the balance of payments is in deficit when a country imports more goods, services, and income than it exports.

current account surplus

The current account of the balance of payments is in surplus when a country exports more goods, services, and income than it imports.

capital account

In the balance of payments, records transactions involving one-time changes in the stock of assets.

financial account

In the balance of payments, transactions that involve the purchase or sale of assets.

and once as a debit. Imagine that you purchase a car produced in Japan by Toyota for \$20,000. Because your purchase represents a payment to another country for goods, it will enter the balance of payments as a debit on the current account. Toyota now has the \$20,000 and must do something with it. If Toyota deposits the money at a U.S. bank, Toyota has purchased a U.S. asset—a bank deposit worth \$20,000—and the transaction will show up as a \$20,000 credit on the financial account. Or Toyota might deposit the cash in a Japanese bank in return for Japanese yen. Now the Japanese bank must decide what to do with the \$20,000. Any action that it takes will ultimately result in a credit for the U.S. balance of payments. For example, if the bank lends the \$20,000 to a Japanese firm that uses it to import personal computers from the United States, then the \$20,000 must be credited to the U.S. balance-of-payments current account. Or the Japanese bank might use the \$20,000 to purchase U.S. government bonds, in which case it will show up as a credit on the U.S. balance-of-payments financial account.

Thus, any international transaction automatically gives rise to two offsetting entries in the balance of payments. Because of this, the sum of the current account balance, the capital account, and the financial account balance should always add up to zero. In practice, this does not always occur due to the existence of “statistical discrepancies,” the source of which need not concern us here (note that, in 2021, the statistical discrepancy amounted to \$108 billion).

DOES THE CURRENT ACCOUNT DEFICIT MATTER?

As discussed earlier, some people express concern when a country is running a deficit on the current account of its balance of payments.⁴¹ In recent years, a number of rich countries, including most notably the United States, have run persistent current account deficits. When a country runs a current account deficit, the money that flows to other countries can then be used by those countries to purchase assets in the deficit country. Thus, when the United States runs a trade deficit with China, the Chinese can use the money that they receive from U.S. consumers to purchase U.S. assets such as stocks, bonds, and the like. Put another way, a deficit on the current account is financed by selling assets to other countries—that is, by increasing liabilities on the financial account. Thus, the persistent U.S. current account deficit is being financed by a steady sale of U.S. assets (stocks, bonds, real estate, and whole corporations) to other countries. In short, countries that run current account deficits become net debtors.

For example, as a result of financing its current account deficit through asset sales, the United States must deliver a stream of interest payments to foreign bondholders, rents to foreign landowners, and dividends to foreign stockholders. One might argue that such payments to foreigners drain resources from a country and limit the funds available for investment within the country. Because investment within a country is necessary to stimulate economic growth, a persistent current account deficit can choke off a country’s future economic growth. This potential issue is the basis of the argument that persistent deficits are bad for an economy.

However, things are not this simple. For one thing, in an era of global capital markets, money is efficiently directed toward its highest value uses, and over the past quarter of a century, many of the highest value uses of capital have been in the United States. So even though capital is flowing out of the United States in the form of payments to foreigners, much of that capital finds its way right back into the country to fund productive investments in the United States. In short, it is not clear that the current account deficit chokes off U.S. economic growth. In fact, notwithstanding the 2008–2009 recession, the U.S. economy has grown substantially over the past 30 years, despite running a persistent current account deficit and despite financing that deficit by selling U.S. assets to foreigners (although the U.S. economy shrank significantly in 2020 due to the impact of the COVID-19 pandemic, it returned to its pre-pandemic size by mid-2021 and grew strongly in 2022). One reason for the persistent long term growth of the U.S. economy is that that foreigners reinvest much of the income earned from U.S. assets and from exports to the United States right back into the United States. This revisionist view, which has gained in popularity in recent years, suggests that a persistent current account deficit might not be the drag on economic growth it was once thought to be.⁴²

Having said this, a nagging fear still lingers that at some point, the appetite that foreigners have for U.S. assets might decline. If foreigners suddenly reduced their investments in the United States, what would happen? In short, instead of reinvesting the dollars that they earn from exports and investment in the United States back into the country, they would sell those dollars for another

currency, European euros, Japanese yen, or Chinese yuan, for example, and invest in euro-, yen-, and yuan-denominated assets instead. Such investment would lead to a fall in the value of the dollar on foreign exchange markets, and that in turn would increase the price of imports and lower the price of U.S. exports, making them more competitive, which should reduce the overall level of the current account deficit. Thus, in the long run, the persistent U.S. current account deficit could be corrected via a reduction in the value of the U.S. dollar. The concern is that such adjustments may not be smooth. Rather than a controlled decline in the value of the dollar, the dollar might suddenly lose a significant amount of its value in a very short time, precipitating a “dollar crisis.”⁴³ Because the U.S. dollar is the world’s major reserve currency and is held by many foreign governments and banks, any dollar crisis could deliver a body blow to the world economy and at the very least trigger a global economic slowdown. That would not be a good thing.

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Part 3 The Global Trade and Investment Environment

7

GOVERNMENT POLICY AND INTERNATIONAL TRADE

Learning Objectives

After reading this chapter, you will be able to:

- LO7-1** Identify the policy instruments used by governments to influence international trade flows.
- LO7-2** Understand why governments sometimes intervene in international trade.
- LO7-3** Summarize and explain the arguments against strategic trade policy.
- LO7-4** Describe the development of the world trading system and the current trade issue.
- LO7-5** Explain the implications for managers of developments in the world trading system.



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opening case

AMERICA'S INFLATION REDUCTION ACT AND ELECTRIC VEHICLE SUBSIDIES

For decades the United States championed free trade and criticized the use of subsidies by foreign governments to support their domestic producers and give them an advantage in international markets. In the 1980s and 1990s, frequent targets of American criticism included the European Union (for subsidizing Airbus) and Japan (which used subsidies to help its emerging semiconductor firms compete more effectively in global markets). In the 2000s, attention turned to China, a country that used subsidies to support state-owned enterprises and help private enterprises such as the telecom equipment company Huawei gain share from established producers in the global marketplace. From the mid 1990s onward, the United States used the World Trade Organization and antidumping measures to seek redress against countries whose governments were using subsidies. After Donald Trump became president in 2017, America switched to a more confrontational approach, erecting tariff barriers to protect domestic industries from foreign competitors and increasing its own subsidies to support domestic producers. To date, the Biden administration has continued with this more nationalistic approach.

The latest example of America's emerging industrial policy can be found in the Inflation Reduction Act (IRA), which was passed into law in August 2022. A major objective of the IRA was to accelerate the switch toward green technologies, including the adoption of electric vehicles (EVs) and renewable electricity generation facilities based on wind and solar power. Sold to the public as a way of fighting climate change, the IRA included a raft of government subsidies that could amount to \$370 billion in direct and indirect support to North American producers over the next 10 years. America's trading partners were quick to point out that these subsidies would distort international trade and put more efficient foreign producers at a disadvantage. In other words, they constituted a restraint on trade.

For example, consider EV production. Under the IRA, consumers in the United States will be able to get a tax credit of up to \$7,500 if they purchase an EV that is assembled in North America (this is an indirect subsidy to producers since it boosts demand). To qualify for the credit, the vehicle must cost under \$50,000 (\$80,000 for SUVs and pickup trucks), and the buyer's income must be under \$300,000 for married couples or \$150,000 for single people. The goal of these targeted consumer subsidies is to encourage America's middle class to purchase EVs produced in North America. The focus on North America, rather than just the United States, reflects the fact that the United States has a free trade agreement with Canada and Mexico (the USMCA) and that many North American producers have production facilities and supply chains that encompass all three countries.

In addition to the finished vehicle, the IRA contains rules requiring that battery components also be sourced in North America. These rules will be phased in over time. The goal is to reduce dependence on China for the supply of electric batteries, particularly the lithium ion batteries used to power most EVs. This will not be easy given that China had 80 percent of the world's advanced battery making capacity in 2021 and that 60 percent of the lithium refining capacity in the world is in China.

Trade experts in Europe were quick to raise the alarm, arguing that the subsidies contained in the IRA would put their own EV companies at a competitive disadvantage and were also in violation of WTO rules that prohibit subsidies involving local content requirements. French President Emmanuel Macron called the IRA a "killer for our industry." For European auto companies, the IRA increased their incentive to invest in North American production facilities, even if the costs of producing in North America are higher. This implies potential job losses in Europe.

In the pre-Trump era, countries adversely impacted by America's actions would have taken their complaint to the World Trade Organization. However, the WTO dispute settlement apparatus has not been operational since the Trump administration refused to appoint new members to the seven-person Appellate Body. The WTO Appellate Body, which can issue binding rulings on trade disputes, must have at least three members to issue a report. America has veto power over the appointment of new members. The Trump administration vetoed the appointment of new members as old members retired, effectively rendering the body nonoperational once the membership fell below three. The Biden administration has continued to block the appointment of new members, cognizant perhaps that the Appellate Body might rule against some Biden administration policies such as the IRA.

Faced with this new reality, there is growing evidence that European and Asian governments will respond to the IRA by increasing their own subsidies to EV and battery producers. In Europe, there is growing political support for relaxing EU state aid rules so that governments can target more support toward "strategic industries." In South Korea, the environment ministry has reportedly informed automobile companies that subsidies for EVs could be limited to firms that operate their own service centers in the country, which would exclude most foreign automobile companies. The cumulative result of such policies could be a wave of national and regional subsidies targeted at encouraging the localization of EV and battery production. Defenders of pre-Trump era trade policies argue that such an outcome would mean lower gains from trade and weaker economic growth across the board. For the time being, however, those who argue for a return to the old order have been relegated to the sidelines.

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Introduction

The review of the classical trade theories of Smith, Ricardo, and Heckscher–Ohlin in Chapter 6 showed that in a world without trade barriers, trade patterns are determined by the relative productivity of different factors of production in different countries. Countries will specialize in products they can make most efficiently, while importing products they can produce less efficiently. Chapter 6 also laid out the intellectual case for free trade. Remember, **free trade** refers to a situation in which a government does not attempt to restrict or influence what its citizens can buy from or sell to another country. As we saw in Chapter 6, the theories of Smith, Ricardo, and Heckscher–Ohlin predict that the consequences of free trade include both static economic gains (because free trade supports a higher level of domestic consumption and more efficient utilization of resources) and dynamic economic gains (because free trade stimulates economic growth and the creation of wealth).

This chapter looks at the political reality of international trade. Although many nations are nominally committed to free trade, they tend to intervene in international trade to protect the interests of politically important groups, to promote the interests of key domestic producers, or for reasons of national security. For example, the U.S. government has a long history of intervening in the steel industry, imposing import tariffs to protect domestic producers from market share losses due to the importation of less expensive foreign steel. The last three U.S. presidents—Bush, Obama, and Trump—have all authorized tariffs on foreign steel imports. The current resident of the White House, President Biden, partly relaxed some of the tariffs on steel imports from key European allies imposed by Trump but has kept in place tariffs on steel imports from other nations. Similarly, as described in the opening case, in August 2022, the Biden administration signed the Inflation Reduction Act into law. Among other things, this Act provided for close to \$370 billion in subsidies over 10 years to North American producers of electric vehicles, lithium ion batteries, and renewables such as wind and solar energy. The IRA also contained local content requirements, mandating, for example, that to benefit from subsidies, a vehicle would need to be assembled in North America and use batteries that were produced in North America. One goal of the IRA is to reduce dependence on China as primary source for lithium ion batteries and (potentially) electric vehicles. Another is to create more jobs in the North American automobile industry. A third goal is to accelerate the adoption of green technology as a way of combating climate change.

This chapter starts by describing the range of policy instruments that governments use to intervene in international trade. A detailed review of governments' various political and economic motives for intervention follows. In the third section of this chapter, we consider how the case for free trade stands up in view of the various justifications given for government intervention in international trade. Then we look at the emergence of the modern international trading system, which is based on the **General Agreement on Tariffs and Trade (GATT)** and its successor, the World Trade Organization (WTO). The GATT and WTO are the creations of a series of multinational treaties. We also consider the current shift in American policy toward bilateral trade agreements, managed trade, and greater protectionism. The final section of this chapter discusses the implications of this material for management practice.

free trade

The absence of barriers to the free flow of goods and services between countries.

General Agreement on Tariffs and Trade (GATT)

International treaty that committed signatories to lowering barriers to the free flow of goods across national borders and led to the WTO.

Instruments of Trade Policy



LO7-1

Identify the policy instruments used by governments to influence international trade flows.

Trade policy uses eight main instruments: tariffs, bans, subsidies, import quotas, voluntary export restraints, local content requirements, administrative policies, and antidumping duties. Tariffs are the oldest and simplest instrument of trade policy. As we shall see later in this chapter, they

import tariff

A tax levied on imports of goods or services.

specific tariff

A tariff levied as a fixed charge for each unit of good imported.

ad valorem tariff

A tariff levied as a proportion of the value of an imported good.

are also the instrument that the GATT and its successor, the WTO, have been most successful in limiting. A fall in tariff barriers in recent decades has been accompanied by a rise in nontariff barriers, such as subsidies, quotas, voluntary export restraints, and antidumping duties.

IMPORT TARIFFS

An **import tariff** is a tax levied on imports (or exports). Tariffs fall into two categories. **Specific tariffs** are levied as a fixed charge for each unit of a good imported (e.g., \$3 per barrel of oil). **Ad valorem tariffs** are levied as a proportion of the value of the imported good. In most cases, import tariffs are placed on imports to protect domestic producers from foreign competition by raising the price of imported goods. However, tariffs also produce revenue for the government. Until the income tax was introduced, for example, the U.S. government received most of its revenues from tariffs.



Did You Know?

Did you know that one of the first arguments for protectionist trade policies was proposed by Alexander Hamilton in 1792?

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Import tariffs are paid by the importer (and export tariffs by the exporter). Thus, the 25 percent ad valorem tariff placed on imports of foreign steel by President Trump in 2018 are paid for not by foreign steel producers, but by the American importers of foreign steel. These import tariffs are, in effect, a tax on American steel consumers. The important thing to understand about an import tariff is who suffers and who gains. The government gains because the tariff increases government revenues. Domestic producers gain because the tariff affords them some protection against foreign competitors by increasing the cost of imported foreign goods. Consumers lose because they must pay more for certain imports.

For example, the idea of the 2018 tariff on imports of foreign steel was to protect domestic steel producers from cheap imports of foreign steel and create more steelmaking jobs in the United States. Research suggests that only about 1,000 additional steelmaking jobs were created in the first 18 months the tariffs were in effect. However, about half of the cost increase in steel was passed onto steel consumers in the form of higher prices. This raised the costs of manufacturing companies that made heavy use of steel in their own products, decreasing their international competitiveness. As a result, research has estimated that about 75,000 jobs were lost in steel-consuming manufacturing companies during the first 18 months the tariffs were in place.¹ This finding suggests that the loss to consumers of steel from higher input prices significantly outweigh any benefits U.S. steelmakers may have gained from protection from foreign producers.

In general, two conclusions can be derived from economic analysis of the effect of import tariffs.² First, tariffs are generally pro-producer and anti-consumer. While they protect producers from foreign competitors, this restriction of supply also raises domestic prices. For example, a study by Japanese economists calculated that tariffs on imports of foodstuffs, cosmetics, and chemicals into Japan cost the average Japanese consumer about \$890 per year in the form of higher prices. Almost all studies find that import tariffs impose significant costs on domestic consumers in the form of higher prices. Second, import tariffs reduce the overall efficiency of the world economy. They reduce efficiency because a protective tariff encourages domestic firms to produce products at home that could be produced more efficiently abroad. The consequence is an inefficient utilization of resources and a failure to realize the gains from trade.³

EXPORT TARIFFS AND BANS

An **export tariff** is a tax placed on the export of a good. The goal behind an export tariff is to discriminate *against* exporting in order to ensure there is sufficient supply of a good within a country. For example, in the past, China has placed an export tariff on the export of grain to ensure China has a sufficient supply. Similarly, during its infrastructure building boom, China had an export tariff in place on certain kinds of steel products to ensure a sufficient supply of steel remained within the country. The steel tariffs were removed in late 2015. Because most countries try to encourage exports, export tariffs are relatively rare. Argentina has export tariffs on several of its agricultural products, including wheat, corn, and soybeans, in order to reduce exports and make more products available for domestic consumers at a lower price. This might help domestic consumers in the short run, but the policy has been bad for the Argentina farmers and the economy, the growth of which has been stunted by a lack of export opportunities.

An **export ban** is a policy that partially or entirely restricts the export of a good. One well-known example was the ban on exports of U.S. crude oil production that was enacted by Congress in 1975. At the time, Organization of the Petroleum Exporting Countries (OPEC) was restricting the supply of oil to drive up prices and punish Western nations for their support of Israel during conflicts between Arab nations and Israel. The export ban in the United States was seen as a way to ensure a sufficient supply of domestic oil at home, thereby helping to keep the domestic price down and boosting national security. The ban was lifted in 2015 after lobbying from U.S. oil producers, who believed they could get higher prices for some of their output if they were allowed to sell on world markets. A more recent export ban was the one the Trump administration placed on exports of microprocessors to the Chinese telecommunications giant Huawei in 2019. This case, and its impact upon one U.S. firm, Qualcomm, is discussed in the Management Focus feature.

export tariff

A tax placed on the export of a good.

export ban

A policy that partially or entirely restricts the export of a good.

MANAGEMENT FOCUS

Huawei Export Ban Hits U.S. Firms

In May 2019, U.S. Commerce Secretary Wilbur Ross announced that the largest Chinese telecommunications equipment provider, Huawei, would be put on the United States Entity List. The Entity List is a trade blacklist consisting of foreign persons, entities, or governments that U.S. citizens and firms are restricted from doing business with. Ross's action was prompted by concerns that Huawei poses a national security threat to the United States. The U.S. government has maintained that Huawei's equipment could be used as a backdoor by the Chinese government to spy on Americans, allegations the company has repeatedly denied. The move banned U.S. companies from selling their products to Huawei. Caught up in the ban were major U.S. companies such as Google, whose Android operating systems were used in Huawei smartphones, and Qualcomm, which was a major communications chip supplier to Huawei.

Huawei responded by claiming that the ban was "politically motivated" and lacked justification and that the Trump administration was using Huawei as a bargaining chip in part of a bigger battle to extract more favorable terms in its ongoing trade negotiations with China. The company also argued that "[t]hese actions violate the basic principles of free market competition. They are in no one's interests, including U.S.

companies." Huawei, which reported revenue of over \$100 billion for the first time in 2018—representing a 20 percent increase from 2017—estimated that the ban would reduce its revenues by \$30 billion over the next two years. Huawei's founder and CEO, Ren Zhengfei, noted that Huawei would now start to look at creating different versions of its products that did not rely on American components. "In the next two years, we are going to do a lot of switchover of different product versions that will take time and that will take time to ramp up, and it will take some time to test whether that works," Ren said. "After that step, we will be stronger."*

Just how significant this ban could be for Huawei's U.S. suppliers became clear in 2021 when reports surfaced that Qualcomm's market share of China's smartphone chip market plunged in 2020 due to the sanctions on Huawei. According to reports, Qualcomm's shipments in China shrank 48.1 percent year-on-year in 2020. The company's market share in China fell to 25.4 percent in 2020 versus 37.9 percent in 2019. For its part, Qualcomm acknowledged that the export ban had hurt its business. Qualcomm's market share in China began to decline through 2020 as Huawei ramped up its focus on its own chip division called HiSilicon. Huawei's high-end devices use the company's Kirin chips, designed by Huawei and manufactured by Taiwan's TSMC.

*A. Kharpal, "Huawei Slashes Revenue Forecast amid Continuing U.S. Pressure," CNBC, June 17, 2019.

In May 2020, the United States reacted to Huawei's strategy of shifting production to TSMC by issuing a rule requiring foreign manufacturers that use U.S. chipmaking equipment to get a license from the U.S. government before selling semiconductors to firms such as Huawei. TSMC, like most chipmakers, relies on equipment made by U.S. companies such as Lam Research and Applied Materials to make chips. Huawei's response was to look for other supply sources and to accelerate its own investment in chipmaking capacity. An early winner was Taiwan's MediaTek, a direct competitor to Qualcomm, which has seen its market share in China jump. The Chinese government,

too, responded by increasing its investment in semiconductor manufacturing R&D.

Sources: A. Kharpal, "Huawei Slashes Revenue Forecast amid Continuing U.S. Pressure," CNBC, June 17, 2019; A. Kharpal, "Why New Rules on Selling Chips to Huawei Could Be a Big Blow for the Chinese Giant," CNBC, May 18, 2020; A. Kharpal, "Huawei Calls US Move to Blacklist More Affiliates 'Unjust' and 'Politically Motivated,'" CNBC, August 19, 2019; "Qualcomm's Chip Market Share Plunges in China," CNBC, January 21, 2021; "Chipmaking Is Being Redesigned. Effects Will Be Far Reaching," *The Economist*, January 23, 2021; A. Fitch, "Qualcomm Sales Jump on iPhone, 5G Demand Surge," *The Wall Street Journal*, February 3, 2021.

SUBSIDIES

subsidy

Government financial assistance to a domestic producer.

A **subsidy** is a government payment to a domestic producer. Subsidies take many forms, including cash grants, low-interest loans, tax breaks, and government equity participation in domestic firms. By lowering production costs, subsidies help domestic producers in two areas: (1) competing against foreign imports and (2) gaining export markets. Agriculture tends to be one of the largest beneficiaries of subsidies in most countries. The European Union has been paying out about €54 billion annually (\$64 billion) in farm subsidies. The farm bill that passed the U.S. Congress in 2018 contained subsidies to producers that equal roughly \$25 billion a year for the next 10 years. The Japanese also have a long history of supporting inefficient domestic producers with farm subsidies. According to the World Trade Organization, in mid-2000 countries spent some \$300 billion on subsidies, \$250 billion of which was spent by 21 developed nations.⁴ In response to a severe sales slump following the global financial crisis, between mid-2008 and mid-2009, some developed nations gave \$45 billion in subsidies to their automobile makers. While the purpose of the subsidies was to help them survive a very difficult economic climate, one of the consequences was to give subsidized companies an unfair competitive advantage in the global auto industry. Somewhat ironically, given the government bailouts of U.S. auto companies during the global financial crisis, in 2012 the Obama administration filed a complaint with the WTO arguing that the Chinese were illegally subsidizing exports of autos and auto parts. Details are given in the accompanying Country Focus feature.

COUNTRY FOCUS

Were the Chinese Illegally Subsidizing Auto Exports?

In late 2012, during that year's presidential election campaign, the Obama administration filed a complaint against China with the World Trade Organization. The complaint claimed that China was providing export subsidies to its auto and auto parts industries. The subsidies included cash grants for exporting, grants for R&D, subsidies to pay interest on loans, and preferential tax treatment.

The United States estimated the value of the subsidies to be at least \$1 billion between 2009 and 2011. The complaint also pointed out that during the years 2002 through 2011, the value of China's exports of autos and auto

parts increased more than ninefold from \$7.4 billion to \$69.1 billion. The United States was China's largest market for exports of auto parts during this period. The United States asserted that, to some degree, this growth may have been helped by subsidies. The complaint went on to claim that these subsidies hurt producers of automobiles and auto parts in the United States. This is a large industry in the United States, employing more than 800,000 people and generating some \$350 billion in sales.

While some in the labor movement applauded the move, the response from U.S. auto companies and auto parts producers was muted. One reason for this is that many U.S. producers do business in China and, in all probability, want to avoid retaliation from the Chinese government. GM, for example, has a joint venture and two wholly

owned subsidiaries in China and is doing very well there. In addition, some U.S. producers benefit by purchasing cheap Chinese auto parts, so any retaliatory tariffs imposed on those imports might actually raise their costs.

More cynical observers saw the move as nothing more than political theater. The week before the complaint was filed, the Republican presidential candidate, Mitt Romney, had accused the Obama administration of “failing American workers” by not labeling China a currency manipulator. So perhaps the complaint was in part simply another move on the presidential campaign chessboard.

In February 2014, the United States expanded its complaint with the WTO against China, arguing that the country had an illegal export subsidy program that includes not

only auto parts, but also textiles, apparel and footwear, advanced materials and metals, speciality chemicals, medical products, and agriculture. In 2016, after pressure from the WTO and U.S., China agreed to eliminate a wide range of subsidies for its exporters. Michael Froman, the U.S. Trade Representative, announced the deal, calling it “a win for Americans employed in seven diverse sectors that run the gamut from agriculture to textiles.”

Sources: J. Healey, “U.S. Alleges Unfair China Auto Subsidies in WTO Action,” *USA Today*, September 17, 2012; M. A. Memoli, “Obama to Tell WTO That China Illegally Subsidizes Auto Imports,” *Los Angeles Times*, September 17, 2012; V. Needham, “US Launches Trade Case against China’s Export Subsidy Program,” *The Hill*, February 11, 2014; D. J. Lynch, “China Eliminates Subsidies for Its Exporters,” *Financial Times*, April 14, 2016.

The main gains from subsidies accrue to domestic producers, whose international competitiveness may be increased as a result. Advocates of strategic trade policy (which, as you will recall from Chapter 6, is an outgrowth of the new trade theory) favor subsidies to help domestic firms achieve a dominant position in those industries in which economies of scale are important and the world market is not large enough to profitably support more than a few firms. Large commercial jet aircraft and leading-edge semiconductors are two such businesses. According to this argument, subsidies can help a firm achieve a first-mover advantage in an emerging industry. If this is achieved, further gains to the domestic economy arise from the employment and tax revenues that a major global company can generate. However, government subsidies must be paid for, typically by taxing individuals and corporations.

Whether subsidies generate national benefits that exceed their national costs is debatable. In practice, many subsidies are not that successful at increasing the international competitiveness of domestic producers. Rather, they tend to protect the inefficient and promote excess production. One study estimated that if advanced countries abandoned subsidies to farmers, global trade in agricultural products would be 50 percent higher and the world as a whole would be better off by \$160 billion.⁵ Another study estimated that removing all barriers to trade in agriculture (both subsidies and tariffs) would raise world income by \$182 billion.⁶ This increase in wealth arises from the more efficient use of agricultural land.

IMPORT QUOTAS AND VOLUNTARY EXPORT RESTRAINTS

An **import quota** is a direct restriction on the quantity of some good that may be imported into a country. The restriction is usually enforced by issuing import licenses to a group of individuals or firms. For example, the United States has a quota on cheese imports. The only firms allowed to import cheese are certain trading companies, each of which is allocated the right to import a maximum number of pounds of cheese each year. In some cases, the right to sell is given directly to the governments of exporting countries.

A common hybrid of a quota and a tariff is known as a tariff rate quota. Under a **tariff rate quota**, a lower tariff rate is applied to imports within the quota than those over the quota. For example, as illustrated in Figure 7.1, an ad valorem tariff rate of 10 percent might be levied on 1 million tons of rice imports into South Korea, after which an out-of-quota rate of 80 percent might be applied. Thus, South Korea might import 2 million tons of rice, 1 million at a 10 percent tariff rate and another 1 million at an 80 percent tariff. Tariff rate quotas are common in agriculture, where their goal is to limit imports over quota. In May 2022, the Biden administration placed a tariff rate quota on imports of steel from the United Kingdom and EU. Under this agreement, the United States allowed 500,000 tons of steel products to enter the United States tariff free. For imports in excess of 500,000 tons, however, a 25 percent import tariff would be imposed.

import quota

A direct restriction on the quantity of a good that can be imported into a country.

tariff rate quota

Lower tariff rates applied to imports within the quota than those over the quota.

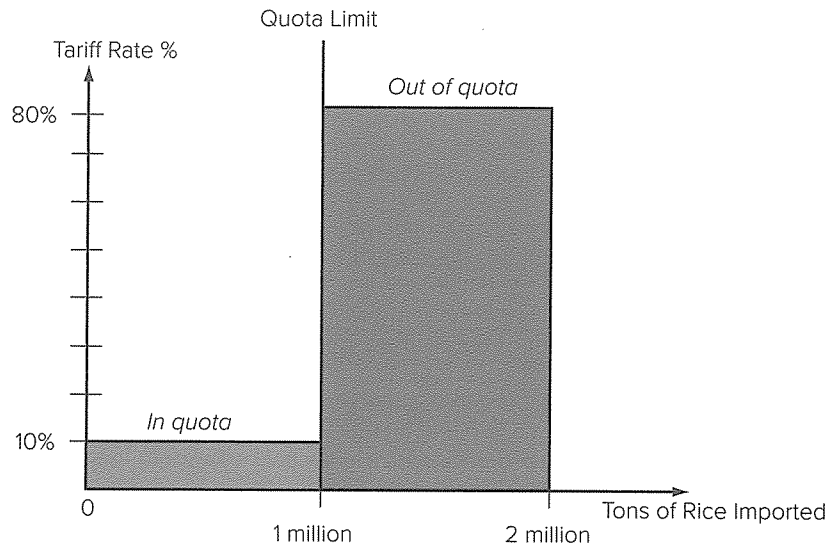


FIGURE 7.1 Hypothetical tariff rate quota.

voluntary export restraint (VER)

A quota on trade imposed from the exporting country's side, instead of the importer's; usually imposed at the request of the importing country's government.

A variant on the import quota is the voluntary export restraint. A **voluntary export restraint (VER)** is a quota on trade imposed by the exporting country, typically at the request of the importing country's government. For example, in 2012 Brazil imposed what amounts to voluntary export restraints on shipments of vehicles from Mexico to Brazil. The two countries have a decade-old free trade agreement, but a surge in vehicles heading to Brazil from Mexico prompted Brazil to raise its protectionist walls. Mexico agreed to quotas on Brazil-bound vehicle exports for three years.⁷ Foreign producers agree to VERs because they fear more damaging punitive tariffs or import quotas might follow if they do not. Agreeing to a VER is seen as a way to make the best of a bad situation by appeasing protectionist pressures in a country.

As with tariffs and subsidies, both import quotas and VERs benefit domestic producers by limiting import competition. As with all restrictions on trade, quotas do not benefit consumers. An import quota or VER always raises the domestic price of an imported good. When imports are limited to a low percentage of the market by a quota or VER, the price is bid up for that limited foreign supply. The extra profit that producers make when supply is artificially limited by an import quota is referred to as a **quota rent**.

quota rent

Extra profit producers make when supply is artificially limited by an import quota.

If a domestic industry lacks the capacity to meet demand, an import quota can raise prices for *both* the domestically produced and the imported good. This happened in the U.S. sugar industry, in which a tariff rate quota system has long limited the amount foreign producers can sell in the U.S. market. According to one study, import quotas have caused the price of sugar in the United States to be as much as 40 percent greater than the world price.⁸ These higher prices have translated into greater profits for U.S. sugar producers, which have lobbied politicians to keep the lucrative agreement. They argue U.S. jobs in the sugar industry will be lost to foreign producers if the quota system is scrapped.

local content requirement (LCR)

A requirement that some specific fraction of a good be produced domestically.

LOCAL CONTENT REQUIREMENTS

A **local content requirement (LCR)** is a requirement that some specific fraction of a good be produced domestically. The requirement can be expressed either in physical terms (e.g., 75 percent of component parts for this product must be produced locally) or in value terms (e.g., 75 percent of the value of this product must be produced locally). Local content regulations have been widely used by developing countries to shift their manufacturing base from the simple assembly of products whose parts are manufactured elsewhere into the local manufacture of component parts. They have also been used in developed countries to try to protect local jobs and industry from foreign competition. For example, a little-known law in the United States, the Buy America Act, specifies that government agencies must give preference to American products when putting contracts for equipment out to bid unless the foreign products have a significant price advantage. The law specifies a product as "American" if 51 percent of the materials by value are produced domestically.

This amounts to a local content requirement. If a foreign company, or an American one for that matter, wishes to win a contract from a U.S. government agency to provide some equipment, it must ensure that at least 51 percent of the product by value is manufactured in the United States.

Local content regulations provide protection for a domestic producer of parts in the same way an import quota does: by limiting foreign competition. The aggregate economic effects are also the same; domestic producers benefit, but the restrictions on imports raise the prices of imported components. In turn, higher prices for imported components are passed on to consumers of the final product in the form of higher final prices. So as with all trade policies, local content regulations tend to benefit producers and not consumers.

ADMINISTRATIVE POLICIES

In addition to the formal instruments of trade policy, governments of all types sometimes use informal or administrative policies to restrict imports and boost exports. **Administrative trade policies** are bureaucratic rules designed to make it difficult for imports to enter a country. It has been argued that the Japanese are the masters of this trade barrier. In recent decades, Japan's formal tariff and nontariff barriers have been among the lowest in the world. However, critics charge that the country's informal administrative barriers to imports more than compensate for this. For example, Japan's car market has been hard for foreigners to crack. In 2016, only 6 percent of the 4.9 million cars sold in Japan were foreign and only 1 percent were U.S. cars. American carmakers have argued for decades that Japan makes it difficult to compete by setting up regulatory hurdles, such as vehicle parts standards, that don't exist anywhere else in the world. The Trans-Pacific Partnership (TPP) negotiated by the Obama administration tried to address this issue. As part of the TPP deal, the United States would have reduced tariffs on imports of Japanese light trucks in return for Japan adopting U.S. standards on auto parts, which would have made it easier to import and sell American cars in Japan. However, President Donald Trump pulled the United States out of the TPP in January 2017.⁹

administrative trade policies

Administrative policies, typically adopted by government bureaucracies, that can be used to restrict imports or boost exports.

ANTIDUMPING POLICIES

In the context of international trade, **dumping** is variously defined as selling goods in a foreign market at below their costs of production or as selling goods in a foreign market at below their "fair" market value. The difference between these two definitions is that the fair market value of a good is normally judged to be greater than the costs of producing that good because the former includes a "fair" profit margin. Dumping is viewed as a method by which firms unload excess production in foreign markets. Some dumping may be the result of predatory behavior, with producers using substantial profits from their home markets to subsidize prices in a foreign market with a view to driving indigenous competitors out of that market. Once this has been achieved, so the argument goes, the predatory firm can raise prices and earn substantial profits.

dumping

Selling goods in a foreign market for less than their cost of production or below their "fair" market value.

Antidumping policies are designed to punish foreign firms that engage in dumping. The ultimate objective is to protect domestic producers from unfair foreign competition. Although antidumping policies vary from country to country, the majority are similar to those used in the United States. If a domestic producer believes that a foreign firm is dumping production in the U.S. market, it can file a petition with two government agencies: the Commerce Department and the International Trade Commission (ITC). If a complaint has merit, the Commerce Department may impose an antidumping duty on the offending foreign imports (antidumping duties are often called **countervailing duties**). These duties, which represent a special tariff, can be fairly substantial and stay in place for up to five years.

antidumping policies

Designed to punish foreign firms that engage in dumping and thus protect domestic producers from unfair foreign competition.

countervailing duties

Antidumping duties.



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The Case for Government Intervention



LO7-2

Understand why governments sometimes intervene in international trade.

Now that we have reviewed the various instruments of trade policy that governments can use, it is time to look at the case for government intervention in international trade. Arguments for government intervention take two paths: political and economic. Political arguments for intervention are concerned with protecting the interests of certain groups within a nation (normally producers), often at the expense of other groups (normally consumers), or with achieving some political objective that lies outside the sphere of economic relationships, such as protecting the environment or human rights. Economic arguments for intervention are typically concerned with boosting the overall wealth of a nation (to the benefit of all, both producers and consumers).

POLITICAL ARGUMENTS FOR INTERVENTION

Political arguments for government intervention cover a range of issues, including preserving jobs, protecting industries deemed important for national security, retaliating against unfair foreign competition, protecting consumers from “dangerous” products, furthering the goals of foreign policy, and advancing the human rights of individuals in exporting countries.

Protecting Jobs and Industries

Perhaps the most common political argument for government intervention is that it is necessary for protecting jobs and industries from unfair foreign competition. Competition is most often viewed as unfair when producers in an exporting country are subsidized in some way by their government. For example, it has been repeatedly claimed that Chinese enterprises in several industries, including aluminum, steel, and auto parts, have benefited from extensive government subsidies. Such logic was behind the complaint that the Obama administration filed with the WTO against Chinese auto parts producers in 2012 (see the Country Focus, “Were the Chinese Illegally Subsidizing Auto Exports?” in this chapter). More generally, Robert Scott of the Economic Policy Institute has claimed that the growth in the U.S.–China trade deficit between 2001 and 2015 was, to a significant degree, the result of unfair competition, including direct subsidies to Chinese producers and currency manipulations. Scott estimated that as many as 3.4 million U.S. jobs were lost as a consequence.¹⁰ Donald Trump tapped into anxiety about job losses due to unfair trade from China during his successful 2016 presidential run.

On the other hand, critics charge that claims of unfair competition are often overstated for political reasons. For example, President George W. Bush placed tariffs on imports of foreign steel in 2002 as a response to “unfair competition,” but critics were quick to point out that many of the U.S. steel producers that benefited from these tariffs were located in states that Bush needed to win reelection in 2004. A political motive also underlay establishment of the Common Agricultural Policy (CAP) by the European Union. The CAP was designed to protect the jobs of Europe’s politically powerful farmers by restricting imports and guaranteeing prices. However, the higher prices that resulted from the CAP have cost Europe’s consumers dearly. This is true of many attempts to protect jobs and industries through government intervention. For example, the imposition of steel tariffs in 2002 raised steel prices for American consumers, such as automobile companies, making them less competitive in the global marketplace.

Protecting National Security

Countries sometimes argue that protecting certain industries is necessary because they are important for national security. Defense-related industries often get this kind of attention (e.g., aerospace, advanced electronics, and semiconductors). When the Trump administration announced tariffs on imports of foreign steel and aluminum on March 1, 2018, national security issues were cited as a primary justification. This was the first time since 1986 that a national security threat was used to justify tariffs imposed by the United States. In 2017, the United States was importing about 30 percent of steel used in the country, with the largest sources of imports being Canada and Mexico. Interestingly, and counter to the argument of the Trump administration, critics argued that by raising input prices for many U.S. defense contractors, which tend to be big consumers of steel and aluminum, the tariffs would actually harm the U.S. defense industry and have a negative impact on national security. In December 2022, the national security rationale was invoked by the Biden administration when announcing a ban on commercial exports of advanced U.S. technology that it claimed aided China's military and hypersonics programs and enabled human rights violations.¹¹



globalEDGE

Trade Law

Government policy and international trade is the core focus of this chapter. This topic area has far-ranging implications, such as trade policy, free trade, and the world's international trading system. Basically, we are talking about a lot of legalistic aspects starting at the government level and moving all the way to what organizations and even individuals can and cannot do globally when trading. The globalEDGE™ section “Trade Law” (globaleledge.msu.edu/global-resources/trade-law) is a unique compilation of globalEDGE™ partner-designed “compendiums of trade laws,” country- and region-specific trade law, free

online learning modules created for globalEDGE™ on various aspects of trade law, and much more. One fascinating resource related to trade law is the Anti-Counterfeiting and Product Protection Program (A-CAPPP). A-CAPPP includes counterfeiting-related webinars, presentations, and research-related materials and working papers. Do you know what counterfeiting is? Take a look at the “Trade Law” section of globalEDGE™, and especially the A-CAPPP site, to become more familiar with the topic. (Is China really as bad as many in the international community think?)

Retaliating

Some argue that governments should use the threat to intervene in trade policy as a bargaining tool to help open foreign markets and force trading partners to “play by the rules of the game.” The U.S. government has used the threat of punitive trade sanctions to try to get the Chinese government to enforce its intellectual property laws. During the 1990s, lax enforcement of these laws gave rise to massive copyright infringements in China that cost U.S. companies such as Microsoft hundreds of millions of dollars per year in lost sales revenues. After the United States threatened to impose 100 percent tariffs on a range of Chinese imports and after officials from the two countries exchanged harsh words, the Chinese agreed to tighter enforcement of intellectual property regulations.¹² More recently, two of the stated goals of the Trump administration's imposition of 25 percent tariffs on imports of \$370 billion worth of Chinese imports was to get the Chinese to enact tougher intellectual property laws and to encourage them to stop subsidizing state-owned enterprises. A third goal was to reduce the annual trade deficit with China.

If such a politically motivated rationale for government intervention works, it may liberalize trade and bring with it resulting economic gains. The strategy is risky, however. A country that is being pressured may not back down and instead may respond to the imposition of punitive tariffs by raising trade barriers of its own. This is what happened in the case of Trump's China tariffs. The Chinese quickly responded to the Trump administration's initial tariff raising actions in 2018 by placing tariffs on imports of U.S. farm products. Over the next year, U.S. farm exports to China fell by 63 percent from \$15.8 billion to \$5.9 billion. This created considerable hardship for many American farmers. To help the farmers, the Trump administration increased government support of U.S. farmers by \$28 billion between 2018 and 2019, a move that was financed by U.S. taxpayers. The United States and China reached a preliminary deal in early 2020.

This “phase I” deal was a managed trade arrangement, not a free trade deal, under which the Chinese committed to increasing their purchase of U.S. goods and services by at least \$200 billion over the next two years. For its part, the Trump administration agreed to start removing or reducing some of the tariffs placed on Chinese imports. However, research suggests that in its first two years in force, the Chinese only purchased 58 percent of the total goods that it had agreed to buy under the agreement with the Trump administration, falling short of its target by \$200 billion.¹³

Protecting Consumers

Many governments have long had regulations to protect consumers from unsafe products. The indirect effect of such regulations often is to limit or ban the importation of such products. For example, in 2003 several countries, including Japan and South Korea, decided to ban imports of U.S. beef after a single case of mad cow disease was found in Washington State. The ban was designed to protect consumers from what was seen to be an unsafe product. Together, Japan and South Korea accounted for about \$2 billion of U.S. beef sales, so the ban had a significant impact on U.S. beef producers. After two years, both countries lifted the ban, although they placed stringent requirements on U.S. beef imports to reduce the risk of importing beef that might be tainted by mad cow disease (e.g., Japan required that all beef must come from cattle under 21 months of age).

Furthering Foreign Policy Objectives

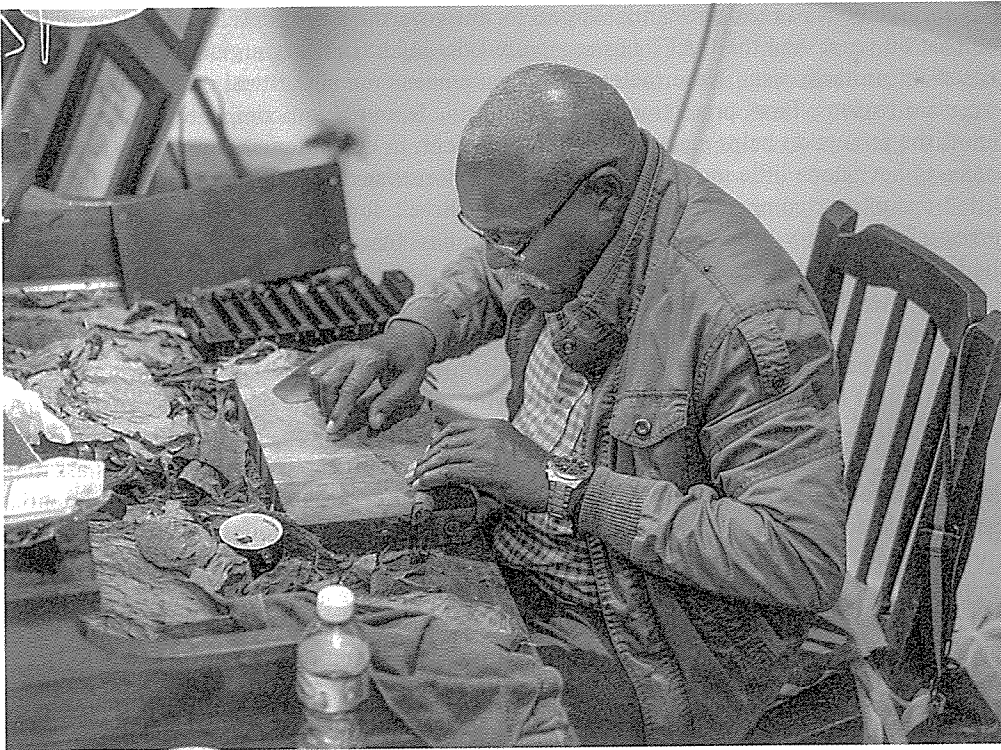
Governments sometimes use trade policy to support their foreign policy objectives.¹⁴ A government may grant preferential trade terms to a country with which it wants to build strong relations. Trade policy has also been used several times to pressure or punish “rogue states” that do not abide by international law or norms. When Russia invaded Ukraine in February 2022, more than 30 countries announced trade sanctions against Russia, including the United States, Canada, France, Germany, the United Kingdom, Japan, and South Korea. Sanctions included cutting imports of oil and natural gas from Russia; blocking financial transactions; and halting exports of key goods to Russia, including semiconductors, communications equipment, and other electronics. While there is growing evidence that these sanctions are starting to take a toll on the Russian economy, they have yet to achieve their objective of getting Russia to withdraw from Ukraine. The sanctions did cause the Russian economy to shrink by around 3 percent in 2022. Forecasts suggest that further contraction is likely in 2023. However, to date Russia has managed to avoid the worst impact of sanctions thanks to countries like China and India, which have stepped up their purchases of Russian oil, which has helped to keep Russia’s economy afloat.¹⁵

In another example, the United States has maintained long-running trade sanctions against Cuba (despite the move by the Obama administration to “normalize” relations with Cuba, these sanctions are still in place). Their principal function is to impoverish Cuba in the hope that the resulting economic hardship will lead to the downfall of Cuba’s communist government and its replacement with a more democratically inclined (and pro-U.S.) regime. The United States has also had trade sanctions in place against Libya and Iran, both of which were accused of supporting terrorist action against U.S. interests and building weapons of mass destruction. In late 2003, the sanctions against Libya seemed to yield some returns when that country announced it would terminate a program to build nuclear weapons. The U.S. government responded by relaxing those sanctions. Similarly, the U.S. government used trade sanctions to pressure the Iranian government to halt its alleged nuclear weapons program. Following a 2015 agreement to limit Iran’s nuclear program, it relaxed some of those sanctions. However, the Trump administration reimposed significant sanctions, arguing that Iran was not adhering to the 2015 agreement.

Other countries can undermine trade sanctions. The U.S. sanctions against Cuba, for example, did not stop other Western countries from trading with Cuba. The U.S. sanctions have done little more than help create a vacuum into which other trading nations, such as Canada and Germany, have stepped. Similarly, as noted earlier, to date Russia has been able to avoid the worse consequences of trade sanctions by switching the sales of its energy products to China and India.

Protecting Human Rights

Protecting and promoting human rights in other countries is an important element of foreign policy for many democracies. Governments sometimes use trade policy to try to improve the



The famous cigar maker Jose Castelar Cairo, better known as El Cueto, about to roll a cigar in Havana, Cuba.

Esben Hansen/123RF

human rights policies of trading partners. For example, in the 1980s and 1990s, Western governments used trade sanctions against South Africa as a way of pressuring that nation to drop its apartheid policies, which were seen as a violation of basic human rights. Similarly, the U.S. government long had trade sanctions in place against the nation of Myanmar, primarily due to the poor human rights practices in that nation. In late 2012, the United States removed trade sanctions against Myanmar in response to democratic reforms in the country. However, a military coup in early 2021 ousted the country's democratically elected government, and it is now possible that some trade sanctions will be reimposed.

ECONOMIC ARGUMENTS FOR INTERVENTION

With the development of the new trade theory and strategic trade policy (see Chapter 6), the economic arguments for government intervention have undergone a renaissance in recent years. Until the early 1980s, most economists saw little benefit in government intervention and strongly advocated a free trade policy. This position has changed at the margins with the development of strategic trade policy, although as we will see in the next section, there are still strong economic arguments for sticking to a free trade stance.

The Infant Industry Argument

The **infant industry argument** is by far the oldest economic argument for government intervention. Alexander Hamilton proposed it in 1792. According to this argument, many developing countries have a potential comparative advantage in manufacturing, but new manufacturing industries cannot initially compete with established industries in developed countries. To allow manufacturing to get a toehold, the argument is that governments should temporarily support new industries (with tariffs, import quotas, and subsidies) until those industries have grown strong enough to meet international competition.

infant industry argument

New industries in developing countries must be temporarily protected from international competition to help them reach a position where they can compete on world markets with the firms of developed nations.

This argument has had substantial appeal for the governments of developing nations during the past 60 years, and the GATT has recognized the infant industry argument as a legitimate reason for protectionism. Nevertheless, many economists remain critical of this argument for two main reasons. First, protection of manufacturing from foreign competition does no good unless the protection helps make the industry efficient. In case after case, however, protection seems to have done little more than foster the development of inefficient industries that have little hope of ever competing in the world market. Brazil, for example, built the world's tenth largest auto industry behind tariff barriers and quotas. Once those barriers were removed in the late 1980s, however, foreign imports soared, and the industry was forced to face up to the fact that after 30 years of protection, the Brazilian auto industry was one of the world's most inefficient.¹⁶

Second, the infant industry argument relies on an assumption that firms are unable to make efficient long-term investments by borrowing money from the domestic or international capital market. Consequently, governments have been required to subsidize long-term investments. Given the development of global capital markets over the past 30 years, this assumption no longer looks as valid as it once did. Today, if a developing country has a potential comparative advantage in a manufacturing industry, firms in that country should be able to borrow money from the capital markets to finance the required investments. Given financial support, firms based in countries with a potential comparative advantage have an incentive to endure the necessary initial losses in order to make long-run gains without requiring government protection. Many Taiwanese and South Korean firms did this in industries such as textiles, semiconductors, machine tools, steel, and shipping. Thus, given efficient global capital markets, the only industries that would require government protection would be those that are not worthwhile.

Strategic Trade Policy

Some new trade theorists have proposed the strategic trade policy argument.¹⁷ We reviewed the basic argument in Chapter 6 when we considered the new trade theory. The new trade theory argues that in industries in which the existence of substantial economies of scale implies that the world market will profitably support only a few firms, countries may predominate in the export of certain products simply because they have firms that were able to capture first-mover advantages. The long-term dominance of Boeing in the commercial aircraft industry has been attributed to such factors.

The **strategic trade policy** argument has two components. First, theorists argue that by appropriate actions, a government can help raise national income if it can somehow ensure that the firm or firms that gain first-mover advantages in an industry are domestic rather than foreign enterprises. Thus, according to the strategic trade policy argument, a government should use subsidies to support promising firms that are active in newly emerging industries. Advocates of this argument point out that the substantial R&D grants that the U.S. government gave Boeing in the 1950s and 1960s probably helped tilt the field of competition in the newly emerging market for passenger jets in Boeing's favor. (Boeing's first commercial jet airliner, the 707, was derived from a military plane.) Similar arguments have been made with regard to Japan's rise to dominance in the production of liquid crystal display screens (used in computers). Although these screens were invented in the United States, the Japanese government, in cooperation with major electronics companies, targeted this industry for research support in the late 1970s and early 1980s. The result was that Japanese firms, not U.S. firms, subsequently captured first-mover advantages in this market.

The second component of the strategic trade policy argument is that it might pay a government to intervene in an industry by helping domestic firms overcome the barriers to entry created by foreign firms that have already reaped first-mover advantages. This argument underlies government support of Airbus, Boeing's major competitor. Formed in 1966 as a consortium of four companies from Great Britain, France, Germany, and Spain, Airbus had less than 5 percent of the world commercial aircraft market when it began production in the mid-1970s. By 2022, it was splitting the market with Boeing. How did Airbus achieve this? According to the U.S. government, the answer is an \$18 billion subsidy from the governments of Great Britain, France, Germany, and Spain. Without this subsidy, Airbus would never have been able to break into the world market.

strategic trade policy

Government policy aimed at improving the competitive position of a domestic industry and/or domestic firm in the world market.

If these arguments are correct, they support a rationale for government intervention in international trade. Governments should target technologies that may be important in the future and use subsidies to support development work aimed at commercializing those technologies. Furthermore, government should provide export subsidies until the domestic firms have established first-mover advantages in the world market. Government support may also be justified if it can help domestic firms overcome the first-mover advantages enjoyed by foreign competitors and emerge as viable competitors in the world market (as in the Airbus and liquid crystal display examples). In this case, a combination of home-market protection and export-promoting subsidies may be needed. One could argue that the subsidies for electric vehicle and battery production contained in the 2022 U.S. Inflation Reduction Act are consistent with the recommendations of strategic trade policy, and are designed to help United States counter threats from China in this sector (see the opening case).



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The Revised Case for Free Trade



Summarize and explain the arguments against strategic trade policy.

The strategic trade policy arguments of the new trade theorists suggest an economic justification for government intervention in international trade. This justification challenges the rationale for unrestricted free trade found in the work of classic trade theorists such as Adam Smith and David Ricardo. In response to this challenge to economic orthodoxy, a number of economists—including some of those responsible for the development of the new trade theory, such as Paul Krugman—point out that although strategic trade policy looks appealing in theory, in practice it may be unworkable. This response to the strategic trade policy argument constitutes the revised case for free trade.¹⁸

RETALIATION AND TRADE WAR

Krugman argues that a strategic trade policy aimed at establishing domestic firms in a dominant position in a global industry is a beggar-thy-neighbor policy that boosts national income at the expense of other countries. A country that attempts to use such policies will probably provoke retaliation. In many cases, the resulting trade war between two or more interventionist governments will leave all countries involved worse off than if a hands-off approach had been adopted in the first place. If the U.S. government were to respond to the Airbus subsidy by increasing its own subsidies to Boeing, for example, the result might be that the subsidies would cancel each other out. In the process, both European and U.S. taxpayers would end up supporting an expensive and pointless trade war, and both Europe and the United States would be worse off. The 2022 Inflation Reduction Act might have brought us close to this point in the production of electric vehicles and lithium ion batteries since the EU, South Korea, and China all appear to be considering responding by increasing their own state support of electric vehicle and battery producers (see the opening case).

Krugman may be right about the danger of a strategic trade policy leading to a trade war. The problem, however, is how to respond when one's competitors are already being supported by government subsidies; that is, how should Boeing and the United States respond to the subsidization of Airbus? According to Krugman, the answer is probably not to engage in retaliatory action but to help establish rules of the game that minimize the use of trade-distorting subsidies. This is what the World Trade Organization seeks to do. We should also note that antidumping policies can be used to target competitors supported by subsidies who are selling goods at prices that are below their costs of production.

DOMESTIC POLICIES

Governments do not always act in the national interest when they intervene in the economy; politically important interest groups often influence them. The European Union's support for the Common Agricultural Policy (CAP), which arose because of the political power of French and German farmers, is an example. The CAP benefits inefficient farmers and the politicians who rely on the farm vote but not consumers in the EU, who end up paying more for their food-stuffs. Thus, a further reason for not embracing strategic trade policy, according to Krugman, is that such a policy is almost certain to be captured by special-interest groups within the economy, which will distort it to their own ends. Krugman concludes that in the United States,

To ask the Commerce Department to ignore special-interest politics while formulating detailed policy for many industries is not realistic; to establish a blanket policy of free trade, with exceptions granted only under extreme pressure, may not be the optimal policy according to the theory but may be the best policy that the country is likely to get.¹⁹



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Development of the World Trading System



Describe the development of the world trading system and the current trade issue.

Economic arguments support unrestricted free trade. While many governments have recognized the value of these arguments, they have been unwilling to unilaterally lower their trade barriers for fear that other nations might not follow suit. Consider the problem that two neighboring countries, say, Brazil and Argentina, face when deciding whether to lower trade barriers between them. In principle, the government of Brazil might favor lowering trade barriers, but it might be unwilling to do so for fear that Argentina will not do the same. Instead, the government might fear that the Argentines will take advantage of Brazil's low barriers to enter the Brazilian market while continuing to shut Brazilian products out of their market through high trade barriers. The Argentinean government might believe that it faces the same dilemma. The essence of the problem is a lack of trust. Both governments recognize that their respective nations will benefit from lower trade barriers between them, but neither government is willing to lower barriers for fear that the other might not follow.²⁰

Such a deadlock can be resolved if both countries negotiate a set of rules to govern cross-border trade and lower trade barriers. But who is to monitor the governments to make sure they are playing by the trade rules? And who is to impose sanctions on a government that cheats? Both governments could set up an independent body to act as a referee. This referee could monitor trade between the countries, make sure that no side cheats, and impose sanctions on a country if it does cheat in the trade game.

While it might sound unlikely that any government would compromise its national sovereignty by submitting to such an arrangement, since World War II an international trading framework has evolved that has exactly these features. For its first 50 years, this framework was known as the General Agreement on Tariffs and Trade (GATT). Since 1995, it has been known as the World Trade Organization (WTO). Here, we look at the evolution and workings of the GATT and WTO.

FROM SMITH TO THE GREAT DEPRESSION

As noted in Chapter 6, the theoretical case for free trade dates to the late eighteenth century and the work of Adam Smith and David Ricardo. Free trade as a government policy was first officially embraced by Great Britain in 1846, when the British Parliament repealed the Corn Laws. The Corn Laws placed a high tariff on imports of foreign grain. The objectives of the Corn Laws tariff were to raise government revenues and to protect British grain producers. Members had made annual motions in Parliament in favor of free trade since the 1820s, when David Ricardo was a member. However, agricultural protection was withdrawn only as a result of a protracted debate when the effects of a harvest failure in Great Britain were compounded by the imminent threat of famine in Ireland. Faced with considerable hardship and suffering among the populace, Parliament narrowly reversed its long-held position.

During the next 80 years or so, Great Britain, as one of the world's dominant trading powers, pushed the case for trade liberalization, but the British government was a voice in the wilderness. Its major trading partners did not reciprocate the British policy of unilateral free trade. The only reason Britain kept this policy for so long was that, as the world's largest exporting nation, it had far more to lose from a trade war than did any other country.

By the 1930s, the British attempt to stimulate free trade was buried under the economic rubble of the Great Depression. Economic problems were compounded in 1930, when the U.S. Congress passed the Smoot-Hawley tariff. Aimed at avoiding rising unemployment by protecting domestic industries and diverting consumer demand away from foreign products, the **Smoot-Hawley Act** erected an enormous wall of tariff barriers. Almost every industry was rewarded with its "made-to-order" tariff. The Smoot-Hawley Act had a damaging effect on employment abroad. Other countries reacted by raising their own tariff barriers. U.S. exports tumbled in response, and the world slid further into the Great Depression.²¹

Smoot-Hawley Act
Enacted in 1930 by the U.S. Congress, this Act erected a wall of tariff barriers against imports into the United States.

1947–1979: GATT, TRADE LIBERALIZATION, AND ECONOMIC GROWTH

Economic damage caused by the beggar-thy-neighbor trade policies that the Smoot-Hawley Act ushered in exerted a profound influence on the economic institutions and ideology of the post-World War II world. The United States emerged from the war both victorious and economically dominant. After the debacle of the Great Depression, opinion in the U.S. Congress had swung strongly in favor of free trade. Under U.S. leadership, the GATT was established in 1947.

The GATT was a multilateral agreement whose objective was to liberalize trade by eliminating tariffs, subsidies, import quotas, and the like. From its foundation in 1947 until it was superseded by the WTO, the GATT's membership grew from 19 to more than 120 nations. The GATT did not attempt to liberalize trade restrictions in one fell swoop; that would have been impossible. Rather, tariff reduction was spread over eight rounds.

In its early years, the GATT was, by most measures, very successful. For example, the average tariff declined by nearly 92 percent in the United States between the Geneva Round of 1947 and the Tokyo Round of 1973–1979. Consistent with the theoretical arguments first advanced by Ricardo and reviewed in Chapter 6, the move toward free trade under the GATT appeared to stimulate economic growth.

1980–1993: PROTECTIONIST TRENDS

During the 1980s and early 1990s, the trading system erected by the GATT came under strain as pressures for greater protectionism increased around the world. Three reasons caused the rise in such pressures during the 1980s. First, the economic success of Japan during that time strained the world trading system (much as the success of China has created strains today). Japan was in ruins when the GATT was created. By the early 1980s, however, it had become the world's second-largest economy and its largest exporter. Japan's success in such industries as automobiles and semiconductors might have been enough to strain the world trading system. Things were made worse by the widespread perception in the West that despite low tariff rates and subsidies, Japanese markets were closed to imports and foreign investment by administrative trade barriers.

Second, the world trading system was strained by the persistent trade deficit in the world's largest economy, the United States. The consequences of the U.S. deficit included painful adjustments in industries such as automobiles, machine tools, semiconductors, steel, and textiles, where domestic producers steadily lost market share to foreign competitors. The resulting unemployment gave rise to renewed demands in the U.S. Congress for protection against imports.

A third reason for the trend toward greater protectionism was that many countries found ways to get around GATT regulations. Bilateral voluntary export restraints (VERs) circumvented GATT agreements, because neither the importing country nor the exporting country complained to the GATT bureaucracy in Geneva—and without a complaint, the GATT bureaucracy could do nothing. Exporting countries agreed to VERs to avoid more damaging punitive tariffs. One of the best-known examples was the automobile VER between Japan and the United States, under which Japanese producers promised to limit their auto imports into the United States as a way of defusing growing trade tensions. According to a World Bank study, 16 percent of the imports of industrialized countries in 1986 were subjected to nontariff trade barriers such as VERs.²²

THE URUGUAY ROUND AND THE WORLD TRADE ORGANIZATION

Against the background of rising pressures for protectionism, in 1986, GATT members embarked on their eighth round of negotiations to reduce tariffs, the Uruguay Round (so named because it was held in Uruguay). This was the most ambitious round of negotiations yet. Until then, GATT rules had applied only to trade in manufactured goods and commodities. In the Uruguay Round, member countries sought to extend GATT rules to cover trade in services. They also sought to write rules governing the protection of intellectual property, to reduce agricultural subsidies, and to strengthen the GATT's monitoring and enforcement mechanisms.

The Uruguay Round dragged on for seven years before an agreement was reached on December 15, 1993. It went into effect July 1, 1995. The Uruguay Round contained the following provisions:

1. Tariffs on industrial goods were to be reduced by more than one-third, and tariffs were to be scrapped on more than 40 percent of manufactured goods.
2. Average tariff rates imposed by developed nations on manufactured goods were to be reduced to less than 4 percent of value, the lowest level in modern history.
3. Agricultural subsidies were to be substantially reduced.
4. GATT fair trade and market access rules were to be extended to cover a wide range of services.
5. GATT rules also were to be extended to provide enhanced protection for patents, copyrights, and trademarks (intellectual property).
6. Barriers on trade in textiles were to be significantly reduced over 10 years.
7. The World Trade Organization was to be created to implement the GATT agreement.

The World Trade Organization

The WTO acts as an umbrella organization that encompasses the GATT along with two related bodies, one on services and the other on intellectual property. The WTO's General Agreement on Trade in Services (GATS) has taken the lead in extending free trade agreements to services.

The WTO's Agreement on Trade-Related Aspects of Intellectual Property Rights (TRIPS) is an attempt to narrow the gaps in the way intellectual property rights are protected around the world and to bring them under common international rules. The WTO has taken over responsibility for arbitrating trade disputes and monitoring the trade policies of member countries. Although the WTO operates on the basis of consensus as the GATT did, in the area of dispute settlement, member countries are no longer able to block adoption of arbitration reports. Arbitration panel reports on trade disputes between member countries are automatically adopted by the WTO unless a consensus exists to reject them. Countries that have been found by the arbitration panel to violate GATT rules may appeal to a permanent Appellate Body, but its verdict is binding. If offenders fail to comply with the recommendations of the arbitration panel, trading partners have the right to compensation or, in the last resort, to impose (commensurate) trade sanctions. Every stage of the procedure is subject to strict time limits. Thus, the WTO was designed to have something that the GATT never had—teeth.²³

The binding verdict of the Appellate Body is central to the WTO's enforcement mechanism. The Appellate Body has seven members who are appointed for staggered four-year terms. Appeals to the Appellate Body have to be heard by a quorum of three members. Several major trading nations, including the United States, have veto power over the appointment of Appellate Body members. The granting of veto power was originally seen as a safeguard against the appointment of members who might let national interests influence their rulings. Unforeseen when the WTO rules were originally negotiated was that the United States under the Trump and Biden administrations would use its veto power to block the appointment of any new members to the Appellate Body, effectively halting its ability to issue rulings once the number of members fell below three.

WTO: EXPERIENCE TO DATE

By 2022, the WTO had 164 members, including China, which joined at the end of 2001, and Russia, which joined in 2012. WTO members collectively account for 98 percent of world trade. Since its formation, the WTO has remained at the forefront of efforts to promote global free trade. Its creators expressed the belief that the enforcement mechanisms granted to the WTO would make it more effective at policing global trade rules than the GATT had been. The great hope was that the WTO might emerge as an effective advocate and facilitator of future trade deals, particularly in areas such as services. The experience so far has been mixed. After a strong early start, since the late 1990s the WTO has been unable to get agreements to further reduce barriers to international trade and investment. The current round of trade talks (the Doha Round) has made very slow progress and is effectively stalled. A shift back toward some limited protectionism occurred following the global financial crisis of 2008–2009.

More recently, the 2016 vote by the British to leave the European Union (Brexit) and the election of Donald Trump, who promised to put “America First,” to the U.S. presidency have suggested that the world may be shifting back toward greater protectionism. These developments have raised a number of serious questions about the future direction of the WTO. Donald Trump expressed ambivalence about the value of the WTO, and under his leadership, the United States stepped back from supporting the institution. Most notably, the United States refused to approve of the appointment of new members to the Appellate Body as their four-year terms expired. To the surprise of some, the Biden administration continued this policy stance. As a consequence, by 2022 there were no more members on the Appellate Body, and it could no longer issue binding rulings. As part of its pivot toward a more nationalistic and protectionist trade policy stance, the United States had effectively pulled the teeth out of the WTO's enforcement mechanism.

WTO as Global Police

The first quarter century in the life of the WTO suggested that its policing and enforcement mechanisms had a positive effect.²⁴ Between 1995 and 2022, 615 trade disputes covering 157 member countries were brought to the WTO.²⁵ This record compares with a total of 196 cases handled by the GATT over almost half a century. Of the cases brought to the WTO, three-fourths have been resolved by informal consultations between the disputing countries. Resolving the remainder has

involved more formal procedures, but until very recently these were largely successful. In general, the countries involved adopted the WTO's recommendations. The fact that countries used the WTO represented an important vote of confidence in the organization's dispute resolution procedures. Unfortunately, as noted earlier, the WTO's Appellate Body can no longer function due to the blocking strategy of the United States. As a consequence, the WTO can no longer issue binding rulings, which has weakened the WTO's enforcement mechanisms. The United States and other countries have taken advantage of this fact to push trade actions that are inconsistent with WTO rules. These include the imposition of tariffs in the ongoing trade war between the United States and China and the growing use of subsidies by several nations to support "strategic" industries such as semiconductors, electric vehicles, and lithium ion battery production.

Expanded Trade Agreements

As explained earlier, the Uruguay Round of GATT negotiations extended global trading rules to cover trade in services. The WTO was given the role of brokering future agreements to open up global trade in services. The WTO was also encouraged to extend its reach to encompass regulations governing foreign direct investment, something the GATT had never done. Two of the first industries targeted for reform were the global telecommunication and financial services industries.

In February 1997, the WTO brokered a deal to get countries to agree to open their telecommunication markets to competition, allowing foreign operators to purchase ownership stakes in domestic telecommunication providers and establishing a set of common rules for fair competition. Most of the world's biggest markets—including the United States, the European Union, and Japan—were fully liberalized by January 1, 1998, when the pact went into effect. All forms of basic telecommunication service are covered, including voice telephone, data, and satellite and radio communications. Many telecommunication companies responded positively to the deal, pointing out that it would give them a much greater ability to offer their business customers one-stop shopping—a global, seamless service for all their corporate needs and a single bill.

This pact was followed in December 1997 by an agreement to liberalize cross-border trade in financial services. The deal covered more than 95 percent of the world's financial services market. Under the agreement, which took effect at the beginning of March 1999, 102 countries pledged to open (to varying degrees) their banking, securities, and insurance sectors to foreign competition. In common with the telecommunication deal, the accord covers not just cross-border trade but also foreign direct investment. Seventy countries agreed to dramatically lower or eradicate barriers to foreign direct investment in their financial services sector. The United States and the European Union (with minor exceptions) are fully open to inward investment by foreign banks, insurance, and securities companies. As part of the deal, many Asian countries made important concessions that allow significant foreign participation in their financial services sectors for the first time.

WTO: UNRESOLVED ISSUES

Since the successes of the 1990s, the World Trade Organization has struggled to make progress on the international trade front. Confronted by a slower growing world economy after 2001, many national governments have been reluctant to agree to a fresh round of policies designed to reduce trade barriers. Political opposition to the WTO has been growing in many nations. Some politicians and nongovernmental organizations blame the WTO, as the public face of globalization, for a variety of ills, including high unemployment, environmental degradation, poor working conditions in developing nations, falling real wage rates among the lower-paid workers in developed nations, and rising income inequality. The rapid rise of China as a dominant trading nation has also played a role here. Reflecting sentiments like those toward Japan 35 years ago, many perceive China as failing to play by the international trading rules, even as it embraced the WTO.

Against this difficult political backdrop, much remains to be done on the international trade front. Four issues at the forefront of the agenda of the WTO are antidumping policies, the high level of protectionism in agriculture, the lack of strong protection for intellectual property rights in many nations, and continued high tariff rates on nonagricultural goods and services in many nations. We shall look at each in turn before discussing the latest round of talks between WTO members aimed at reducing trade barriers, the Doha Round, which began in 2001 and now seems to be stalled.

Antidumping Actions

Antidumping actions proliferated during the 1990s and 2000s. WTO rules allow countries to impose antidumping duties on foreign goods that are being sold more cheaply than at home or below their cost of production when domestic producers can show that they are being harmed. Unfortunately, the rather vague definition of what constitutes “dumping” has proved to be a loophole that many countries are exploiting to pursue protectionism.

Between 1995 and June 2022, WTO members had reported implementation of some 6,541 antidumping actions to the WTO. India initiated the largest number of antidumping actions, some 1,109; the United States initiated 855 over the same period; and the EU, 545. China accounted for 1,542 complaints, South Korea for 484, Taiwan for 334, the United States for 317, and Japan for 238.

Antidumping actions seem to be concentrated in certain sectors of the economy, such as basic metal industries (e.g., aluminum and steel), chemicals, plastics, and machinery and electrical equipment.²⁶ These sectors account for approximately 70 percent of all antidumping actions reported to the WTO. Since 1995, these four sectors have been characterized by periods of intense competition and excess productive capacity, which have led to low prices and profits (or losses) for firms in those industries. We might reasonably hypothesize, therefore, that the high level of antidumping actions in these industries represents an attempt by beleaguered manufacturers to use the political process in their nations to seek protection from foreign competitors, which they claim are engaging in unfair competition.

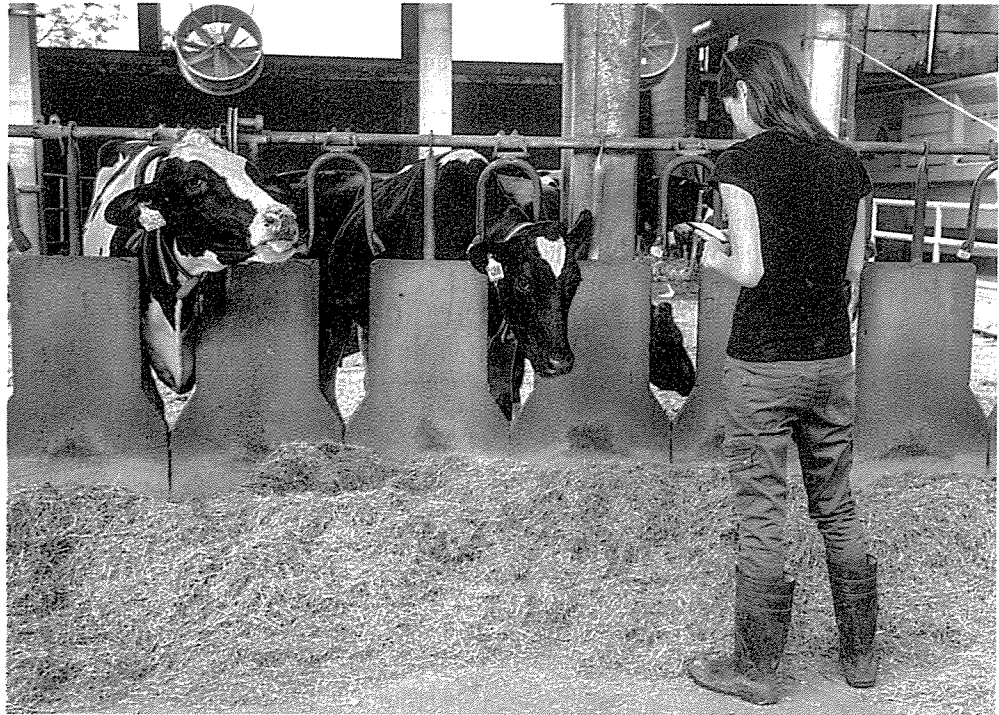
Although some of these claims may have merit, the process can become very politicized as representatives of businesses and their employees lobby government officials to “protect domestic jobs from unfair foreign competition,” and government officials, mindful of the need to get votes in future elections, oblige by pushing for antidumping actions. The WTO is clearly worried by the use of antidumping policies, suggesting that this use reflects persistent protectionist tendencies and pushing members to strengthen the regulations governing the imposition of antidumping duties.

Protectionism in Agriculture

Another focus of the WTO has been the high level of tariffs and subsidies in the agricultural sector of many economies. Tariff rates on agricultural products are generally much higher than tariff rates on manufactured products or services. For example, the average tariff rates on nonagricultural products among developed nations are generally between 3 and 5 percent of value. On agricultural products, however, the average tariff rates are 15 percent for Canada, 12.8 percent for the European Union, 19.3 percent for Japan, and 4.9 percent for the United States.²⁷ The implication is that consumers in countries with high tariffs are paying significantly higher prices than necessary for agricultural products imported from abroad, which leaves consumers with less money to spend on other goods and services.

The historically high tariff rates on agricultural products reflect a desire to protect domestic agriculture and traditional farming communities from foreign competition. In addition to high tariffs, agricultural producers also benefit from substantial subsidies. According to estimates from the Organisation for Economic Co-operation and Development (OECD), government subsidies on average account for about 17 percent of the cost of agricultural production in Canada, 21 percent in the United States, 35 percent in the European Union, and 59 percent in Japan.²⁸ OECD countries spend more than \$300 billion a year in agricultural subsidies.

Not surprisingly, the combination of high tariff barriers and subsidies introduces significant distortions into the production of agricultural products and international trade of those products. The net effect is to raise prices to consumers, reduce the volume of agricultural trade, and encourage the overproduction of products that are heavily subsidized (with the government typically buying the surplus). Because global trade in agriculture currently amounts to around 10 percent of total merchandise trade, the WTO argues that removing tariff barriers and subsidies could significantly boost the overall level of trade, lower prices to consumers, and raise global economic growth by freeing consumption and investment resources for more productive uses. According to estimates from the International Monetary Fund made back in the early 2000s, removal of tariffs and subsidies on agricultural products at that time would raise global economic welfare by \$128 billion annually.²⁹ Others suggested gains as high as \$182 billion.³⁰ More recent estimates are hard to come by, but the comparable figure for today, given economic growth and inflation, would probably be around \$300–\$400 billion.



Production operations at J.M. Larson Dairy.

Bonfanti Diego/Getty Images

The biggest defenders of the existing system have been the advanced nations of the world, which want to protect their agricultural sectors from competition by low-cost producers in developing nations. In contrast, developing nations have been pushing hard for reforms that would allow their producers greater access to the protected markets of the developed nations. Estimates suggest that removing all subsidies on agricultural production alone in OECD countries could return to the developing nations of the world three times more than all the foreign aid they currently receive from the OECD nations.³¹ In other words, free trade in agriculture could help jump-start economic growth among the world's poorer nations and alleviate global poverty.

Protection of Intellectual Property

Another issue that has become increasingly important to the WTO has been protecting intellectual property. The 1995 Uruguay agreement that established the WTO also contained an agreement to protect intellectual property (the Trade-Related Aspects of Intellectual Property Rights, or TRIPS, agreement). The TRIPS regulations oblige WTO members to grant and enforce patents lasting at least 20 years and copyrights lasting 50 years. Rich countries had to comply with the rules within a year. Poor countries, in which such protection was generally much weaker, had a grace period of 5 years, and the very poorest had 10 years. The basis for this agreement was a strong belief among signatory nations that the protection of intellectual property through patents, trademarks, and copyrights must be an essential element of the international trading system. Inadequate protections for intellectual property reduce the incentive for innovation. Because innovation is a central engine of economic growth and rising living standards, advocates have argued that a multilateral agreement is needed to protect intellectual property.

Without such an agreement, it is feared that producers in a country—let's say, India—might market imitations of patented innovations pioneered in a different country—say, the United States. This practice can affect international trade in two ways. First, it reduces the export opportunities in India for the original innovator in the United States. Second, to the extent that the Indian producer is able to export its pirated imitation to additional countries, it also reduces the export opportunities in those countries for the U.S. inventor. Also, one can argue that because the size of the total world market for the innovator is reduced, its incentive to pursue risky and expensive innovations is also reduced. The net effect would be less innovation in the world economy and less economic growth.

Market Access for Nonagricultural Goods and Services

Although the WTO and the GATT have made big strides in reducing the tariff rates on nonagricultural products, much work remains. Although most developed nations have brought their tariff rates on industrial products down to under 4 percent of value, exceptions still remain. In particular, while average tariffs are low, high tariff rates persist on certain imports into developed nations, which limit market access and economic growth. For example, Australia and South Korea, both OECD countries, still have bound tariff rates of 12.5 percent and 8.2 percent, respectively, on imports of transportation equipment (*bound tariff rates* are the highest rate that can be charged, which is often, but not always, the rate that is charged). In contrast, the bound tariff rates on imports of transportation equipment into the United States, European Union, and Japan are 3 percent, 4.1 percent, and 0 percent, respectively. A particular area for concern is high tariff rates on imports of selected goods from developing nations into developed nations.

In addition, tariffs on services remain higher than on industrial goods. The average tariff on business and financial services imported into the United States, for example, is 8.2 percent, into the EU it is 8.5 percent, and into Japan it is 19.7 percent.³² Given the rising value of cross-border trade in services, reducing these figures can be expected to yield substantial gains.

The WTO would like to bring down tariff rates still further and reduce the scope for the selective use of high tariff rates. The ultimate aim is to reduce tariff rates to zero. Although this might sound ambitious, 40 nations have already moved to zero tariffs on information technology goods, so a precedent exists. Empirical work suggests that further reductions in average tariff rates toward zero would yield substantial gains. One estimate by economists at the World Bank suggests that a broad global trade agreement coming out of the Doha negotiations could increase world income by \$263 billion annually, of which \$109 billion would go to poor countries.³³ Another estimate from the OECD suggests a figure closer to \$300 billion annually.³⁴ See the accompanying Country Focus for estimates of the benefits to the U.S. economy from free trade.

Looking farther out, the WTO would like to bring down tariff rates on imports of nonagricultural goods into developing nations. Many of these nations use the infant industry argument to

COUNTRY FOCUS

Estimating the Gains from Trade for the United States

A study published by the Institute for International Economics tried to estimate the gains to the U.S. economy from free trade. According to the study, due to reductions in tariff barriers under the GATT and WTO since 1947, by 2003 the gross domestic product (GDP) of the United States was 7.3 percent higher than would otherwise be the case. The benefits of that amounted to roughly \$1 trillion a year, or \$9,000 extra income for each American household per year.

The same study tried to estimate what would happen if America concluded free trade deals with all its trading partners, reducing tariff barriers on all goods and services to zero. Using several methods to estimate the impact, the study concluded that additional annual gains of between \$450 billion and \$1.3 trillion could be realized. This final march to free trade, according to the authors of the study, could safely be expected to raise incomes of the average American household by an additional \$4,500 per year.

The authors also tried to estimate the scale and cost of employment disruption that would be caused by a move to

universal free trade. Jobs would be lost in certain sectors and gained in others if the country abolished all tariff barriers. Using historical data as a guide, they estimated that 226,000 jobs would be lost every year due to expanded trade, although some two-thirds of those losing jobs would find reemployment after a year. Reemployment, however, would be at a wage that was 13 to 14 percent lower. The study concluded that the disruption costs would total some \$54 billion annually, primarily in the form of lower lifetime wages to those whose jobs were disrupted as a result of free trade. Offset against this, however, must be the higher economic growth resulting from free trade, which creates many new jobs and raises household incomes, creating another \$450 billion to \$1.3 trillion annually in *net* gains to the economy. In other words, the estimated annual gains from trade are far greater than the estimated annual costs associated with job disruption, and more people benefit than lose as a result of a shift to a universal free trade regime.

Source: S. C. Bradford, P. L. E. Grieco, and G. C. Hufbauer, "The Payoff to America from Global Integration," in *The United States and the World Economy: Foreign Policy for the Next Decade*, C. F. Bergsten, ed. (Washington, DC: Institute for International Economics, 2005).

justify the continued imposition of high tariff rates; however, ultimately these rates need to come down for these nations to reap the full benefits of international trade. For example, the bound tariff rates of 53.9 percent on imports of transportation equipment into India and 33.6 percent on imports into Brazil, by raising domestic prices, help protect inefficient domestic producers and limit economic growth by reducing the real income of consumers who must pay more for transportation equipment and related services.

The Doha Round

In 2001, the WTO launched a new round of talks between member states aimed at further liberalizing the global trade and investment framework. For this meeting, it picked the remote location of Doha in the Persian Gulf state of Qatar. The talks were originally scheduled to last three years, although they have already gone on for 21 years and are currently stalled.

The Doha agenda includes cutting tariffs on industrial goods and services, phasing out subsidies to agricultural producers, reducing barriers to cross-border investment, and limiting the use of anti-dumping laws. The talks are currently ongoing. They have been characterized by halting progress punctuated by significant setbacks and missed deadlines. A September 2003 meeting in Cancún, Mexico, broke down, primarily because participants could not agree on how to proceed with reducing agricultural subsidies and tariffs; the EU, United States, and India, among others, proved less than willing to reduce tariffs and subsidies to their politically important farmers, while countries such as Brazil and certain West African nations wanted free trade as quickly as possible. In 2004, both the United States and the EU made a determined push to start the talks again. Since then, however, no progress has been made, and the talks are in deadlock, primarily because of disagreements over how deep the cuts in subsidies to agricultural producers should be. As of 2022, the goal was to reduce tariffs for manufactured and agricultural goods by 60 to 70 percent and to cut subsidies to half of their current level—but getting nations to agree to these goals was proving exceedingly difficult.

BILATERAL AND MULTILATERAL TRADE AGREEMENTS

bilateral or multilateral trade agreements

Reciprocal trade agreements between two or more partners.

In response to the apparent failure of the Doha Round to progress, many nations have pushed forward with **bilateral or multilateral trade agreements**, which are reciprocal trade agreements between two or more partners. For example, in 2014 Australia and China entered into a bilateral free trade agreement. Similarly, in March 2012 the United States entered into a bilateral free trade agreement with South Korea. Under this agreement, 80 percent of U.S. exports of consumer and industrial products became duty free, and 95 percent of bilateral trade in industrial and consumer products will be duty free by 2017 (this agreement was revised in 2018 to further open up the Korean market to American exports). The agreement was estimated to boost U.S. GDP by some \$10 to \$12 billion. Under the Obama administration, the United States pursued two major multilateral trade agreements, one with 11 other Pacific Rim countries, including Australia, New Zealand, Japan, Malaysia, and Chile (the TPP), and another with the European Union. However, following the accession of Donald Trump to the presidency in the United States, the United States withdrew from the TPP (although the remaining 11 nations went ahead with a revised agreement), and the trade agreement being negotiated with the EU was put on ice.

Bilateral and multilateral trade agreements are designed to capture gain from trade beyond those agreements currently attainable under WTO treaties. Bilateral and multilateral trade agreements are allowed under WTO rules, and countries entering into these agreements are required to notify the WTO. As of 2022, some 355 regional or bilateral trade agreements were in force. Reflecting the lack of progress on the Doha Round, the number of such agreements has increased significantly since 2000 when only 94 were in force.

THE WORLD TRADING SYSTEM UNDER THREAT

In 2016, two events challenged the long-held belief that there was a global consensus behind the 70-year push to embrace free trade and lower barriers to the cross-border flow of goods and services. The first was the decision by the British to withdraw from the European Union following a national referendum (Brexit). We discuss Brexit in more detail in Chapter 9, but it is worth noting that the British withdrawal from what is arguably one of the most successful free trade zones in the world is a big setback for those who argue that free trade is a good thing. The second event was the

victory of Donald Trump in the 2016 U.S. presidential election. As discussed in Chapter 6, Trump appeared to hold neo-mercantilist views on trade. He opposed many free trade deals. Indeed, one of his first actions was to pull the United States out of the Trans-Pacific Partnership, a 12-nation free trade zone that was close to ratification. In early 2018, he placed tariffs on imports of solar panels, washing machines, steel, and aluminum into the United States. Trump also renegotiated NAFTA and expressed hostility to the WTO. Most notably, Trump sidestepped the WTO's rules-based arbitration mechanisms in his trade dispute with China. In late 2019, the Trump administration also blocked replacements for two members on the WTO's Appellate Body, which hears appeals in trade disputes. With just one member remaining, the Appellate Body was no longer able to hear new cases, which undermined the WTO's enforcement mechanism. As noted earlier, to date the Biden administration has continued the policies of the Trump administration with regard to the WTO.

The significance of these developments is that heretofore both Britain and the United States have been leaders in the global push toward greater free trade. While Britain still seems committed to free trade, despite the Brexit decision, the position of the United States, the world's largest economy, is less clear. The Biden administration to date has kept most of Trump's trade policies in place, including most notably the tariffs on imports from China. If the United States continues to turn its back on new free trade deals (such as the TPP), other nations could follow. If this happens, the impact on the world economy may be negative, resulting in greater protectionism, slower economic growth, and fewer jobs around the globe.

Writing in 2020, the director general of the WTO noted that in its first 25 years the binding rules governing international trade facilitated dramatic growth in cross-border activity. Since 1995, the dollar value of world trade has nearly quadrupled, while the real volume of world trade has expanded by 2.7 times. This growth far outstrips the two fold increase in world GDP over that period. Over the same period, average tariff rates between member countries have almost halved from 10.5 percent to 6.4 percent. However, he also noted that despite these considerable achievements:

the WTO faces challenges today that are unmatched in our relatively short history. Over the past two years, governments have introduced trade restrictions covering a substantial amount of international trade—affecting \$747 billion in global imports in the past year alone. The rising uncertainty about market conditions is causing businesses to postpone investment, weighing on growth and the future potential of our economies. How WTO member governments face up to these challenges will shape the course of the global economy for decades to come.³⁵

He was, of course, referring to the rise of tariffs, the trade dispute between the world's two largest economies, the United States and China, and the apparent belief of some politicians that bilateral agreements, and managed trade deals, are superior to the rules-based multilateral trade order embodied in the WTO.



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360° View: Managerial Implications



LO7-5

Explain the implications for managers of developments in the world trading system.

TRADE BARRIERS, FIRM STRATEGY, AND POLICY IMPLICATIONS

What are the implications for business practice? Why should the international manager care about the political economy of free trade or about the relative merits of arguments for free trade and protectionism? There are two answers to this question. The first concerns the impact of trade barriers on a firm's strategy. The second concerns the role that business firms can play in promoting free trade or trade barriers.

Trade Barriers and Firm Strategy

To understand how trade barriers affect a firm's strategy, consider first the material in Chapter 6. Drawing on the theories of international trade, we discussed how it makes sense for the firm to disperse its various production activities to those countries around the globe where they can be performed most efficiently. Thus, it may make sense for a firm to design and engineer its product in one country, to manufacture components in another, to perform final assembly operations in yet another country, and then export the finished product to the rest of the world.

Clearly, trade barriers constrain a firm's ability to disperse its productive activities in such a manner. First and most obviously, tariff barriers raise the costs of exporting products to a country (or of exporting partially finished products between countries). This may put the firm at a competitive disadvantage relative to indigenous competitors in that country. In response, the firm may then find it economical to locate production facilities in that country so that it can compete on even footing. Second, quotas may limit a firm's ability to serve a country from locations outside that country. Again, the response by the firm might be to set up production facilities in that country—even though doing so may result in higher production costs.

Such reasoning was one of the factors behind the rapid expansion of Japanese automaking capacity in the United States during the 1980s and 1990s. This followed the establishment of a VER agreement between the United States and Japan that limited U.S. imports of Japanese automobiles. More recently, Donald Trump's threat to impose high tariffs on companies that shift their production to other nations in order to reduce costs—and then export goods back to the United States—forced some enterprises to rethink their outsourcing strategy. In particular, a number of automobile companies, including Ford and General Motors, modified their plans to shift some production to factories in Mexico and announced plans to expand U.S. production.³⁶

Third, to conform to local content regulations, a firm may have to locate more production activities in a given market than it would otherwise. Again, from the firm's perspective, the consequence might be to raise costs above the level that could be achieved if each production activity were dispersed to the optimal location for that activity. And finally, even when trade barriers do not exist, the firm may still want to locate some production activities in a given country to reduce the threat of trade barriers being imposed in the future.

All these effects are likely to raise the firm's costs above the level that could be achieved in a world without trade barriers. The higher costs that result need not translate into a significant competitive disadvantage relative to other foreign firms, however, if the countries imposing trade barriers do so to the imported products of all foreign firms, irrespective of their national origin. But when trade barriers are targeted at exports from a particular nation, firms based in that nation are at a competitive disadvantage to firms of other nations. The firm may deal with such targeted trade barriers by moving production into the country imposing barriers. Another strategy may be to move production to countries whose exports are not targeted by the specific trade barrier.

Finally, the threat of antidumping action limits the ability of a firm to use aggressive pricing to gain market share in a country. Firms in a country also can make strategic use of antidumping measures to limit aggressive competition from low-cost foreign producers. For example, the U.S. steel industry has been very aggressive in bringing antidumping actions against foreign steel-makers, particularly in times of excess capacity due to weak global demand for steel. In 1998 and 1999, the United States faced a surge in low-cost steel imports as a severe recession in Asia left producers there with excess capacity. The U.S. producers filed several complaints with the

International Trade Commission. One argued that Japanese producers of hot rolled steel were selling it at below cost in the United States. The ITC agreed and levied tariffs ranging from 18 to 67 percent on imports of certain steel products from Japan (these tariffs are separate from the steel tariffs discussed earlier).³⁷

Policy Implications

As noted in Chapter 6, business firms are major players on the international trade scene. Because of their pivotal role in international trade, firms can and do exert a strong influence on government policy toward trade. This influence can encourage protectionism, or it can encourage the government to support the WTO and push for open markets and freer trade among all nations. Government policies with regard to international trade can have a direct impact on business.

Consistent with strategic trade policy, examples can be found of government intervention in the form of tariffs, quotas, antidumping actions, and subsidies helping firms and industries establish a competitive advantage in the world economy. In general, however, the arguments contained in this chapter and in Chapter 6 suggest that government intervention has three drawbacks. Intervention can be self-defeating because it tends to protect the inefficient rather than help firms become efficient global competitors. Intervention is dangerous; it may invite retaliation and trigger a trade war. Finally, intervention is unlikely to be well executed, given the opportunity for such a policy to be captured by special-interest groups. Does this mean that business should simply encourage government to adopt a laissez-faire free trade policy?

Most economists would probably argue that the best interests of international business are served by a free trade stance but not a laissez-faire stance. The long-run interests of the business community are probably best served by encouraging the government to aggressively promote greater free trade by, for example, strengthening the WTO. Business probably has much more to gain from government efforts to open protected markets to imports and foreign direct investment than from government efforts to support certain domestic industries in a manner consistent with the recommendations of strategic trade policy.

This conclusion is reinforced by a phenomenon we touched on in Chapter 1—the increasing integration of the world economy and internationalization of production that has occurred over the past two decades. We live in a world where many firms of all national origins increasingly depend on globally dispersed production systems for their competitive advantage. Such systems are the result of freer trade. Freer trade has brought great advantages to firms that have exploited it and to consumers who benefit from the resulting lower prices. Given the danger of retaliatory action, business firms that lobby their governments to engage in protectionism must realize that by doing so, they may be denying themselves the opportunity to build a competitive advantage by constructing a globally dispersed production system. By encouraging their governments to engage in protectionism, their own activities and sales overseas may be jeopardized if other governments retaliate. This does not mean a firm should never seek protection in the form of antidumping actions and the like, but it should review its options carefully and think through the larger consequences.

Key Terms

free trade, p. 207

General Agreement on Tariffs and Trade (GATT), p. 207

import tariff, p. 208

specific tariff, p. 208

ad valorem tariff, p. 208

export tariff, p. 209

export ban, p. 209

subsidy, p. 210

import quota, p. 211

tariff rate quota, p. 211

voluntary export restraint (VER), p. 212

quota rent, p. 212

local content requirement (LCR), p. 212

administrative trade policies, p. 213

dumping, p. 213

antidumping policies, p. 213

countervailing duties, p. 213

infant industry argument, p. 217

strategic trade policy, p. 218

Smoot-Hawley Act, p. 221

bilateral or multilateral trade agreements, p. 228

Summary

This chapter described how the reality of international trade deviates from the theoretical ideal of unrestricted free trade reviewed in Chapter 6. In this chapter, we reported the various instruments of trade policy, reviewed the political and economic arguments for government intervention in international trade, reexamined the economic case for free trade in light of the strategic trade policy argument, and looked at the evolution of the world trading framework. While a policy of free trade may not always be the theoretically optimal policy (given the arguments of the new trade theorists), in practice it is probably the best policy for a government to pursue. In particular, the long-run interests of business and consumers may be best served by strengthening international institutions such as the WTO. Given the danger that isolated protectionism might escalate into a trade war, business probably has far more to gain from government efforts to open protected markets to imports and foreign direct investment (through the WTO) than from government efforts to protect domestic industries from foreign competition. The chapter made the following points:

1. Trade policies such as tariffs, subsidies, antidumping regulations, and local content requirements tend to be pro-producer and anti-consumer. Gains accrue to producers (who are protected from foreign competitors), but consumers lose because they must pay more for imports and domestic goods.
2. There are two types of arguments for government intervention in international trade: political and economic. Political arguments for intervention are concerned with protecting the interests of certain groups, often at the expense of other groups, or with promoting goals with regard to foreign policy, human rights, consumer protection, and the like. Economic arguments for intervention are about boosting the overall wealth of a nation.
3. A common political argument for intervention is that it is necessary to protect jobs. However, political intervention often hurts consumers, and it can be self-defeating. Countries sometimes argue that it is important to protect certain industries for reasons of national security. Some argue that government should use the threat to intervene in trade policy as a

bargaining tool to open foreign markets. This can be a risky policy; if it fails, the result can be higher trade barriers.

4. The infant industry argument for government intervention contends that to let manufacturing get a foothold, governments should temporarily support new industries. In practice, however, governments often end up protecting the inefficient.
5. Strategic trade policy suggests that, with subsidies, government can help domestic firms gain first-mover advantages in global industries where economies of scale are important. Government subsidies may also help domestic firms overcome barriers to entry into such industries.
6. The problems with strategic trade policy are twofold: (a) Such a policy may invite retaliation, in which case all will lose, and (b) strategic trade policy may be captured by special-interest groups, which will distort it to their own ends.
7. The GATT was a product of the postwar free trade movement. The GATT was successful in lowering trade barriers on manufactured goods and commodities. The move toward greater free trade under the GATT appeared to stimulate economic growth.
8. The completion of the Uruguay Round of GATT talks and the establishment of the World Trade Organization in 1995 strengthened the world trading system by extending GATT rules to services, increasing protection for intellectual property, reducing agricultural subsidies, and enhancing monitoring and enforcement mechanisms.
9. Trade barriers act as a constraint on a firm's ability to disperse its various production activities to optimal locations around the globe. One response to trade barriers is to establish more production activities in the protected country.
10. Business may have more to gain from government efforts to open protected markets to imports and foreign direct investment than from government efforts to protect domestic industries from foreign competition.

Critical Thinking and Discussion Questions

1. Do you think governments should consider human rights when granting preferential trading rights to countries? What are the arguments for and against taking such a position?
2. Whose interests should be the paramount concern of government trade policy: the interests of producers (businesses and their employees) or those of consumers?
3. Given the arguments relating to the new trade theory and strategic trade policy, what kind of trade policy should business be pressuring government to adopt?
4. You are an employee of a U.S. firm that produces personal computers in Thailand and then exports them to the United States and other countries for sale. The personal computers were originally produced in Thailand

to take advantage of relatively low labor costs and a skilled workforce. Other possible locations considered at the time were Malaysia and Hong Kong. The U.S. government decides to impose punitive 100 percent ad valorem tariffs on imports of computers from Thailand to punish the country for administrative trade barriers that restrict U.S. exports to Thailand. How should your firm respond? What does this tell you about the use of targeted trade barriers?

5. Reread the Management Focus, “Huawei Export Ban Hits U.S. Firms.” Is the United States justified in banning U.S. companies, or foreign semiconductor makers that use American-made production technology, from doing business with Huawei? What are the possible long-term implications of this policy? Who might benefit? Who might lose? What policy stance would you recommend the Biden administration adopt with regard to this export ban?

globalEDGE Research Task globalEDGE.msu.edu

Use the globalEDGE™ website (globalEDGE.msu.edu) to complete the following exercises:

1. You work for a pharmaceutical company that hopes to provide products and services in New Zealand. Yet management’s current knowledge of this country’s trade policies and barriers is limited. After searching a resource that summarizes the *import and export regulations*, outline the most important foreign trade barriers your firm’s managers must keep in mind while developing a strategy for entry into New Zealand’s pharmaceutical market.
2. The number of member nations of the World Trade Organization has increased considerably in recent years. In addition, some nonmember countries have observer status in the WTO. Such status requires accession negotiations to begin within five years of attaining this preliminary position. Visit the WTO’s website to identify a list of current members and observers. Identify the last five countries that joined the WTO as members. Also, examine the list of current observer countries. Do you notice anything in particular about the countries that have recently joined or have observer status?

CLOSING CASE



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The Jones Act

The Jones Act is the shorthand name for the United States Merchant Marine Act of 1920, whose primary author was Senator Wesley Jones of Washington State. The goal of the legislation was to ensure the existence of a thriving U.S.-owned commercial shipping industry, a topic that had become important during World War I, when blockades underscored the close link between maritime commerce and warfare. Among other things, the law requires that goods transported by ship from one U.S. destination to another be carried on U.S.-owned ships built in the United States and crewed by U.S. legal residents and citizens. When advocating the law, Senator Jones argued it would ensure that, in the case of war, there would always be a large supply of American-made, American-owned, and American-crewed ships ready to supply American commerce, even if conditions became hazardous.

The Jones Act effectively outlaws vessels that are foreign built, owned, and crewed from operating on inland waterways in the United States and from transporting cargo between two U.S. ports—an activity known as *cabotage*. America is hardly alone in having some form of cabotage restriction—many countries have similar laws—but the United States is 1 of only 11 countries that fully excludes foreign vessels. According to an analysis by the OECD, the Jones Act is the most restrictive example of cabotage laws in the world.

Defenders of the Jones Act claim the law is necessary to sustain American jobs and a viable shipping industry. Critics note that by restricting foreign competition, the Jones Act has raised the cost of shipping goods between U.S. ports and shifted demand toward alternatives such as rail and trucking that would otherwise be more expensive. As a consequence of the Jones Act, only 2 percent of U.S. freight travels by sea, compared to 40 percent in the European Union, where cabotage between member states is permitted, and 15 percent in Australia, where vessels do not need to be built domestically to participate in cabotage. Interestingly, when New Zealand relaxed its cabotage restrictions in 1994, the country experienced a 20–25 percent decrease in coastal shipping rates over the next six years.

The central issue is that U.S.-built, owned, and crewed ships are far more expensive to purchase and operate than the typical ship used for international cargo. American-built coastal ships cost between \$190 and \$250 million to build, whereas the cost to build a similar vessel in a foreign shipyard is about \$30 million. The largest producers of cargo ships are to be found in South Korea, China, and Japan. American shipbuilders account for less than 1 percent of total global output. Outside of ships for American coastal routes and inland waterways, little demand exists for the high-cost product of American shipyards. The operating costs of U.S.-owned vessels engaged in foreign commerce are estimated to be 2.7 times greater than that of their foreign competitors, due partially to higher labor and maintenance costs. As a result of their high costs, U.S. vessels carry only about 2 percent of the world's cargo, down from 25 percent some 60 years ago.

If the intention of the Jones Act was to create a thriving domestic maritime industry, the law clearly has not worked. Nor does it seem that the Jones Act has had the desired impact on national security. When U.S. forces were deployed to Saudi Arabia during Operations Desert Shield and Desert Storm, a much larger share of their equipment and supplies (26.6 percent) were carried by foreign-flagged vessels than U.S.-flagged vessels (12.7 percent), and only one of the U.S.-flagged ships was Jones Act compliant.

The Jones Act has very obvious consequences for places such as Alaska, Hawaii, Puerto Rico, and Guam. According to a Federal Reserve Board study, shipping a container from the American mainland to Puerto Rico can cost twice as much as shipping to nearby islands such as Jamaica. This is a major reason for the higher cost of living in Puerto Rico compared to both the mainland and other Caribbean Islands. Furthermore, the Jones Act has had some unintended consequences. It has shifted significant demand to rail and trucking. These modes of transportation would be more expensive than shipping containers by sea if the Jones Act did not exist. One recent study suggested that, without the Jones Act, maritime transportation would be one-third cheaper than rail when moving containers by sea between Long Beach, California, and Memphis, Tennessee, a journey that would involve shipping through the Panama Canal to New Orleans and another 400 miles on a river barge up the lower Mississippi River. Relative to rail, the shipping cost would be reduced from \$5,500 per container to \$3,500. In addition, the environmental costs of ground transportation are higher. Maritime shipping produces approximately 10–40 grams of carbon dioxide to carry one ton of cargo one kilometer. In contrast, rail transport produces 20–150 grams, and trucking produces 60–150 grams. Heavy use of trucking also contributes to traffic congestion and results in more wear on road and bridge infrastructure, raising infrastructure maintenance costs.

Despite these criticisms, the Jones Act seems unlikely to be repealed anytime soon. When Puerto Rico was hit by a hurricane in 2017, President Trump was hesitant to temporarily waive the Jones Act even for a natural disaster, noting that “A lot of people who are in the shipping industry don’t want it lifted.” In the end, Trump relaxed the Jones Act requirements for just 10 days. For its part, the Biden administration has continued to support the Jones Act, citing the need to protect American maritime jobs and national security interests.

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Case Discussion Questions

1. What were the intended consequences of the Jones Act? Do these intentions seem reasonable? How successful has the Jones Act been in achieving its original objectives?
2. What have been the unintended consequences of the Jones Act? How significant have these consequences been?
3. Why is the Jones Act often viewed as a restraint on trade?
4. What do you think would happen if the Jones Act was repealed, and all restrictions on foreign shipping between U.S. ports were removed? What would the implications be for the U.S. economy, for alternative transportation services, for the U.S. shipbuilding and merchant marine industries, and for the environment?
5. Why do you think that the Jones Act is still in place? What does this teach you about the politics of trade policy? What would it take to repeal the Jones Act?

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